INSTRUMENTATION ENGINEERING LABORATORY 2

01068102

LAB 1 : Time Domain Analysis and Design

Instrumentation and Control Engineering,
School of Engineering,
King Mongkut's Institute of Technology Ladkrabang

Assoc. Prof. Dr. Arjin Numsomran

วัตถุประสงค์

1. นักศึกษาสามารถเรียกใช้งานคำสั่งพื้นฐานบน MATLAB ได้
2. นักศึกษาสามารถสร้างและเรียกใช้สคริปและฟังก์ชันของโปรแกรม MATLAB ได้
3. นักศึกษาสามารถเรียกใช้งานคำสั่งพื้นฐานในการวิเคราะห์ระบบควบคุมได้
4. นักศึกษาสามารถวิเคราะห์และออกแบบระบบควบคุมในโดเมนเวลา
5. นักศึกษาสามารถใช้งาน SISOTOOL และ LTIVIEW เพื่อวิเคราะห์และออกแบบระบบควบคุมในโดเมนเวลาได้

หัวข้อที่เกี่ยวข้อง

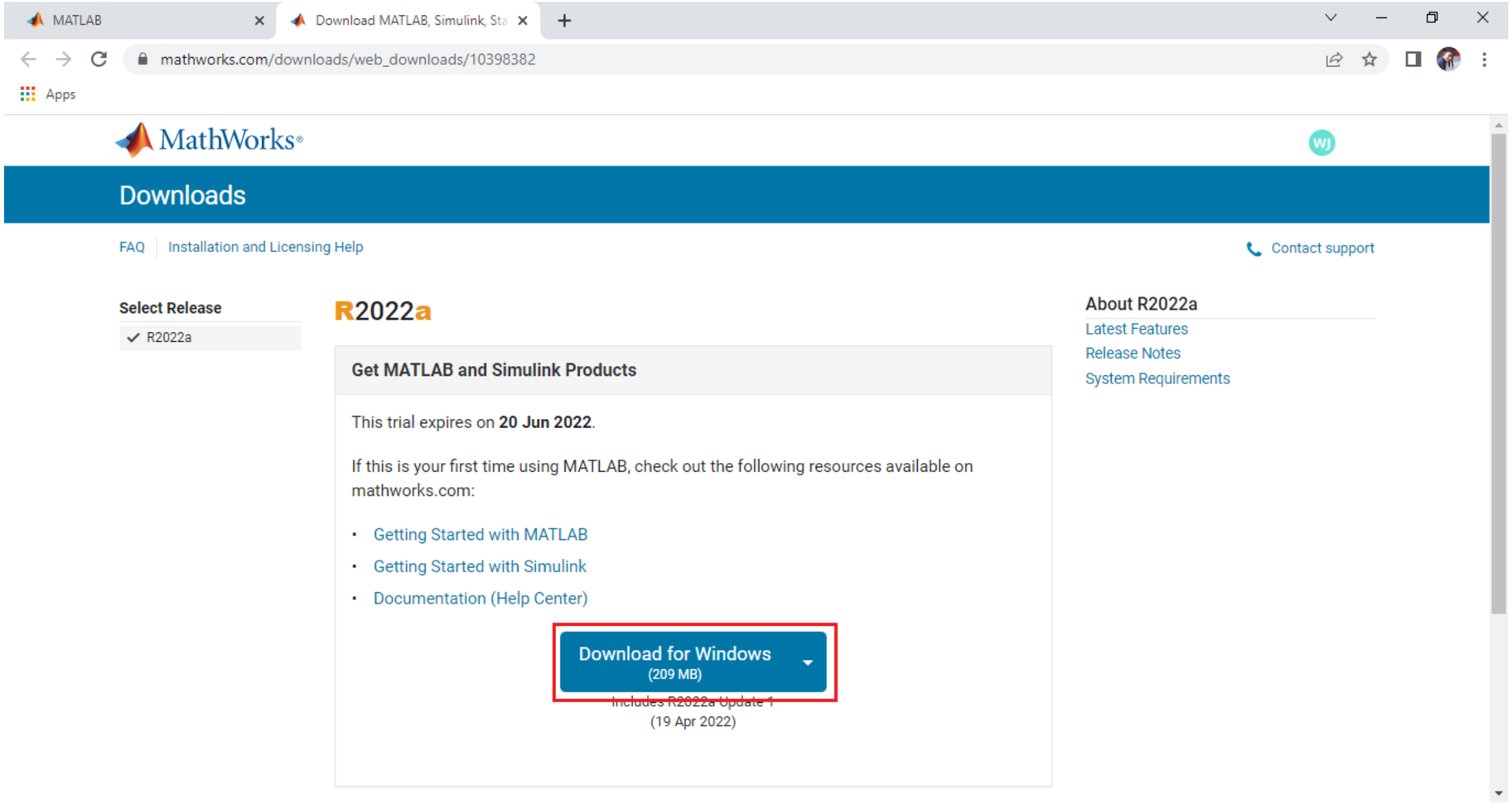
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1. การติดตั้งโปรแกรม MATLAB
2. พื้นฐานการใช้โปรแกรม MATLAB สำหรับระบบควบคุม
3. การวิเคราะผลตอบสนองโดยใช้เครื่องมือ LTIVIEW
4. การออกแบบระบบควบคุมโดยใช้เครื่องมือ SISOTOOL
5. การใช้เครื่องมือ Simulink
6. PID Tuning
7. คำถามท้ายการทดลอง

1.1 การติดตั้งโปรแกรม MATLAB

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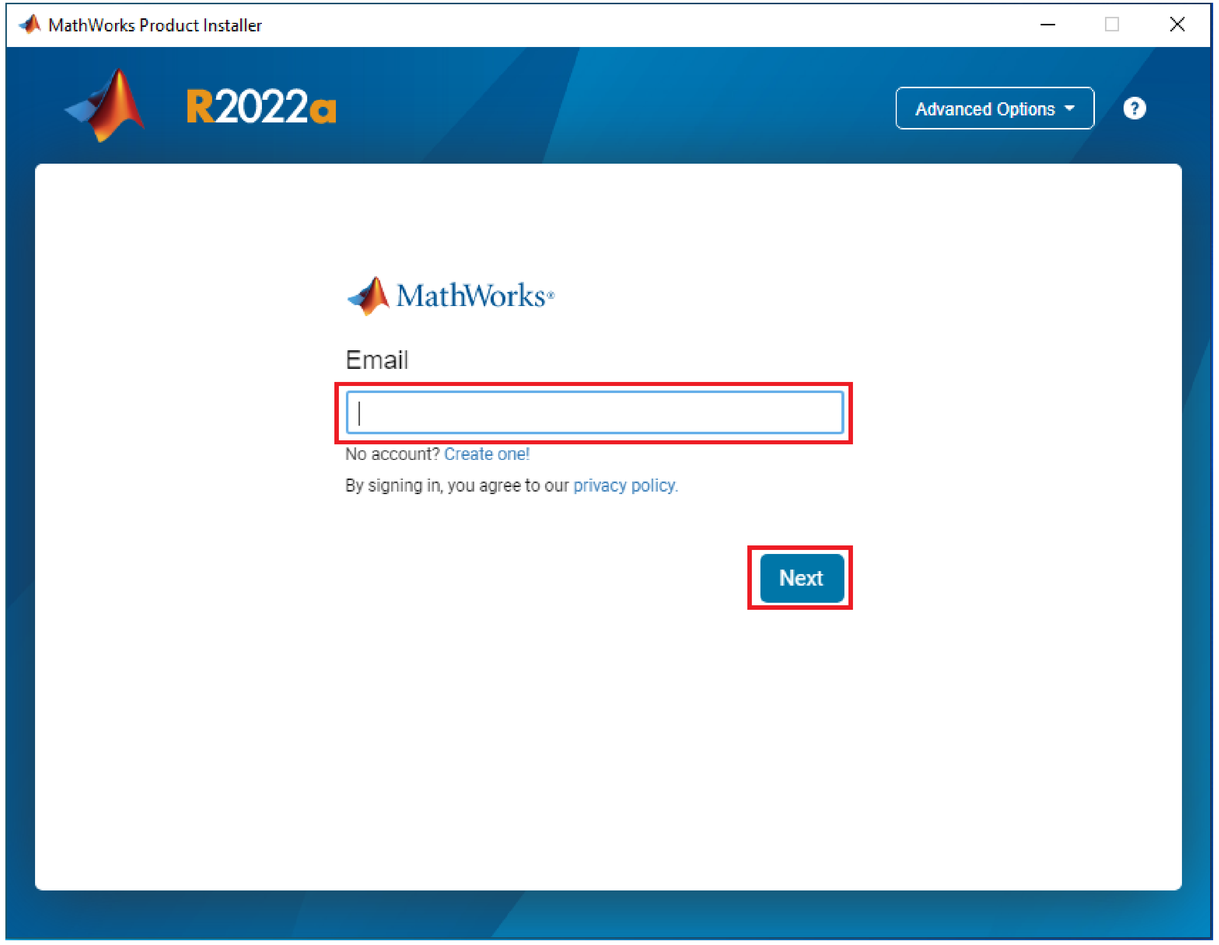
- Download

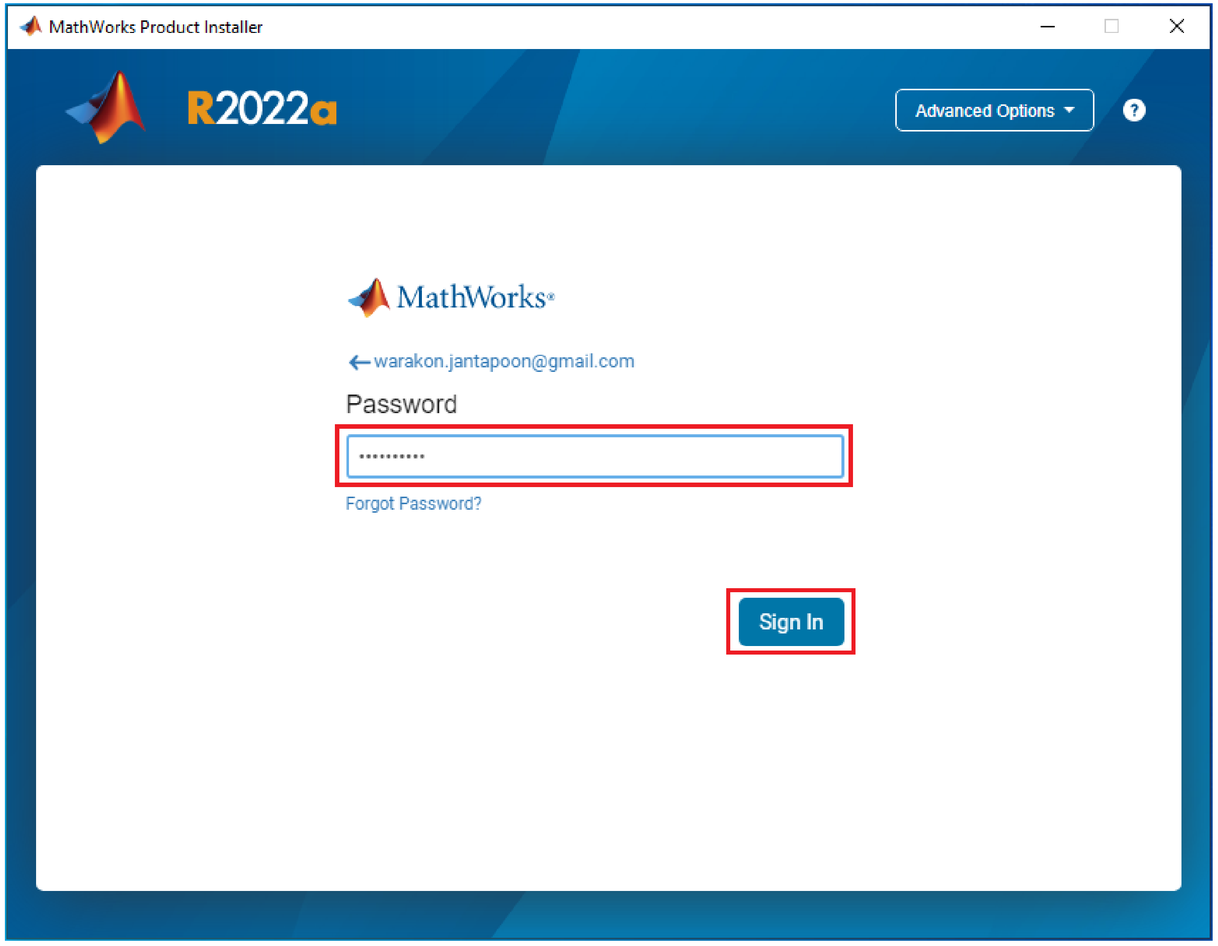


1.1 การติดตั้งโปรแกรม MATLAB

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- Installing (1)

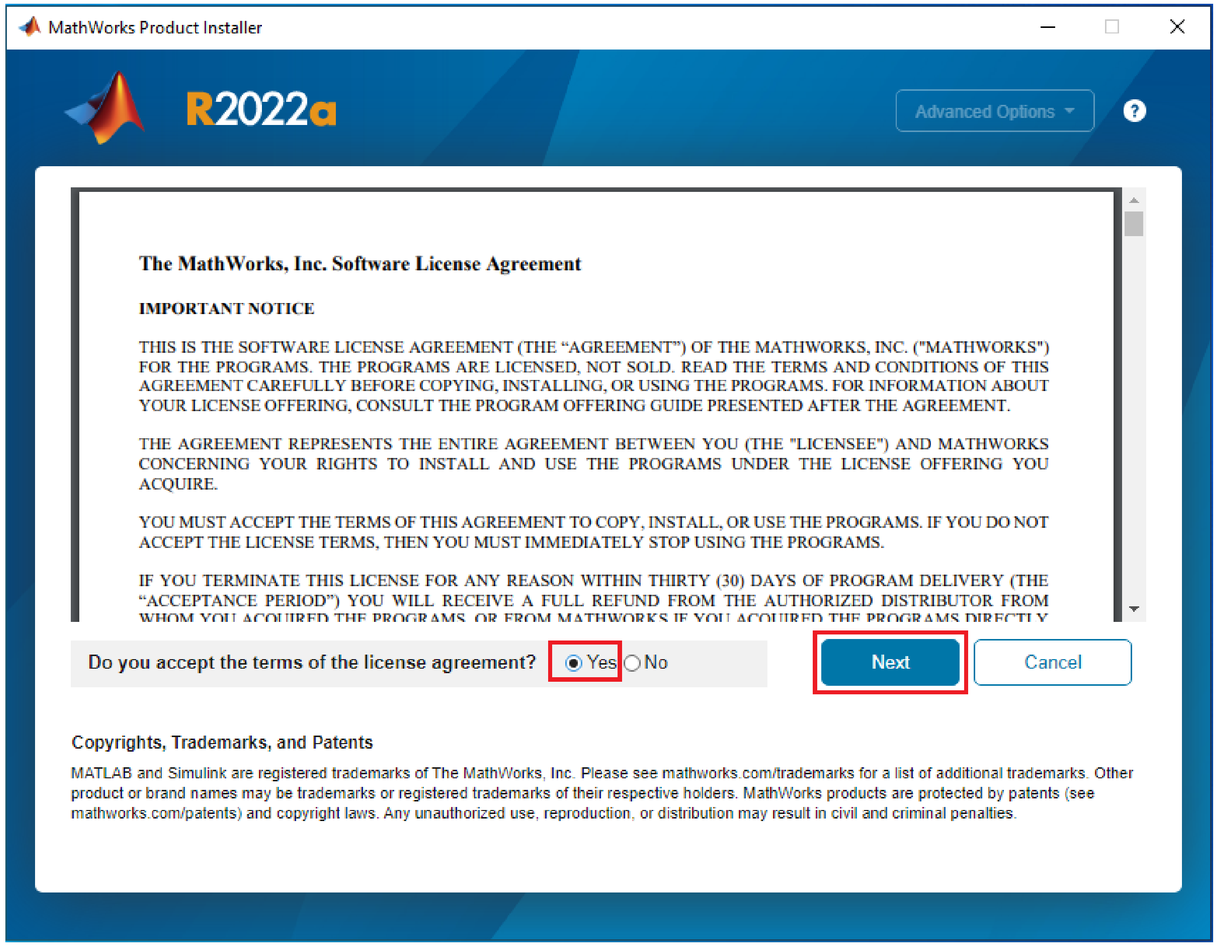


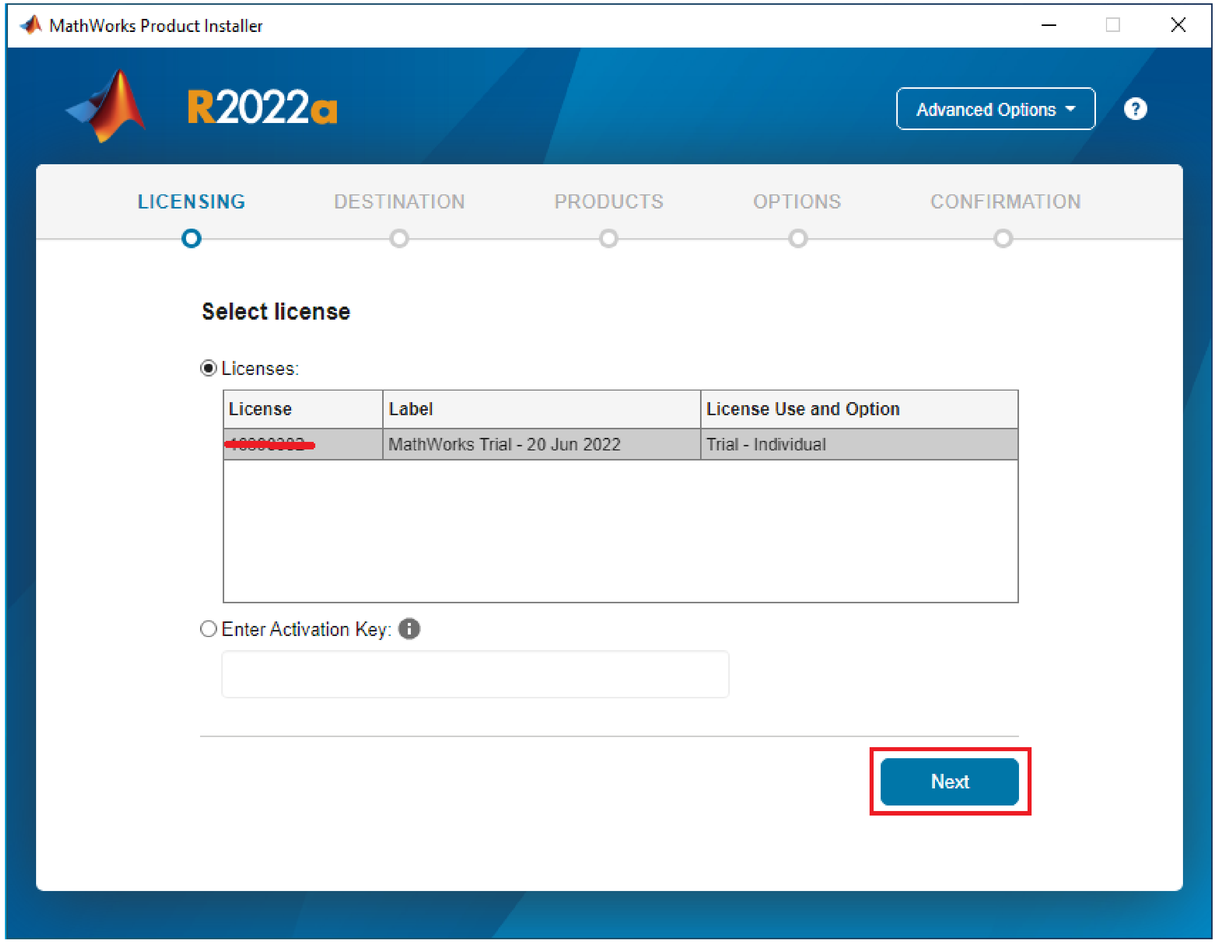


1.1 การติดตั้งโปรแกรม MATLAB

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- Installing (2)

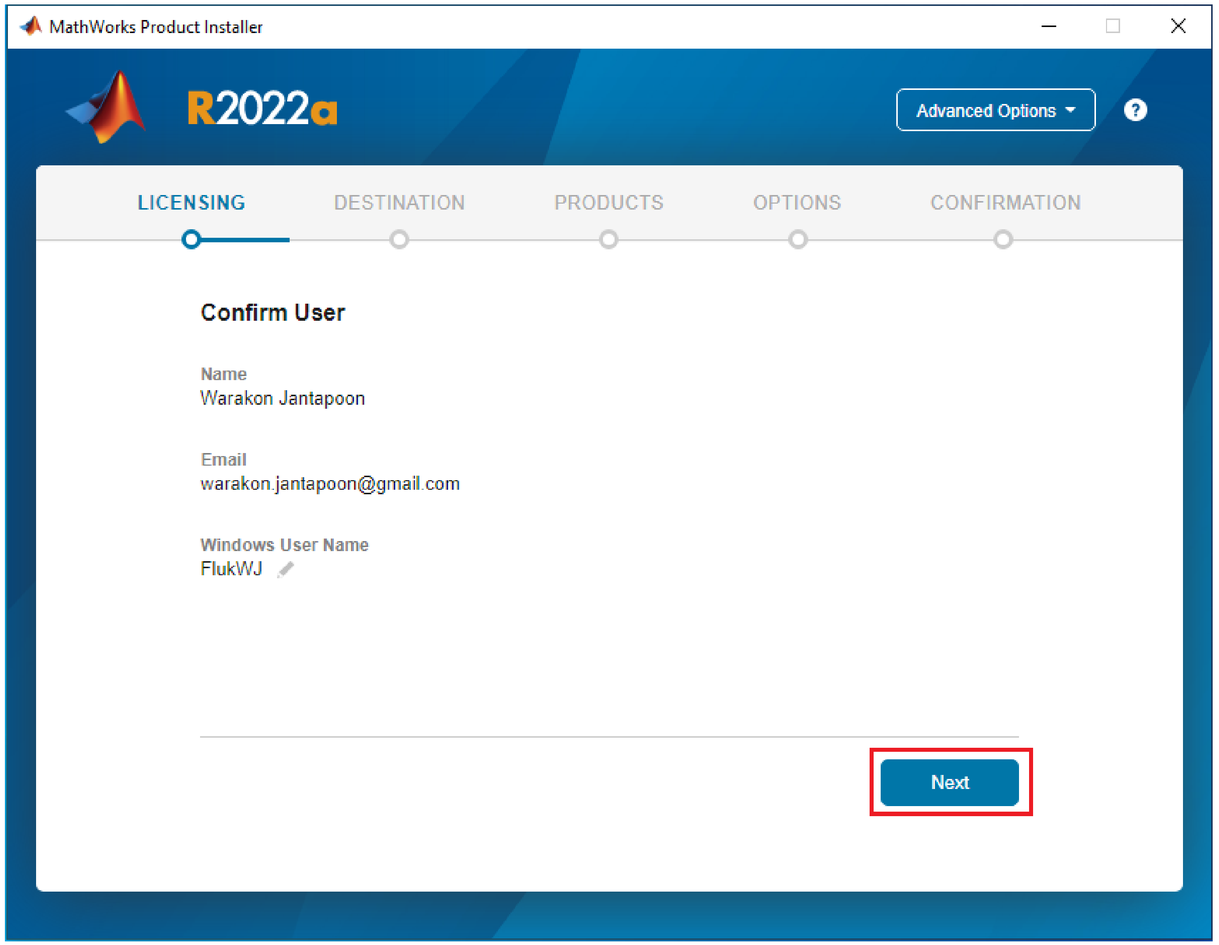


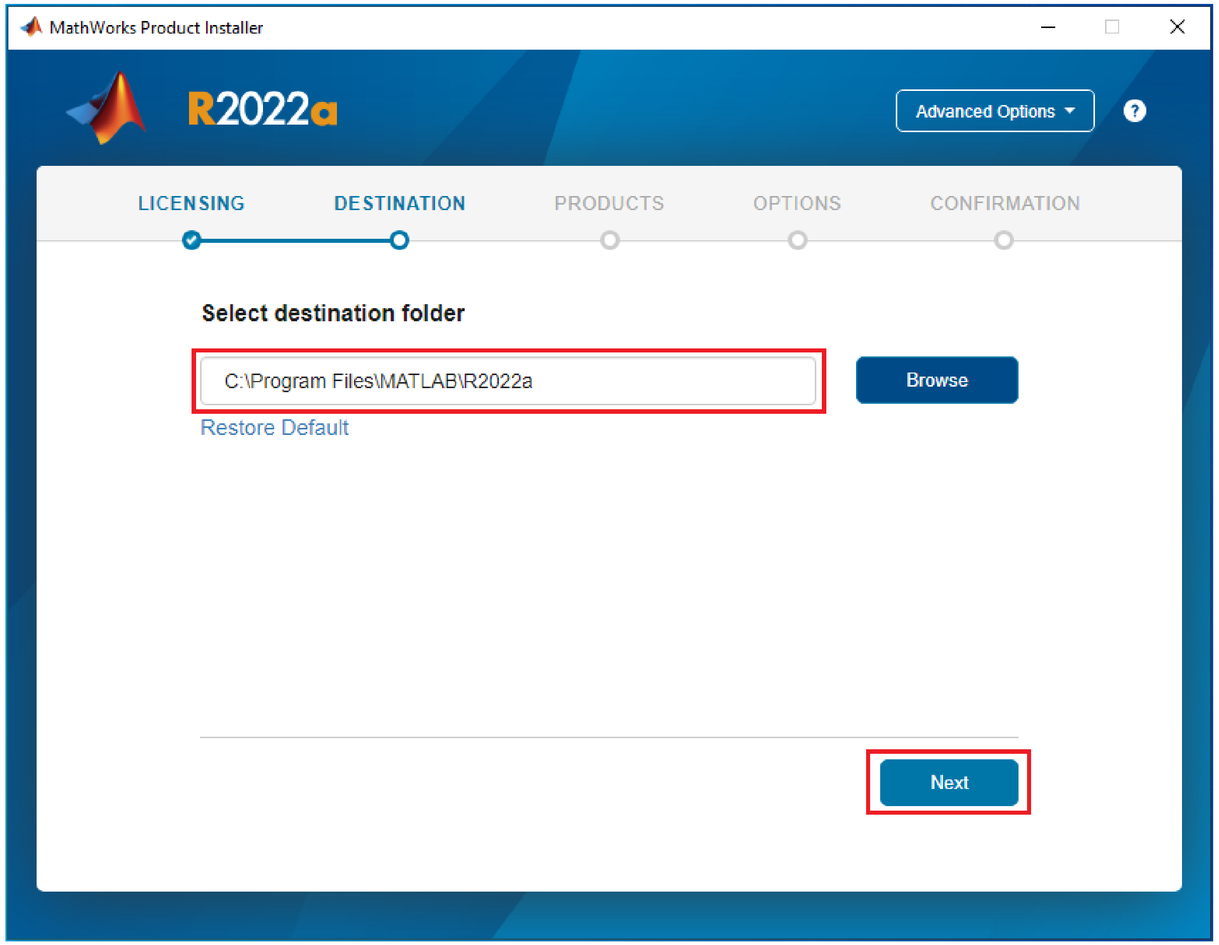


1.1 การติดตั้งโปรแกรม MATLAB

SCHOOL OF ENGINEERING KMITL

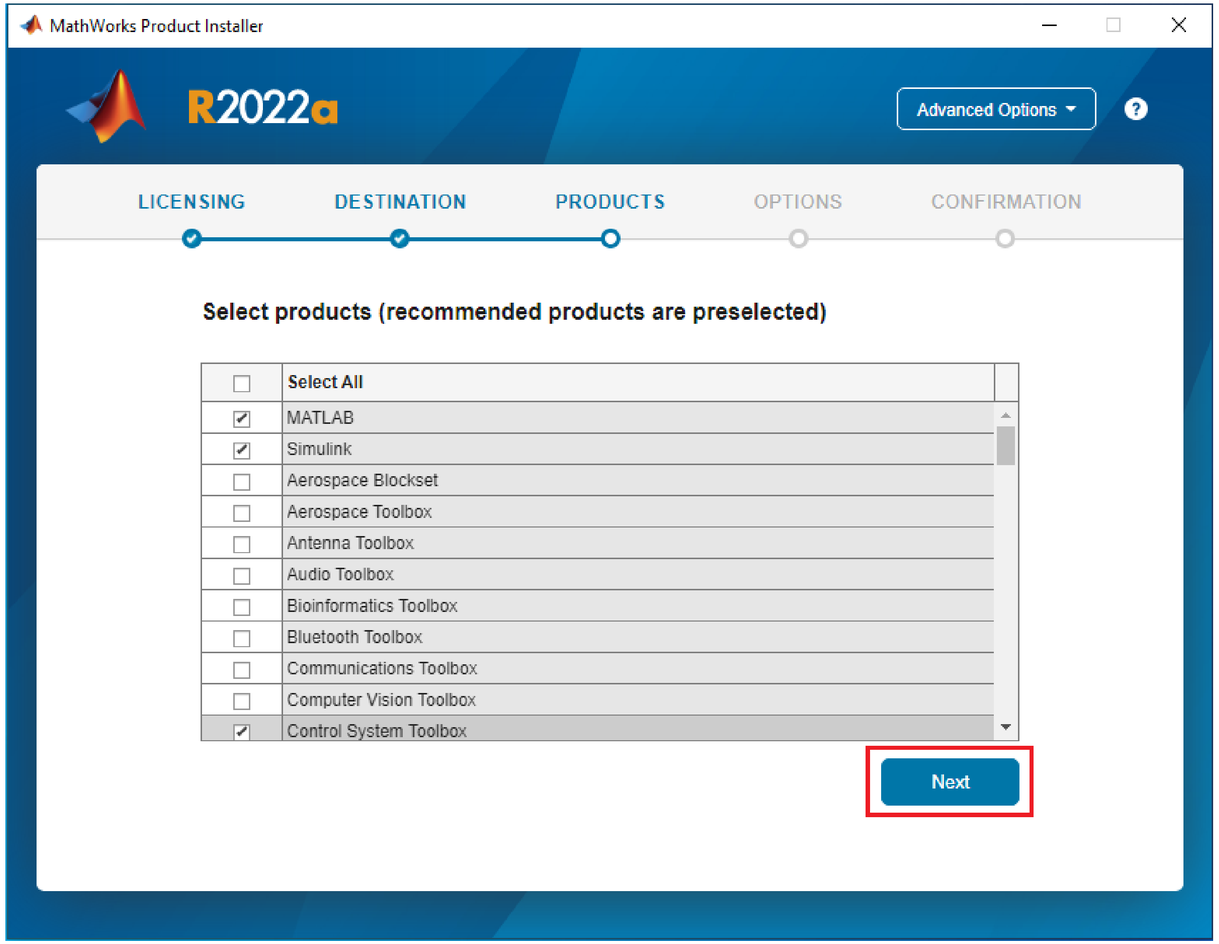
- Installing (3)

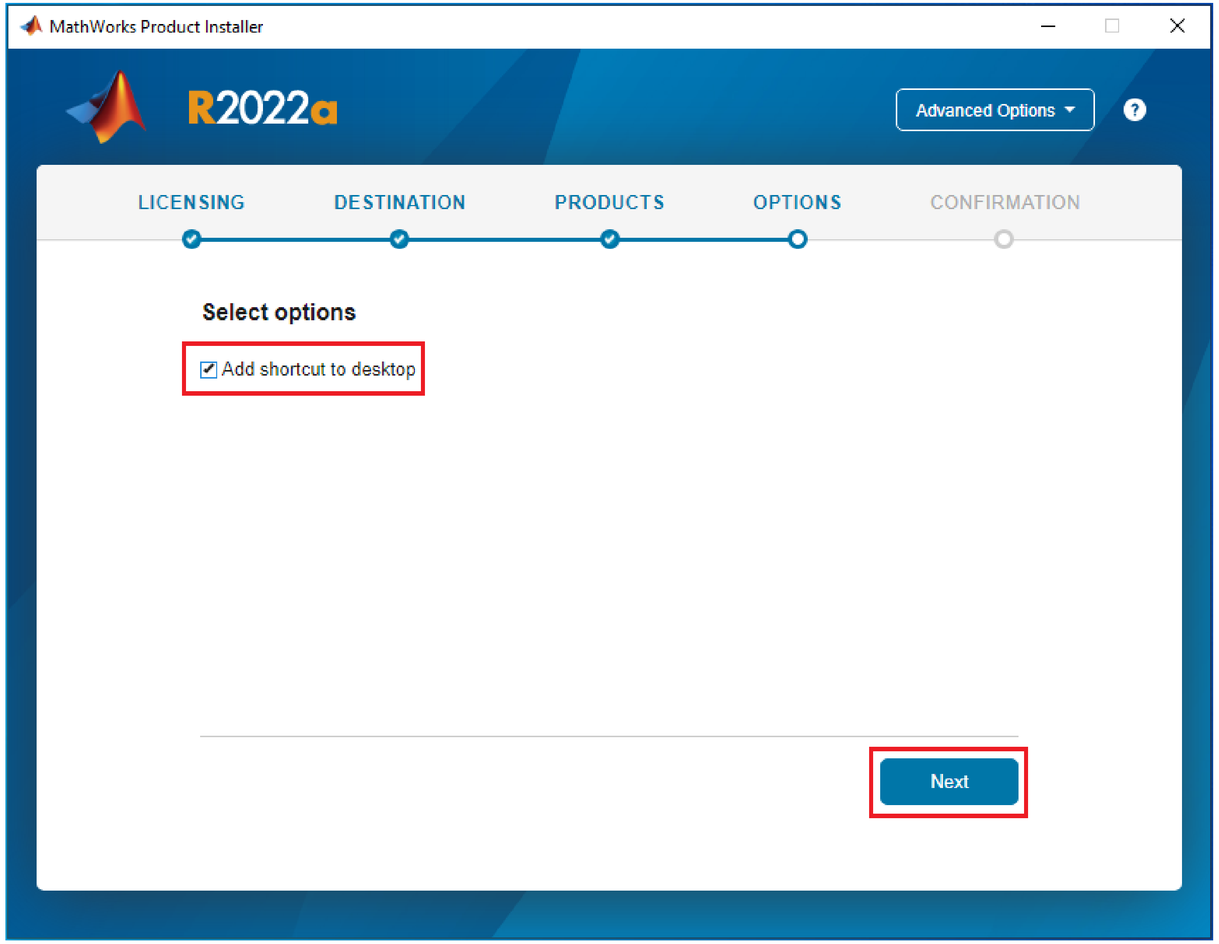




1.1 การติดตั้งโปรแกรม MATLAB

- Installing (4)

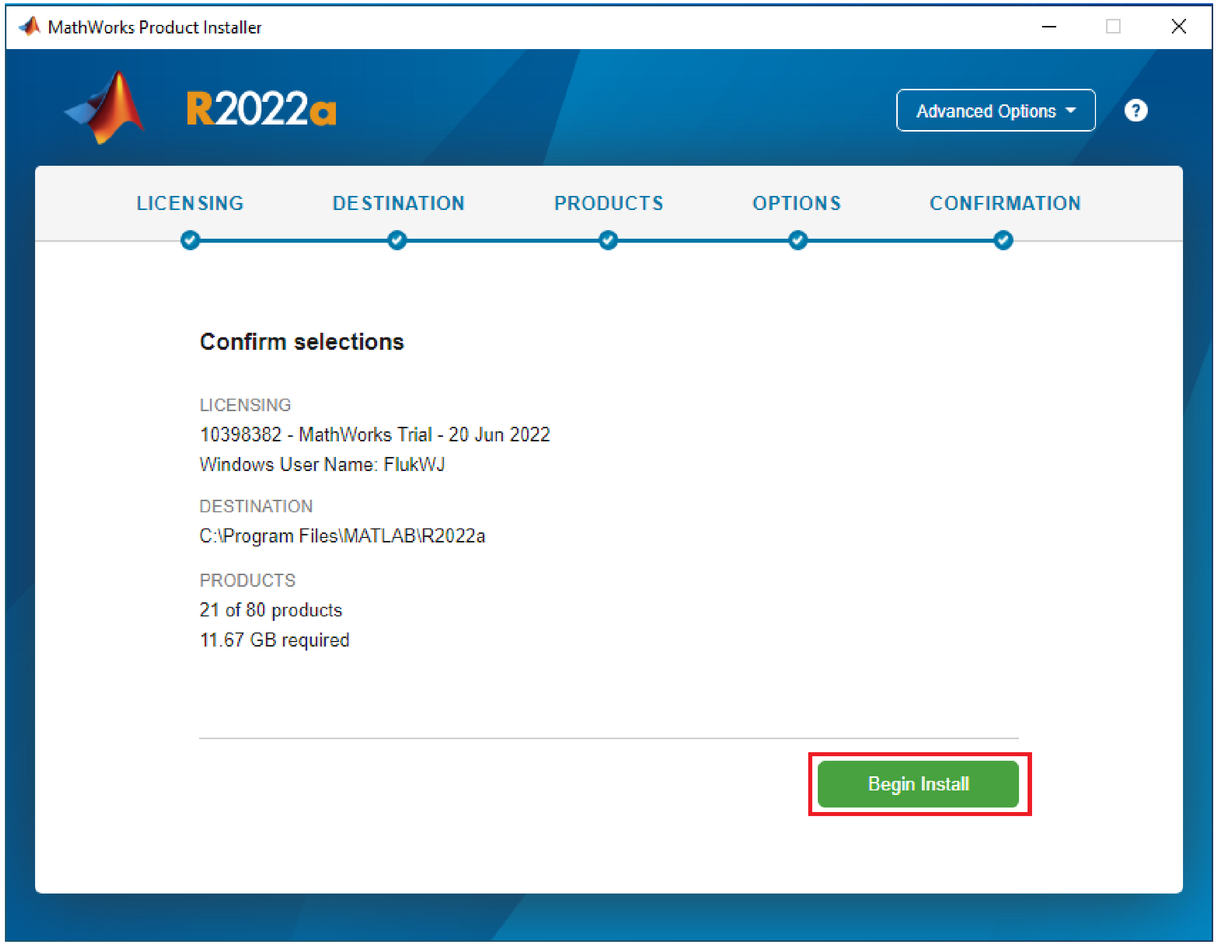


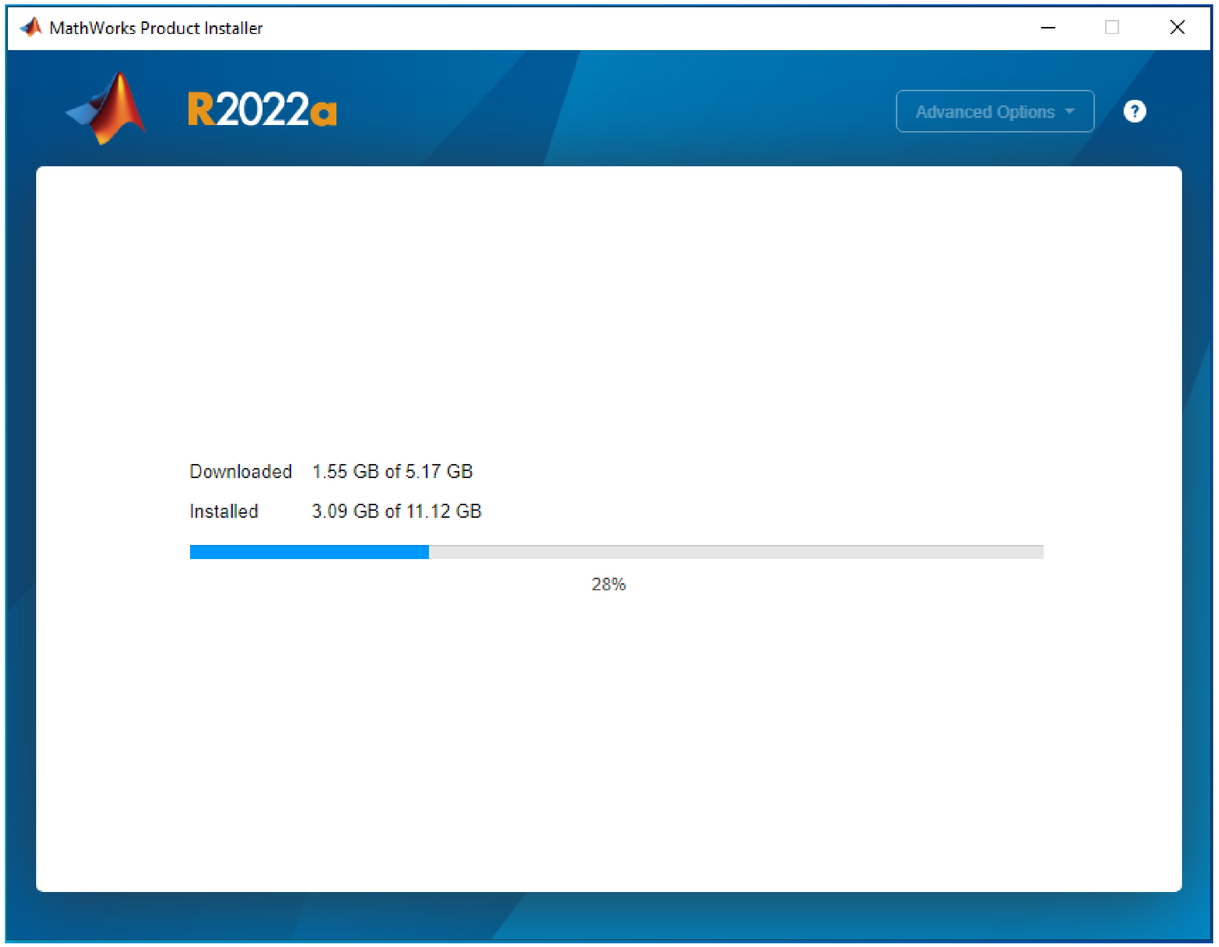


1.1 การติดตั้งโปรแกรม MATLAB

SCHOOL OF ENGINEERING KMITL

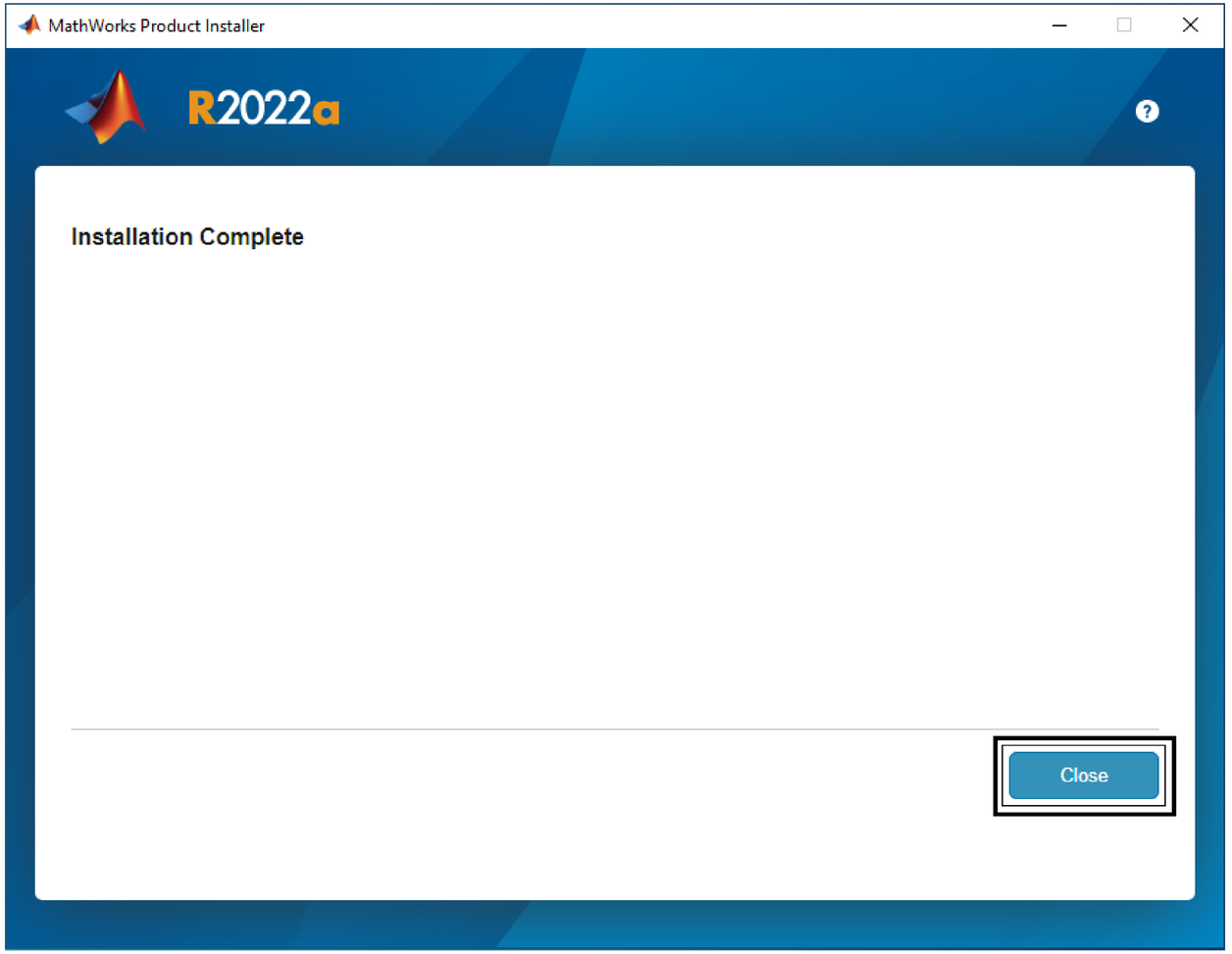
- Installing (5)





1.1 การติดตั้งโปรแกรม MATLAB

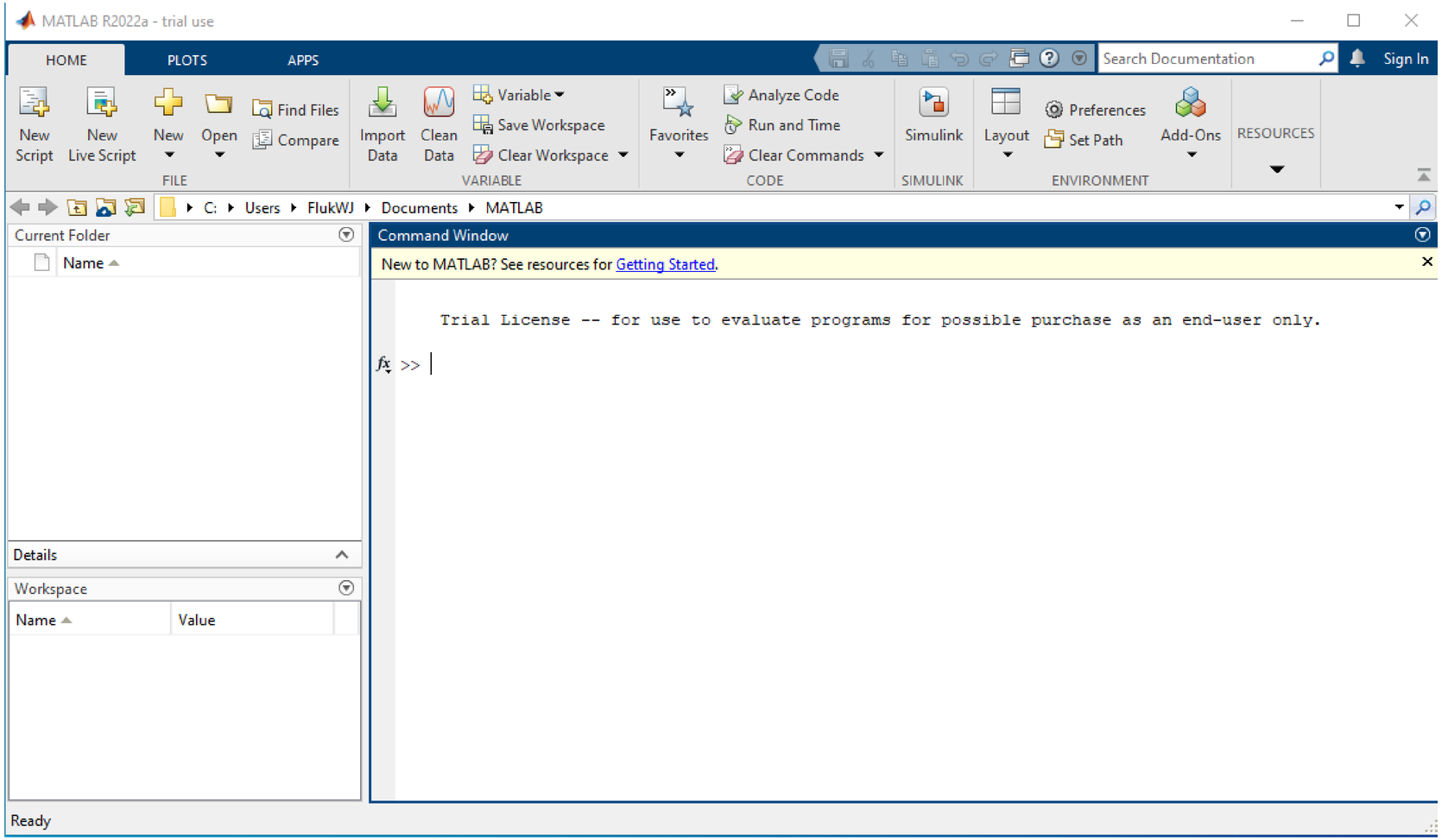
- Installing (6)



1.1 การติดตั้งโปรแกรม MATLAB

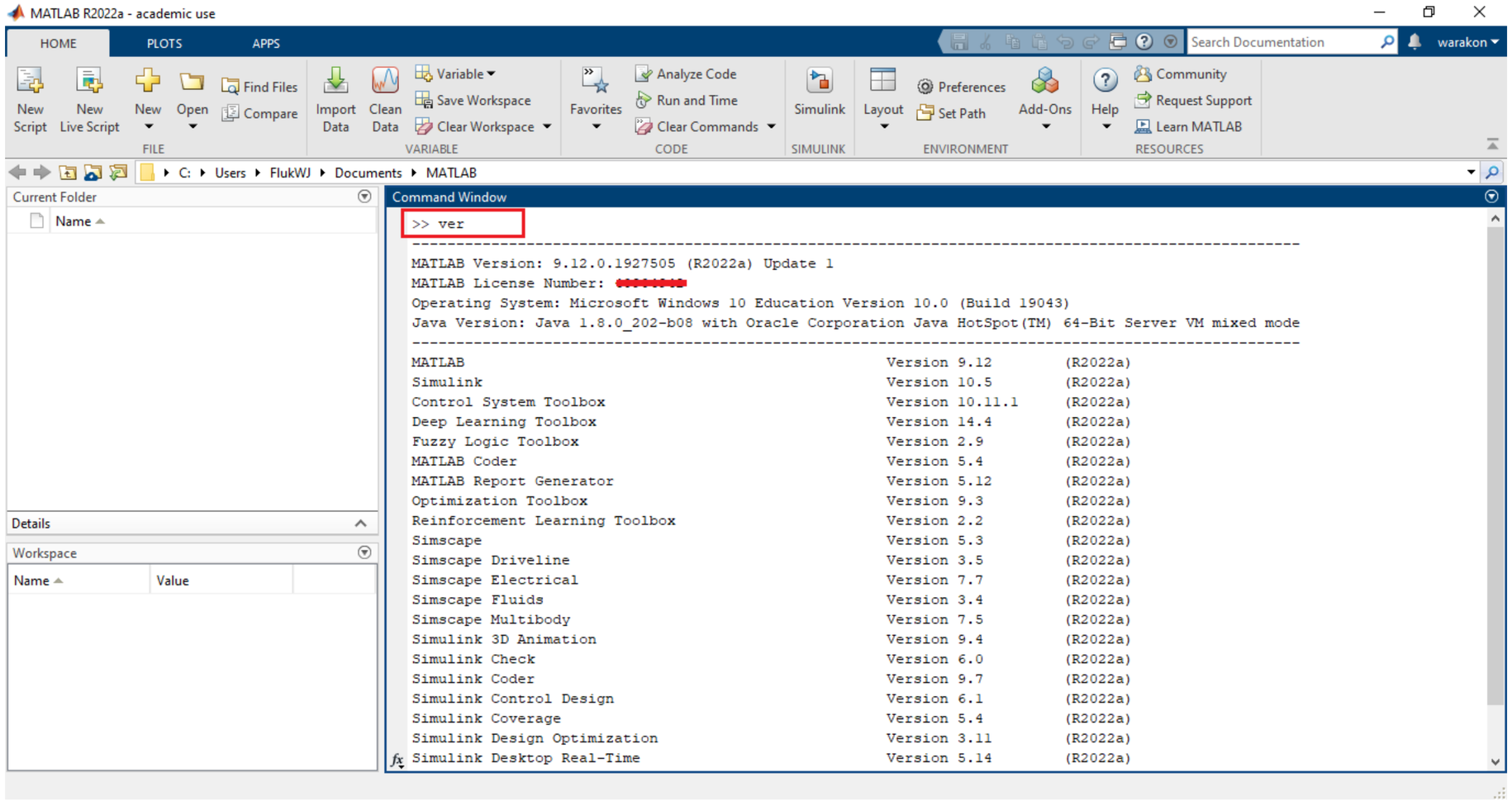
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- Open the MATLAB



1.1 การติดตั้งโปรแกรม MATLAB

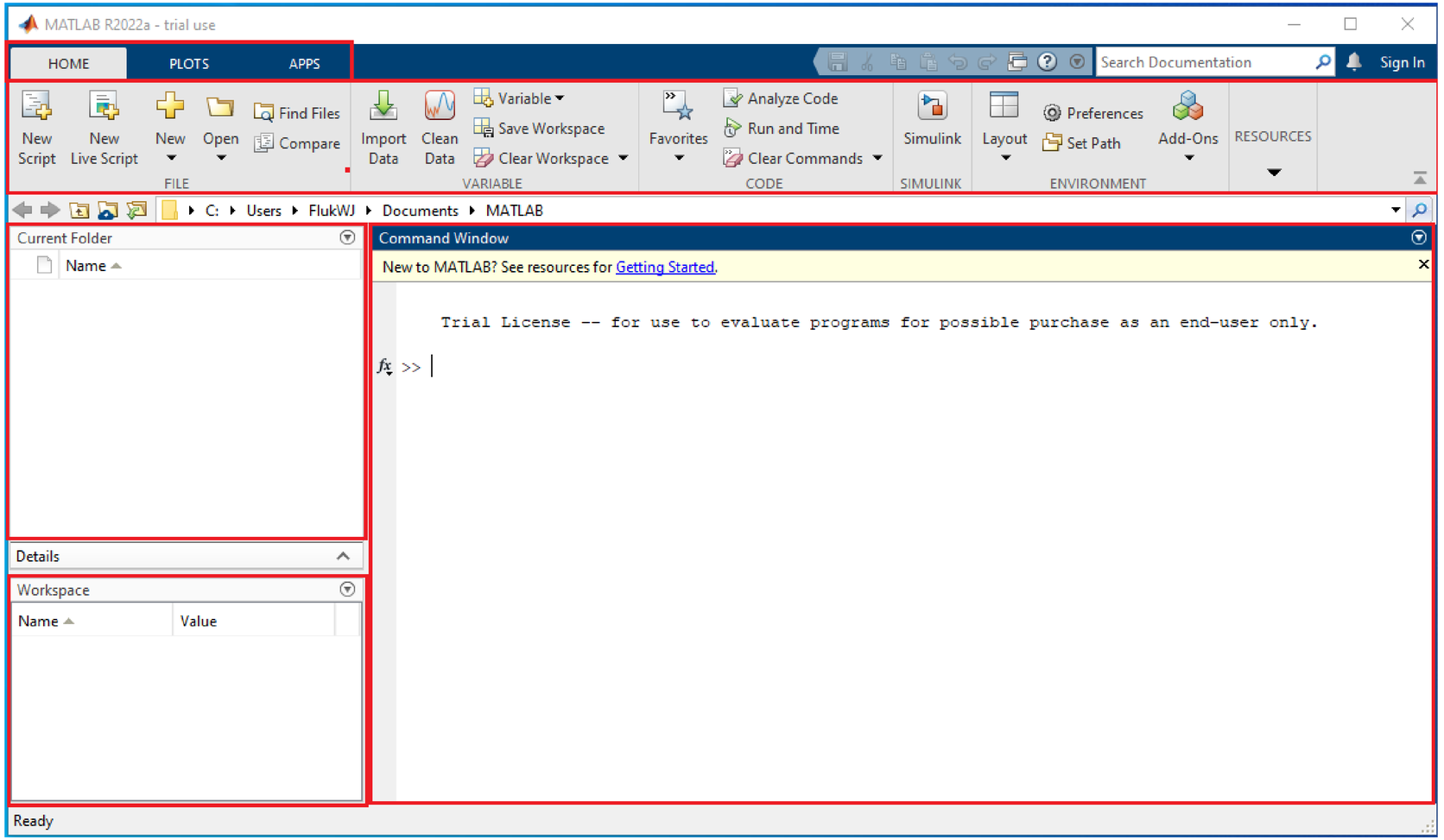
- >>ver



1.2 พื้นฐานการใช้โปรแกรม MATLAB สำหรับระบบควบคุม

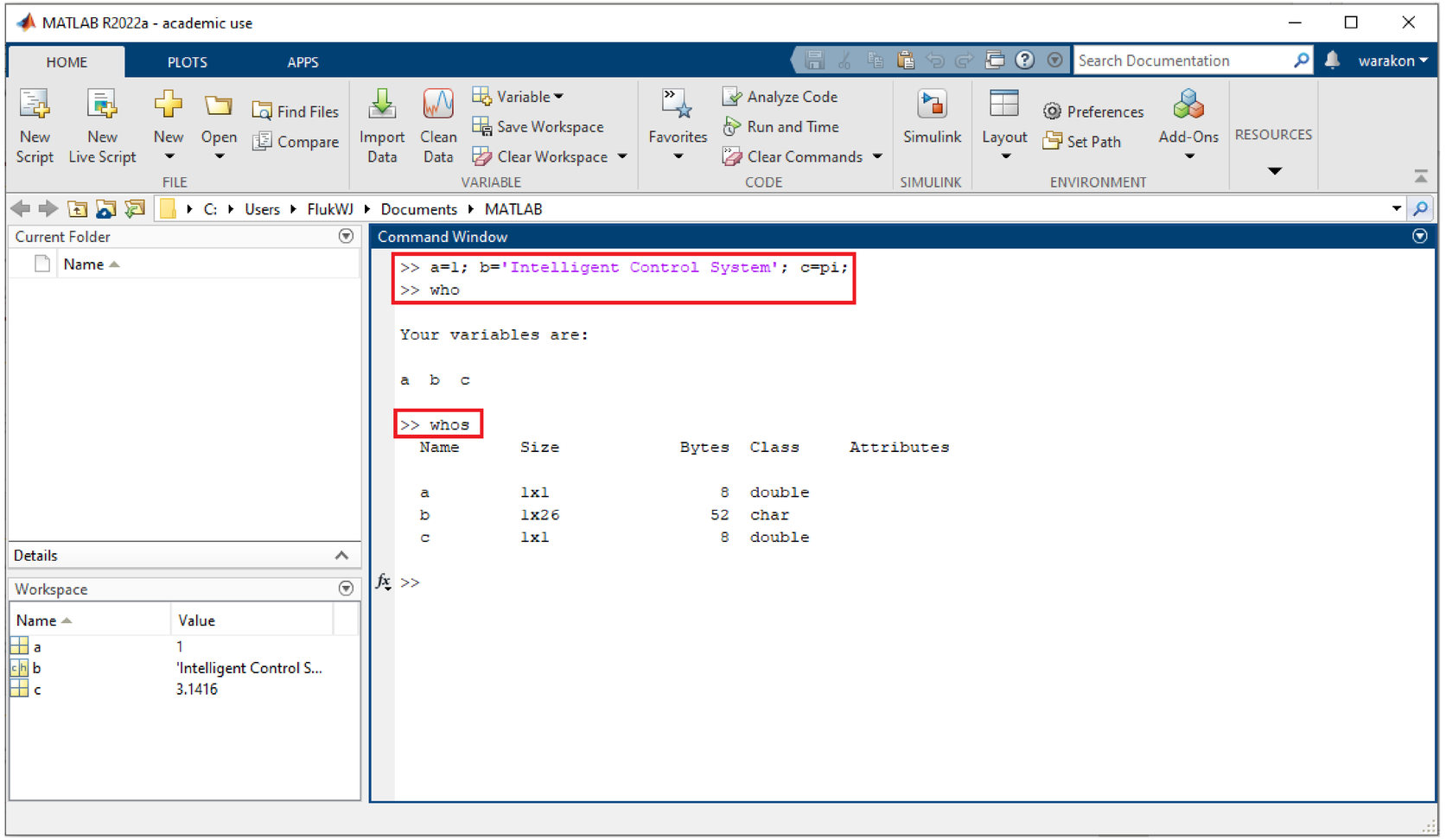
SCHOOL OF ENGINEERING KMITL

- MATLAB structure



1.2 พื้นฐานการใช้โปรแกรม MATLAB สำหรับระบบควบคุม

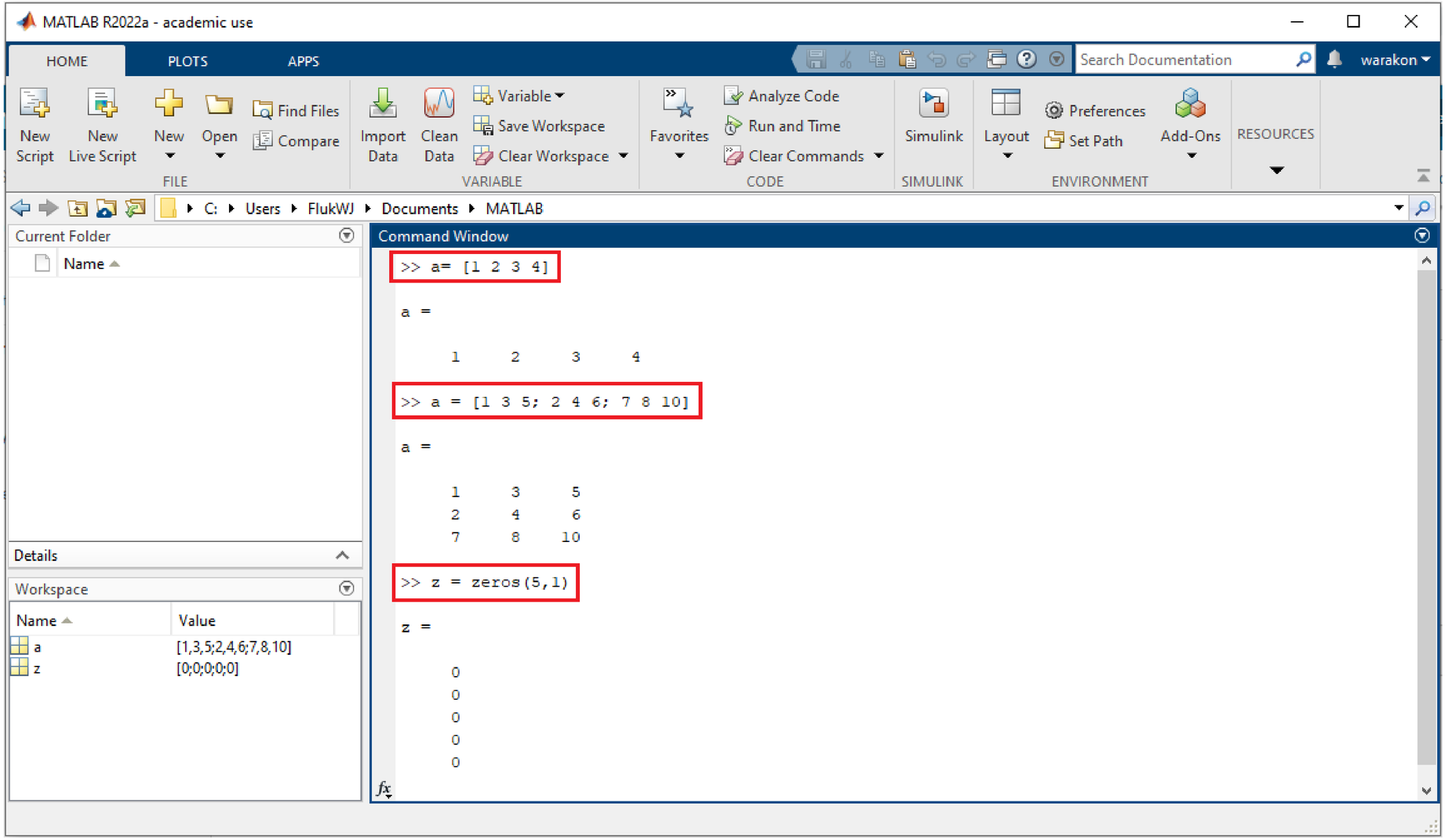
- >> a=1; b='Control System Lab.'; c=pi;
- >> who
- >> whos



1.2 พื้นฐานการใช้โปรแกรม MATLAB สำหรับระบบควบคุม

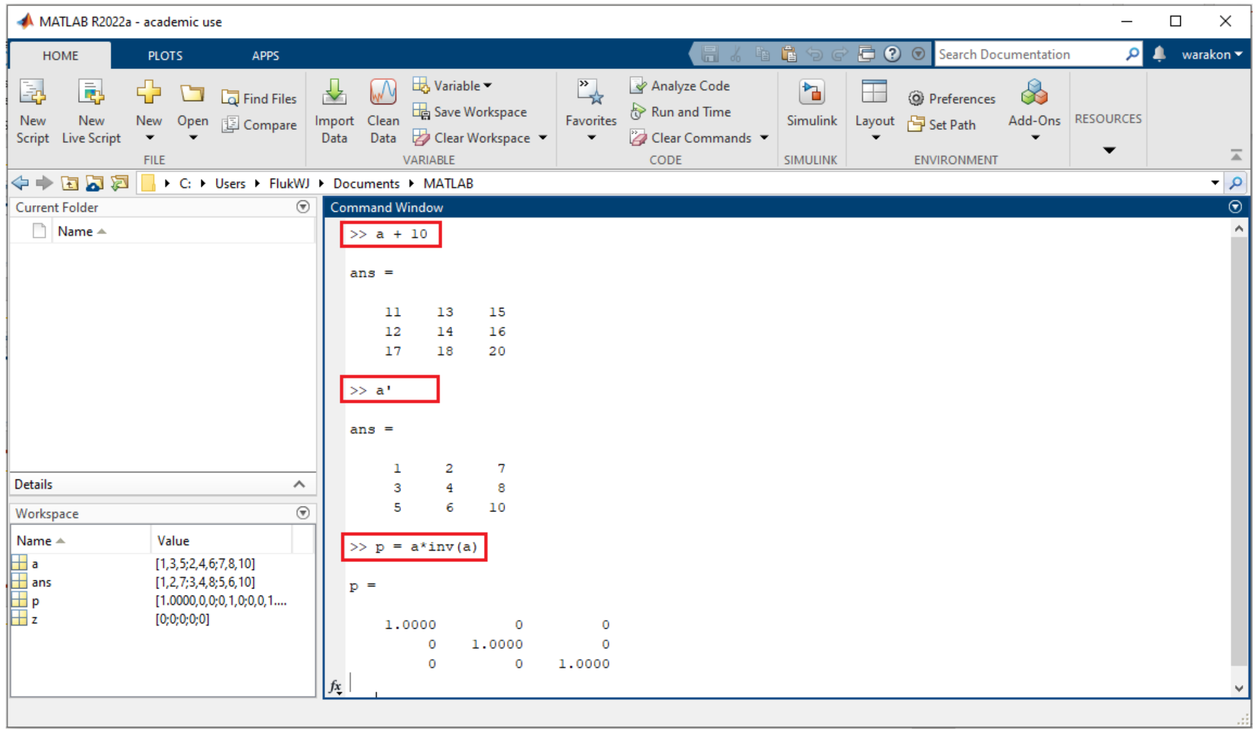
SCHOOL OF ENGINEERING KMITL

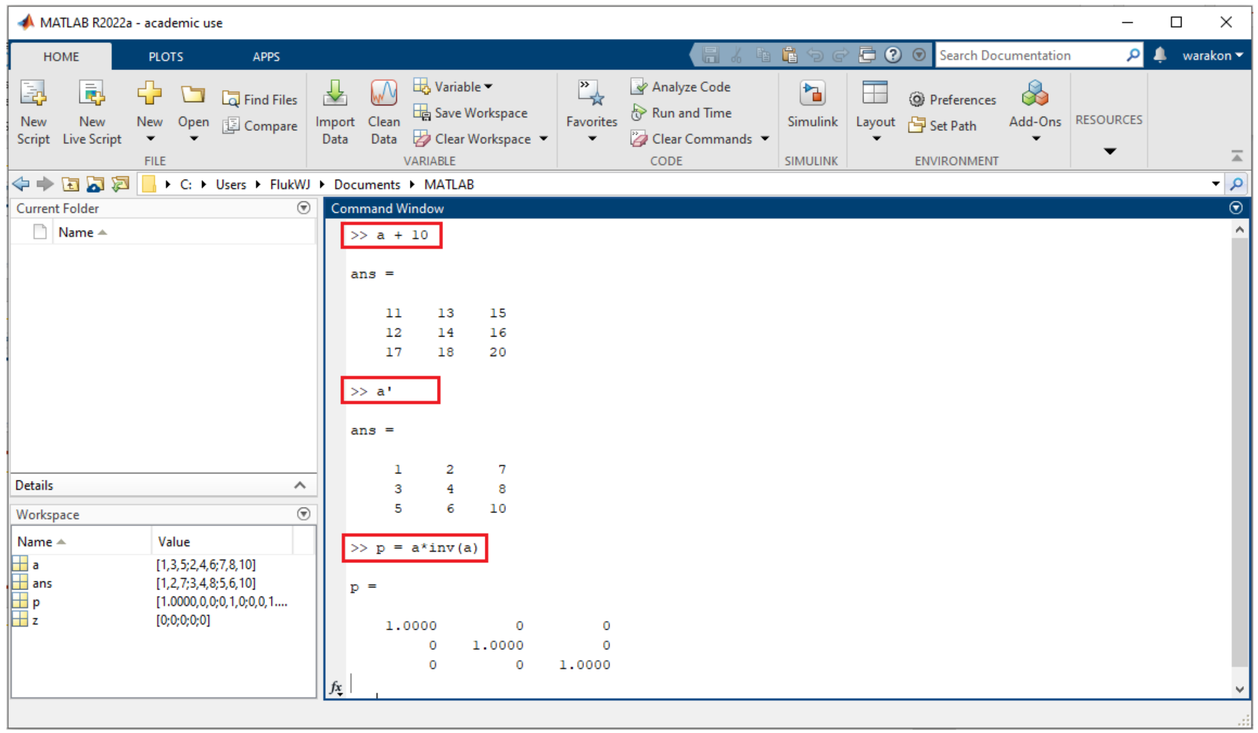
- Matrices and Arrays



1.2 พื้นฐานการใช้โปรแกรม MATLAB สำหรับระบบควบคุม

- Matrix and Array Operations

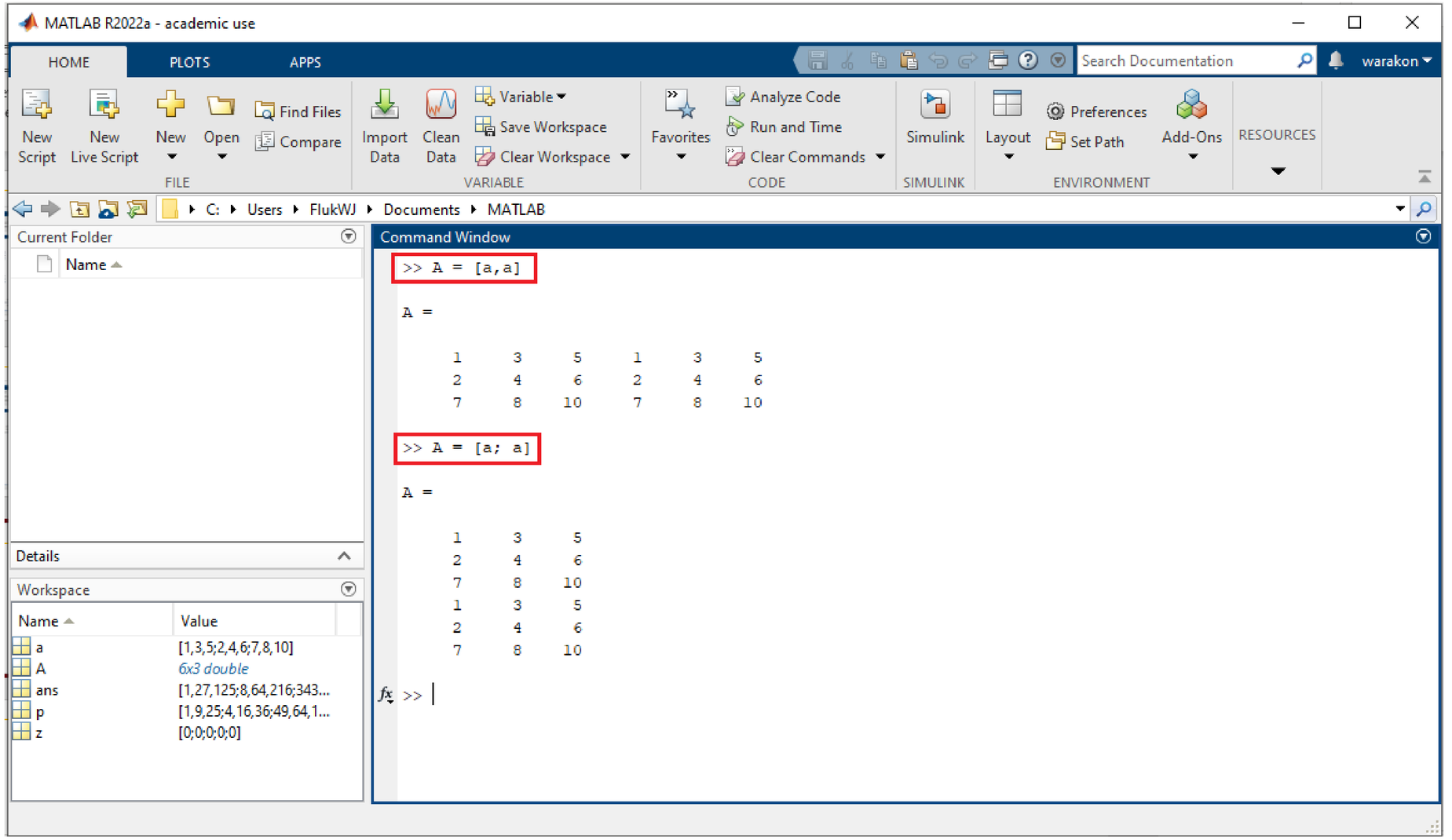




1.2 พื้นฐานการใช้โปรแกรม MATLAB สำหรับระบบควบคุม

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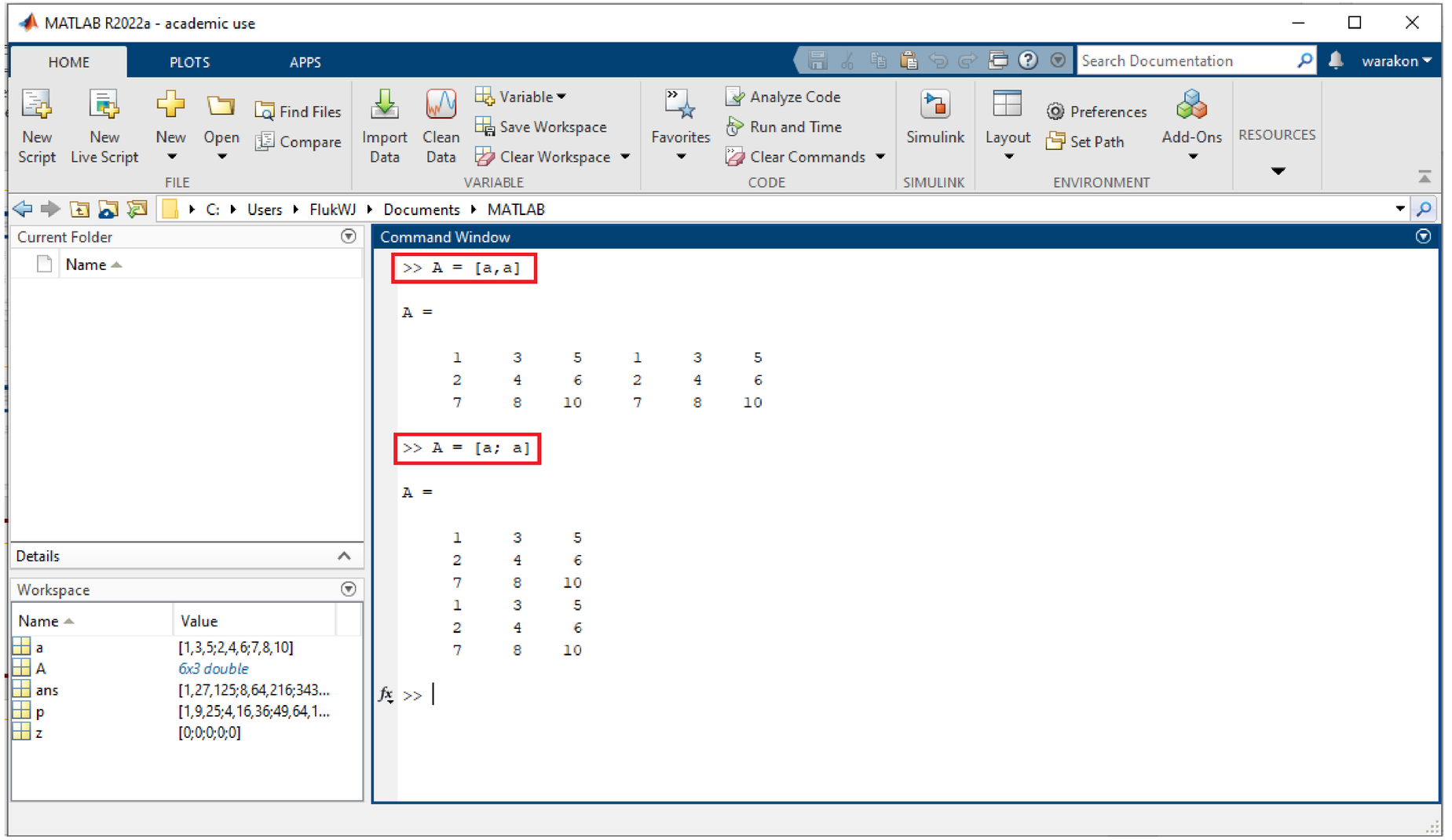
- Concatenation



1.2 พื้นฐานการใช้โปรแกรม MATLAB สำหรับระบบควบคุม

SCHOOL OF ENGINEERING KMITL

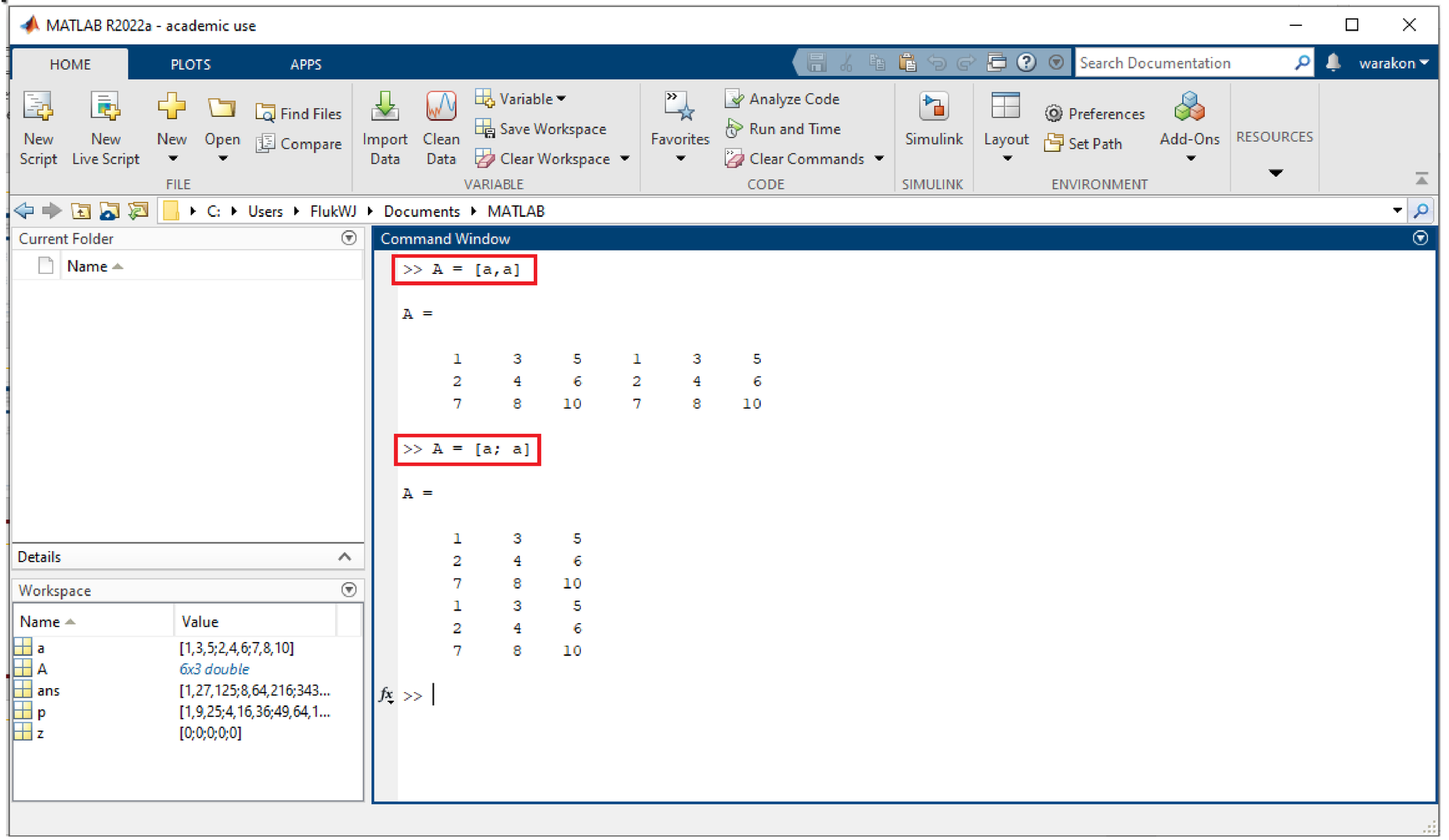
- Concatenation



1.2 พื้นฐานการใช้โปรแกรม MATLAB สำหรับระบบควบคุม

SCHOOL OF ENGINEERING KMITL

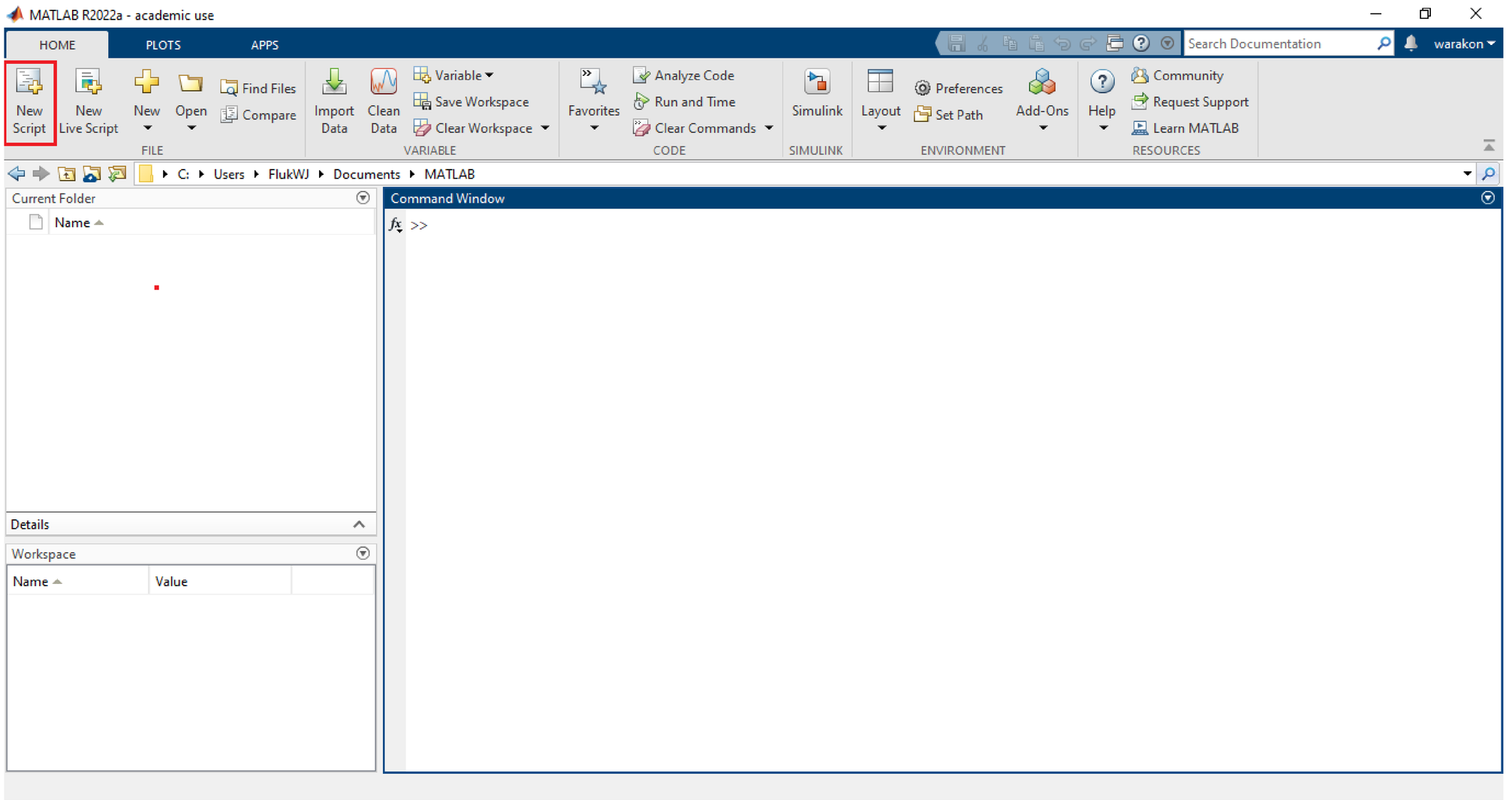
- Complex Numbers



1.2 พื้นฐานการใช้โปรแกรม MATLAB สำหรับระบบควบคุม

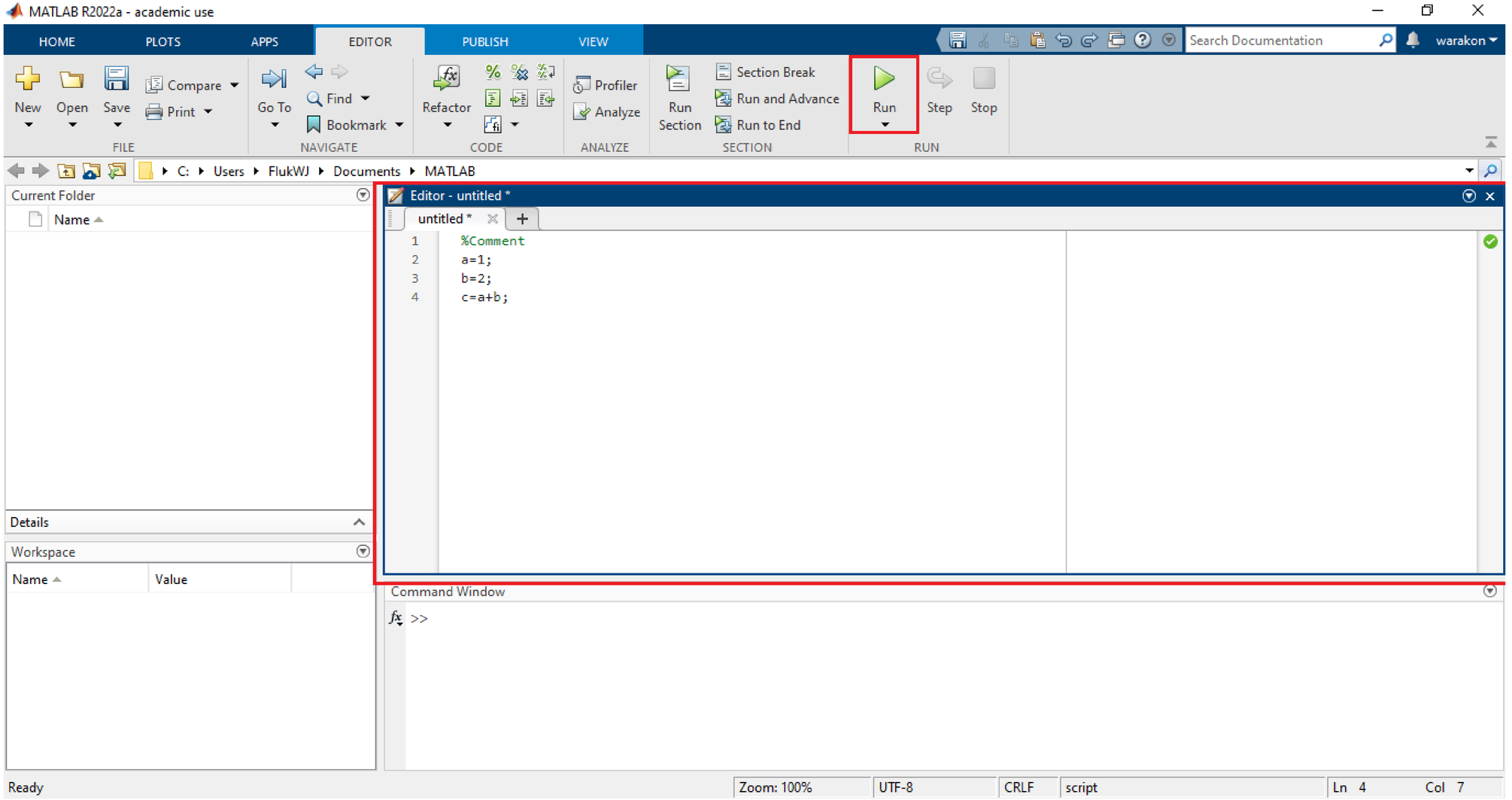
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- Script



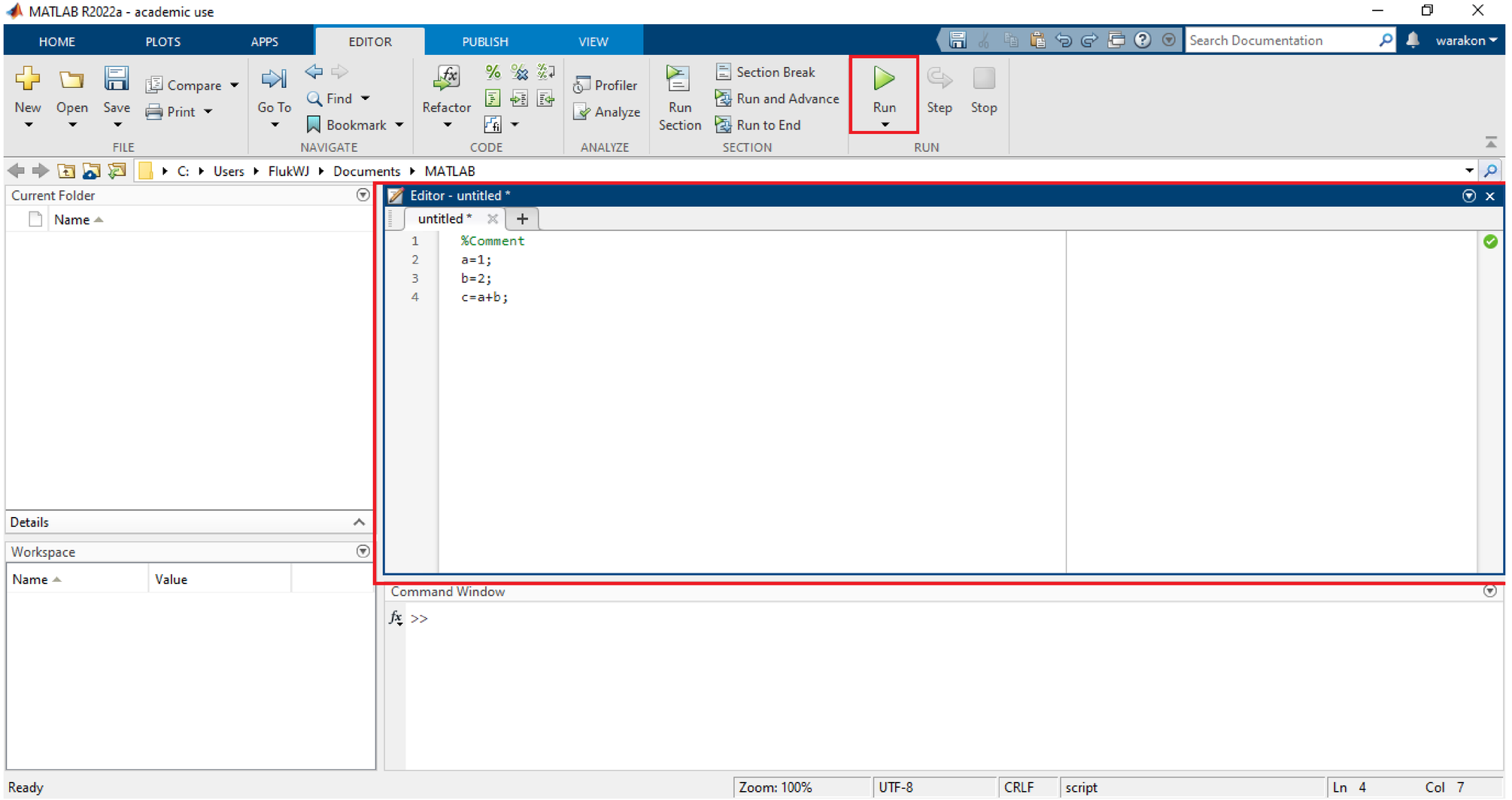
1.2 พื้นฐานการใช้โปรแกรม MATLAB สำหรับระบบควบคุม

- Script



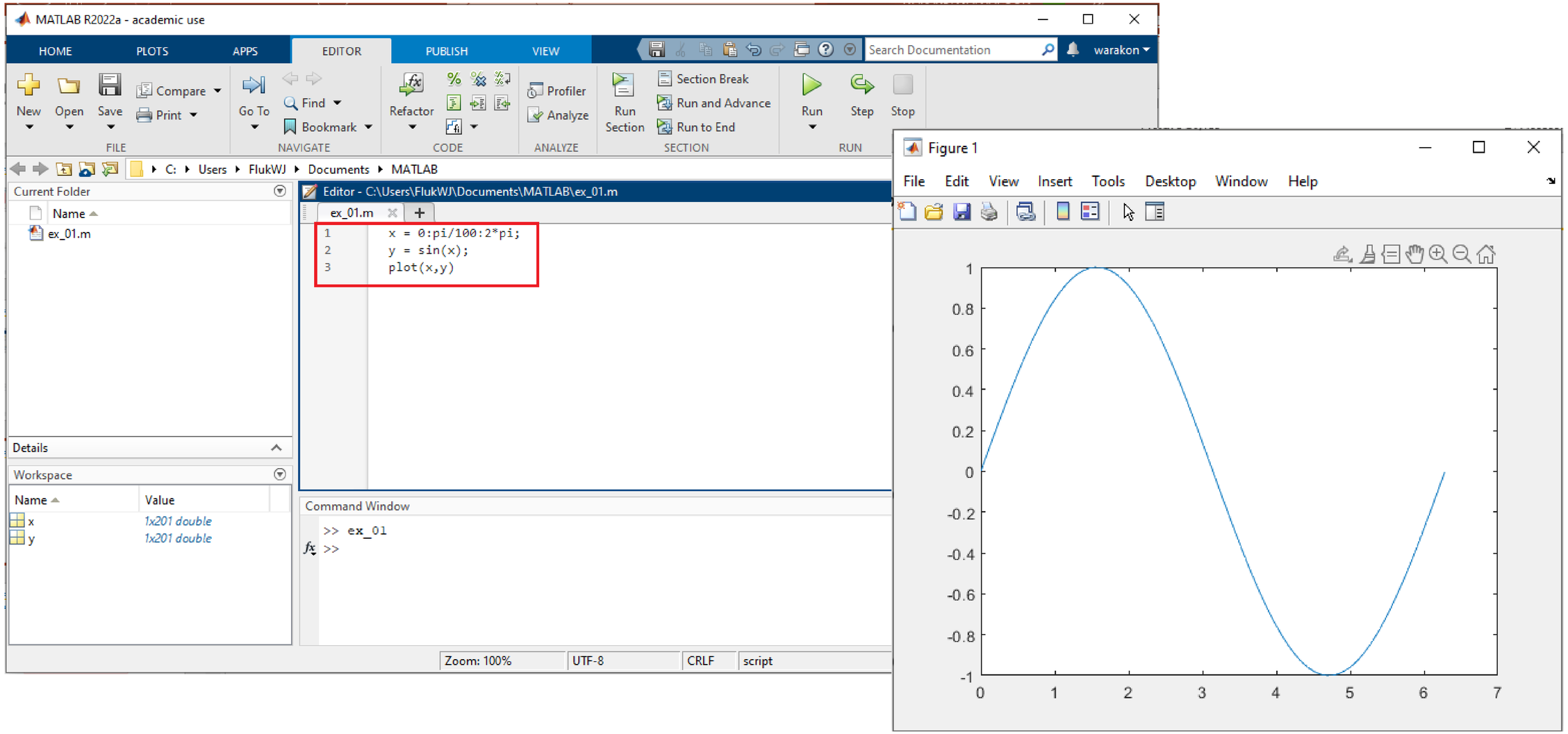
1.2 พื้นฐานการใช้โปรแกรม MATLAB สำหรับระบบควบคุม

- Script



1.2 พื้นฐานการใช้โปรแกรม MATLAB สำหรับระบบควบคุม

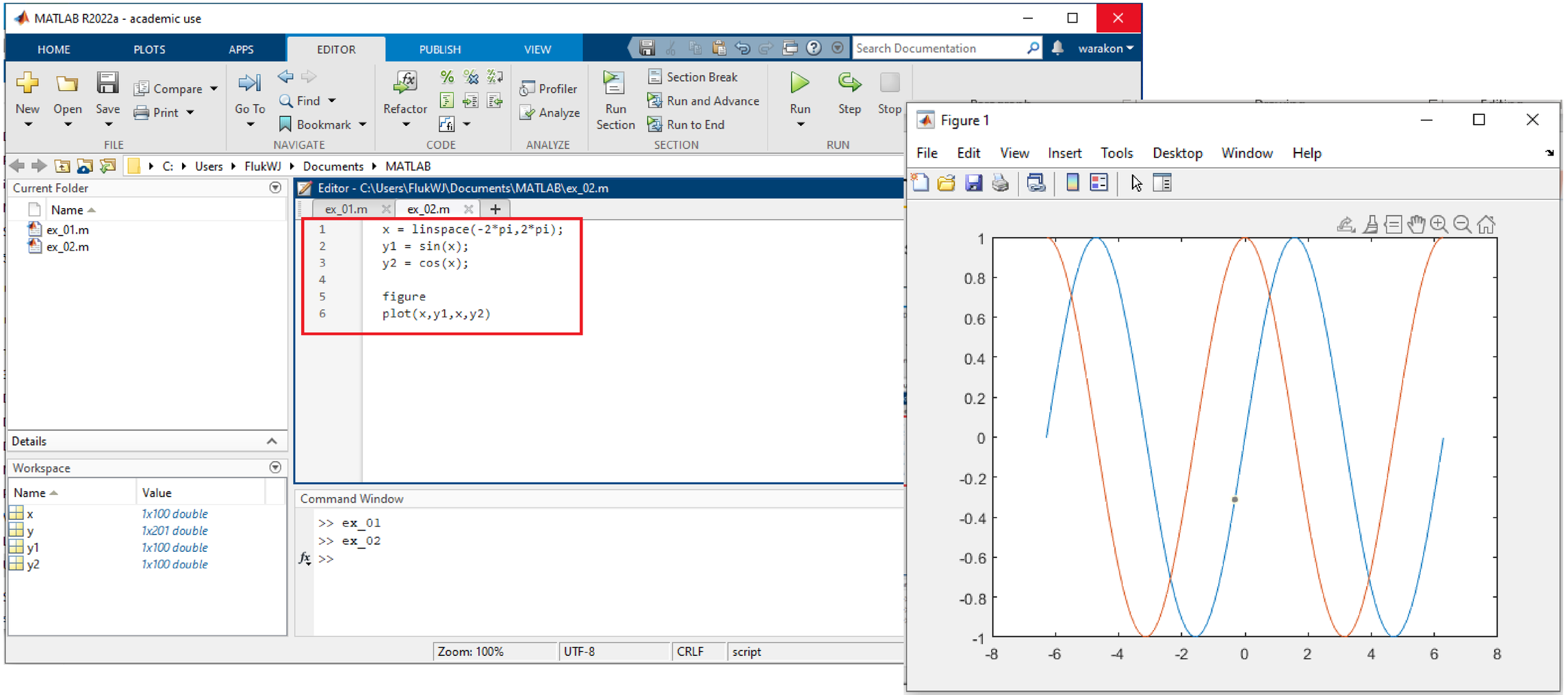
- Create Line Plot



1.2 พื้นฐานการใช้โปรแกรม MATLAB สำหรับระบบควบคุม

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- Plot Multiple Lines

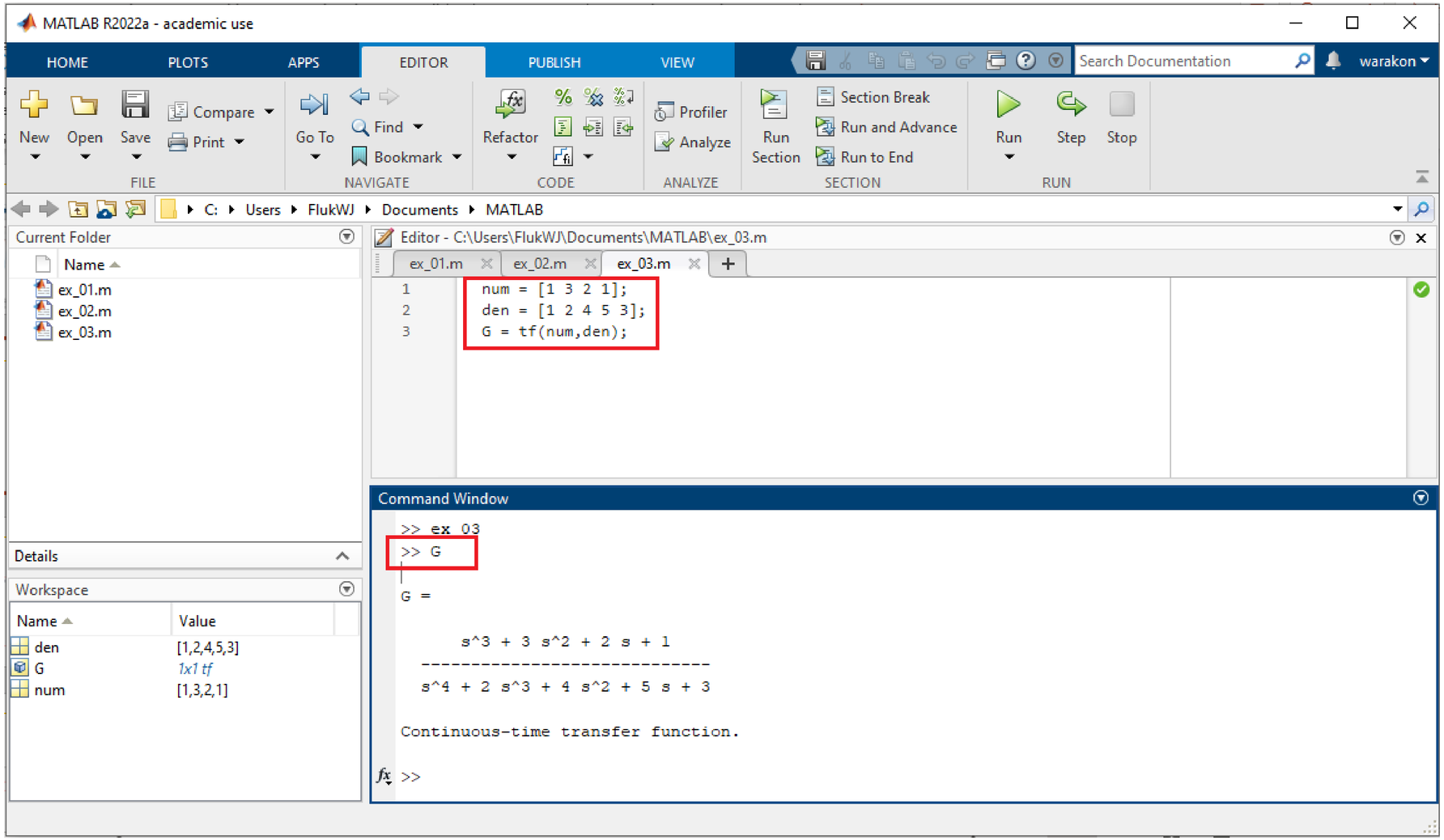


1.2 พื้นฐานการใช้โปรแกรม MATLAB สำหรับระบบควบคุม

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- Transfer Function Model

$$G(s) = \frac{s^3 + 3s^2 + 2s + 1}{s^4 + 2s^3 + 4s^2 + 5s + 3}$$

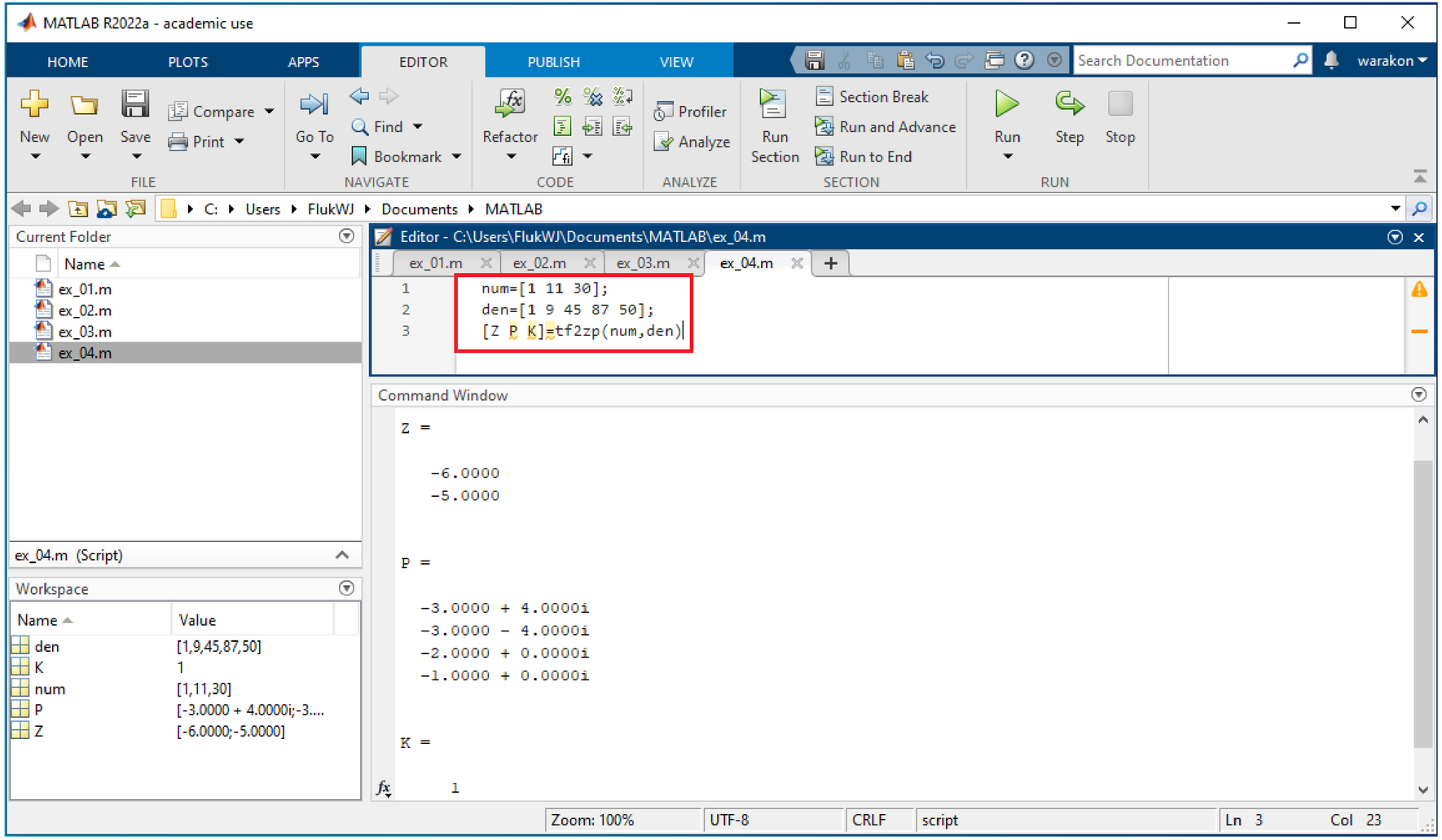


1.2 พื้นฐานการใช้โปรแกรม MATLAB สำหรับระบบควบคุม

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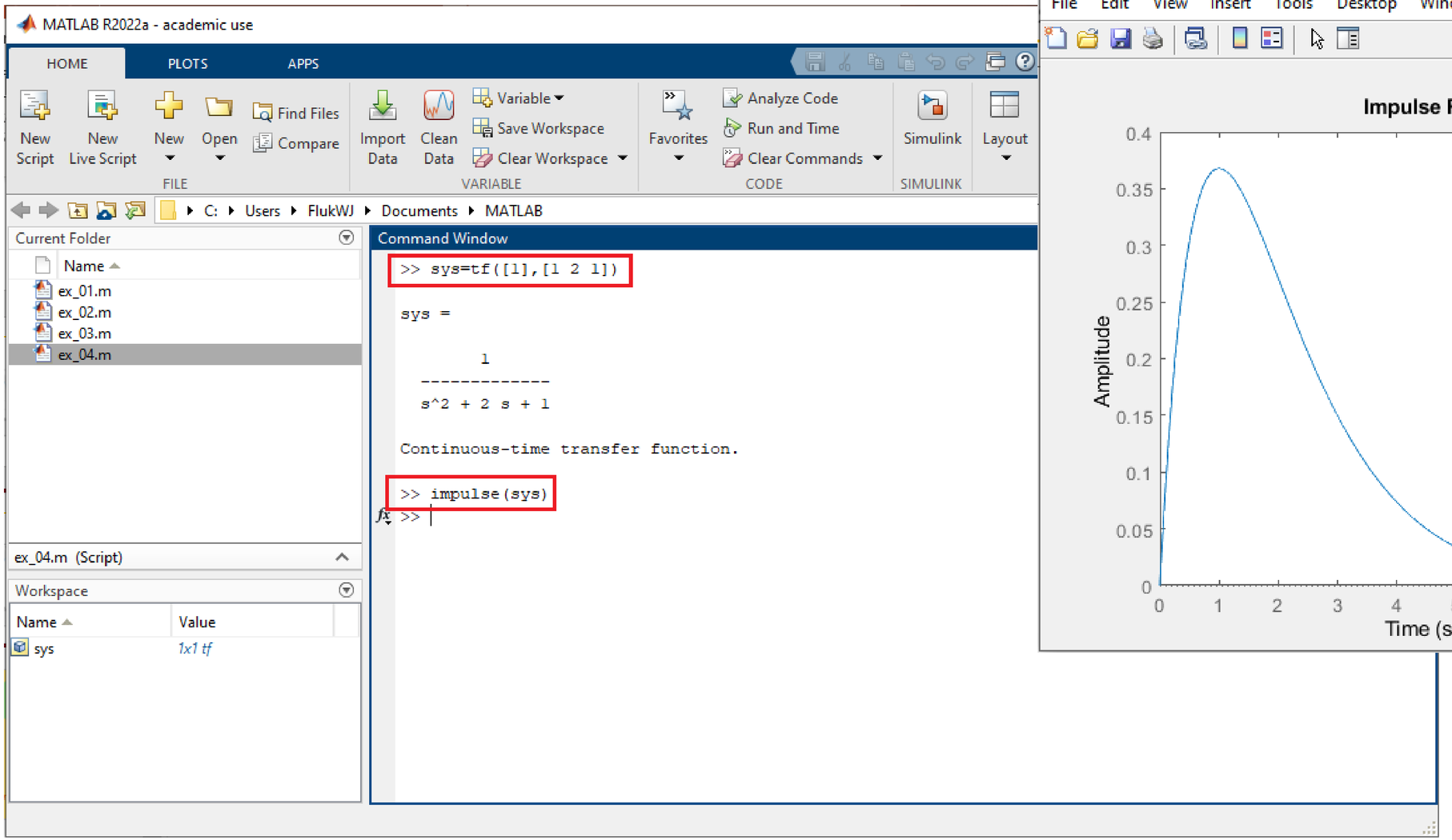
- Zero-Pole-Gain Model

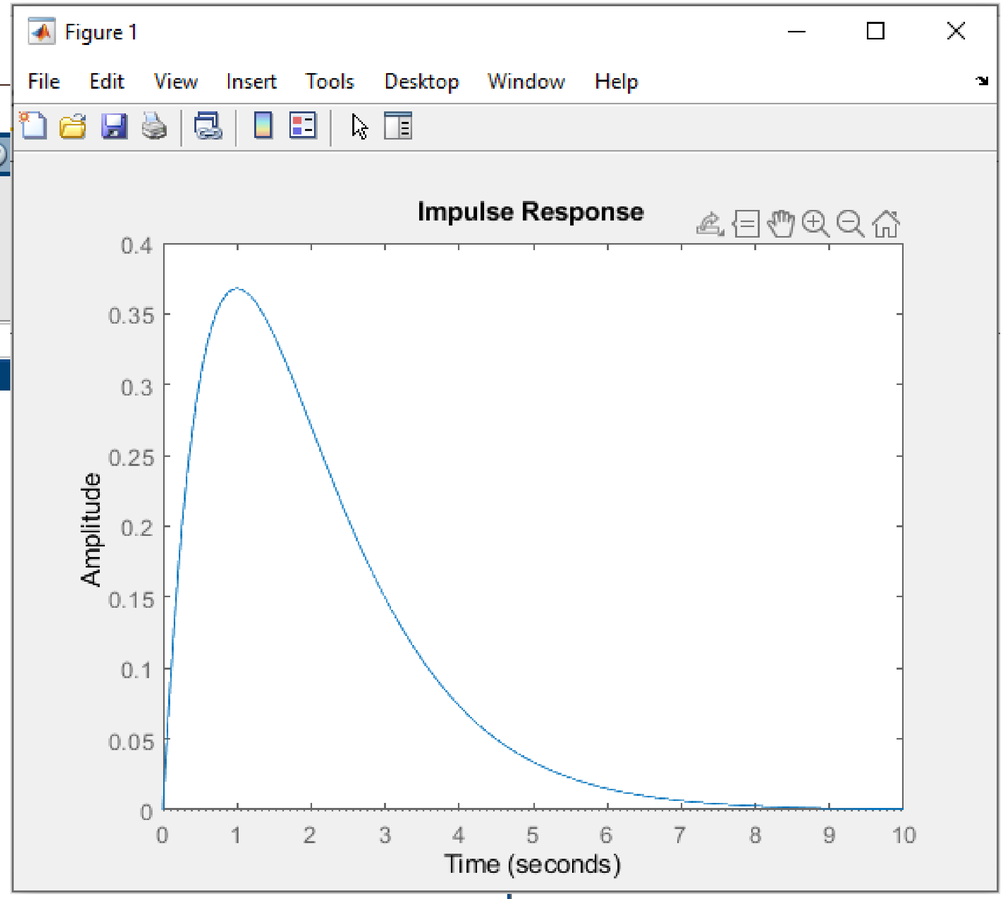
$$G(s) = \frac{s^3 + 11s^2 + 30s}{s^4 + 9s^3 + 45s^2 + 87s + 50} = \frac{s(s+5)(s+6)}{(s+1)(s+2)(s+3+j4)(s+3-j4)}$$



1.2 พื้นฐานการใช้โปรแกรม MATLAB สำหรับระบบควบคุม

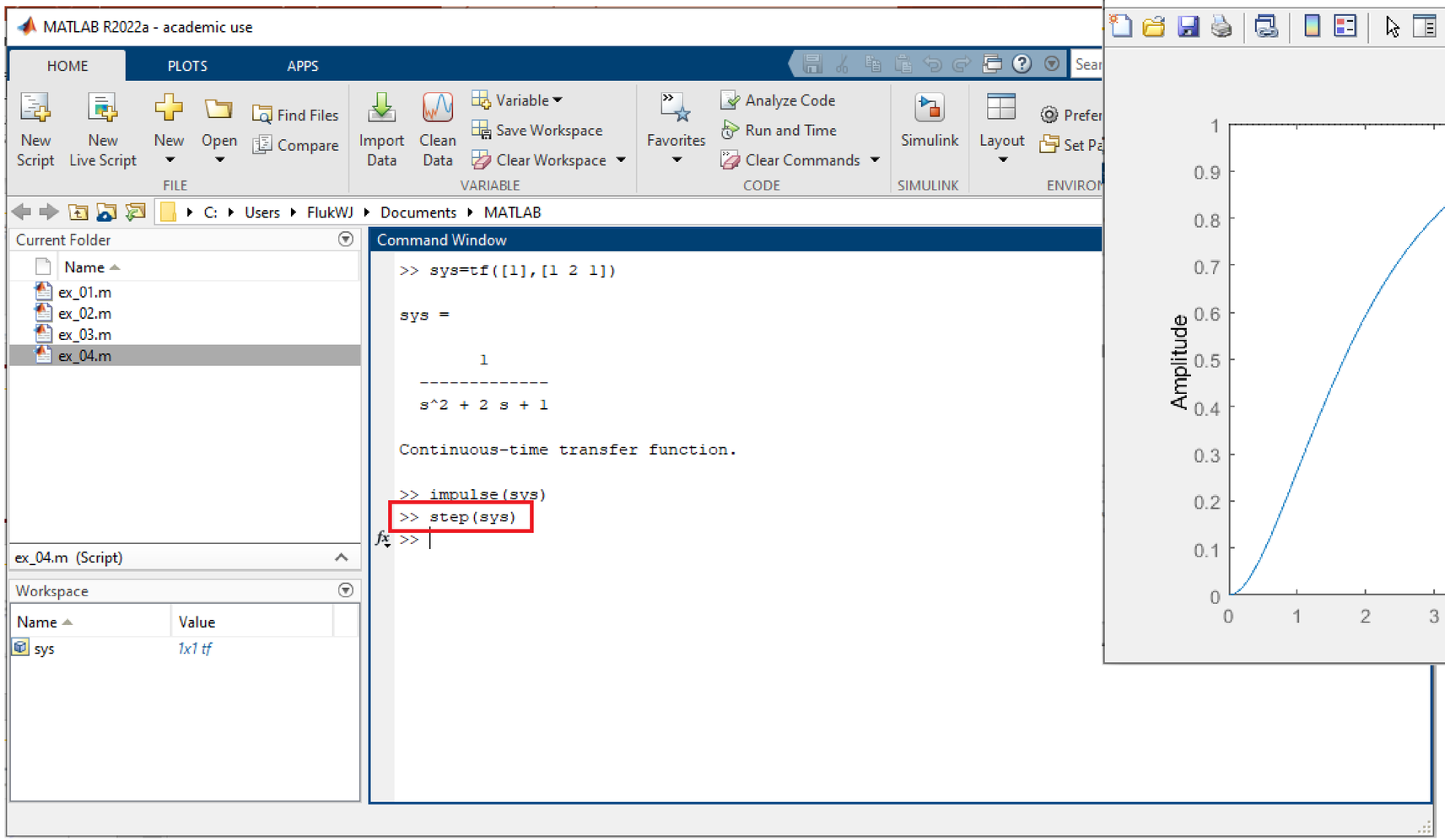
- Unit Impuls $G(s) = \dfrac{1}{s^2 + 2s + 1}$

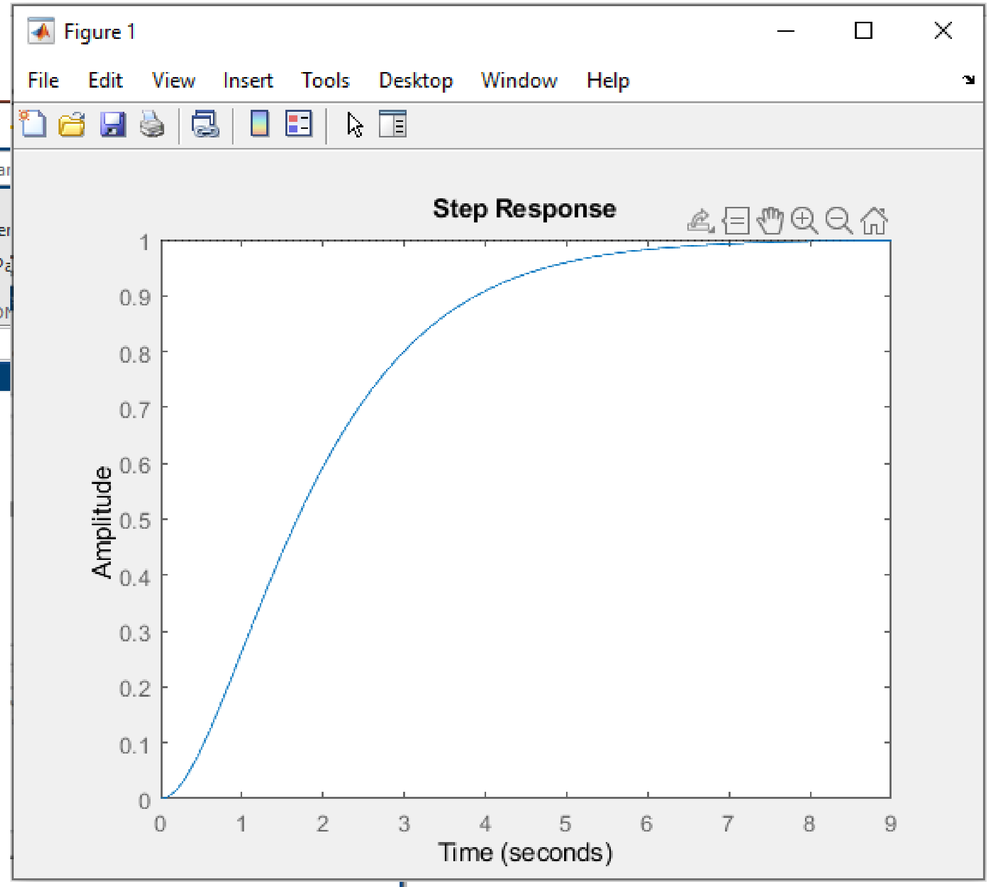




1.2 พื้นฐานการใช้โปรแกรม MATLAB สำหรับระบบควบคุม

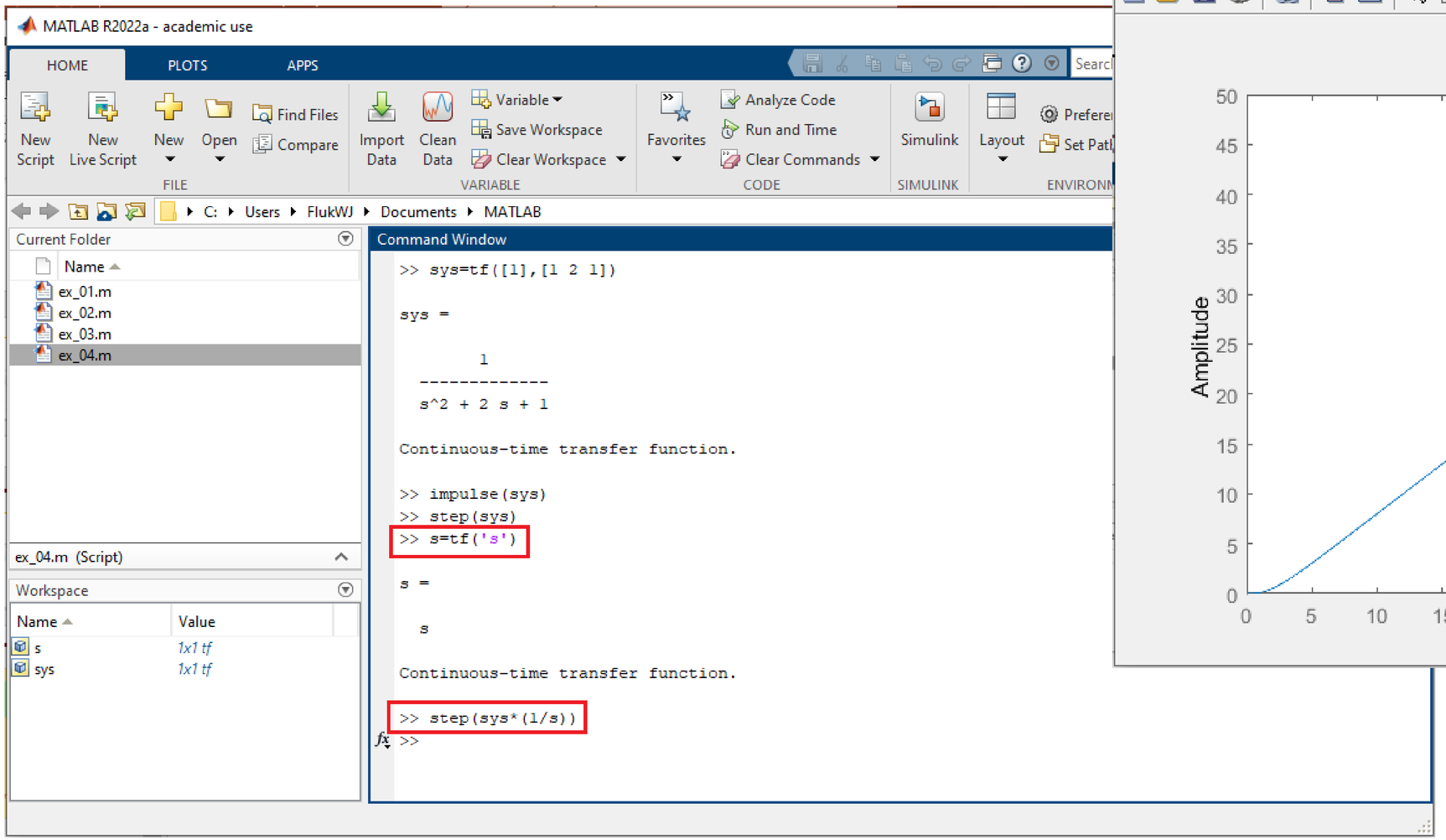
- Unit Step $G(s) = \frac{1}{s^2 + 2s + 1}$

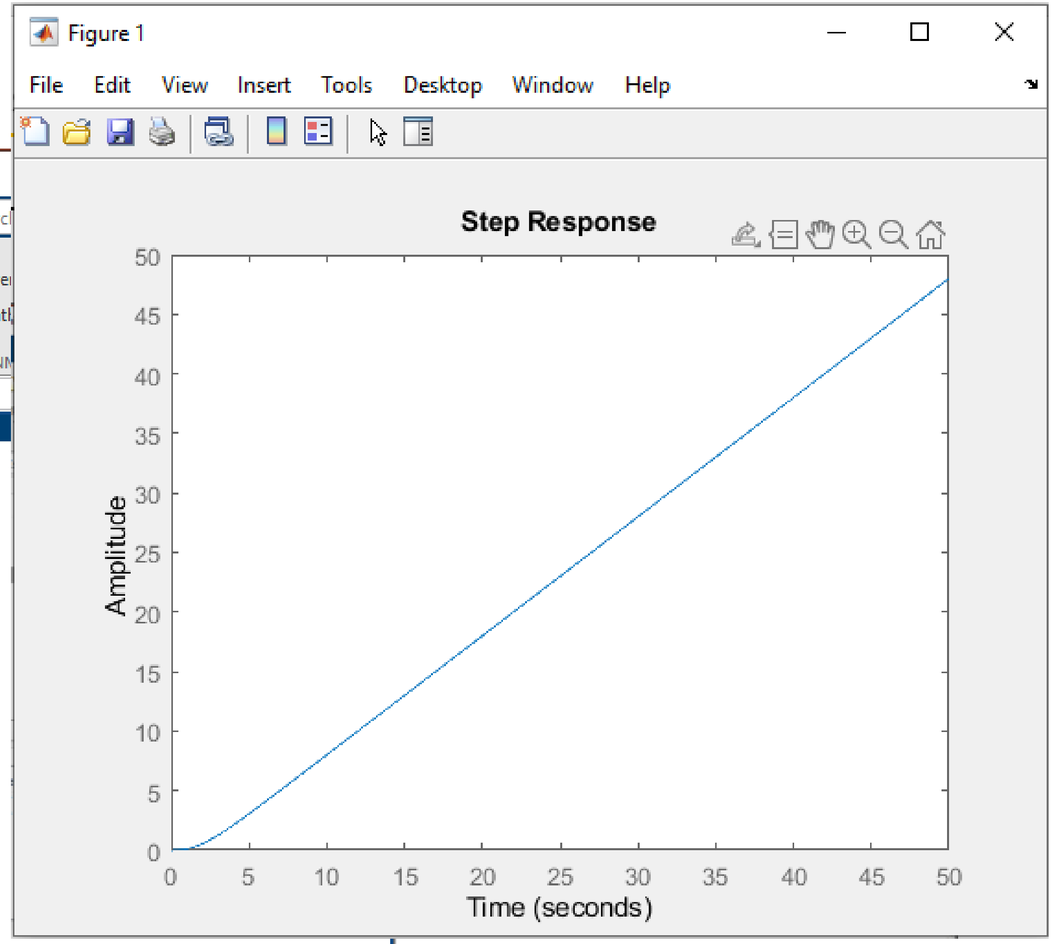




1.2 พื้นฐานการใช้โปรแกรม MATLAB สำหรับระบบควบคุม

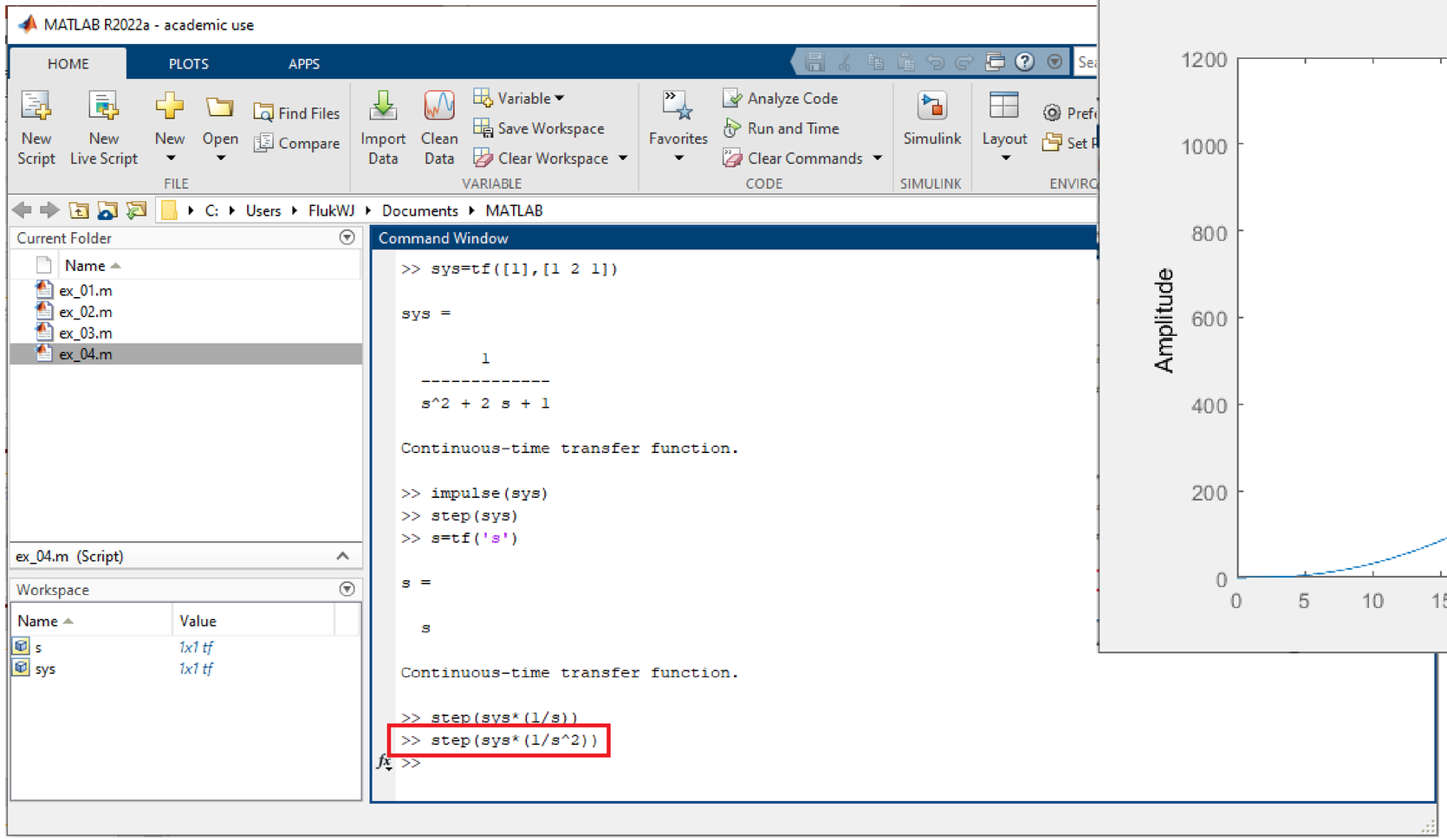
- Unit Ramp $G(s) = \frac{1}{s^2 + 2s + 1}$

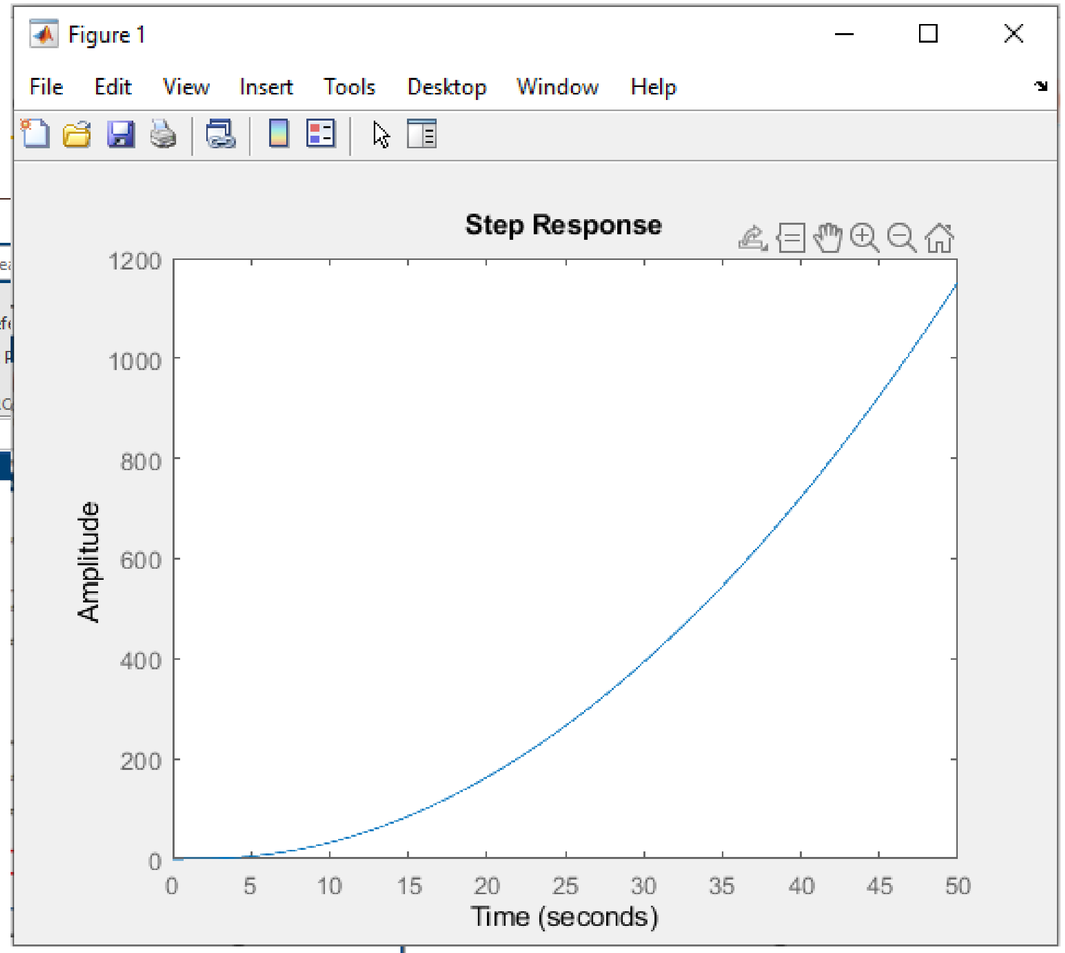




1.2 พื้นฐานการใช้โปรแกรม MATLAB สำหรับระบบควบคุม

- Unit Ramp $G(s) = \frac{1}{s^2 + 2s + 1}$

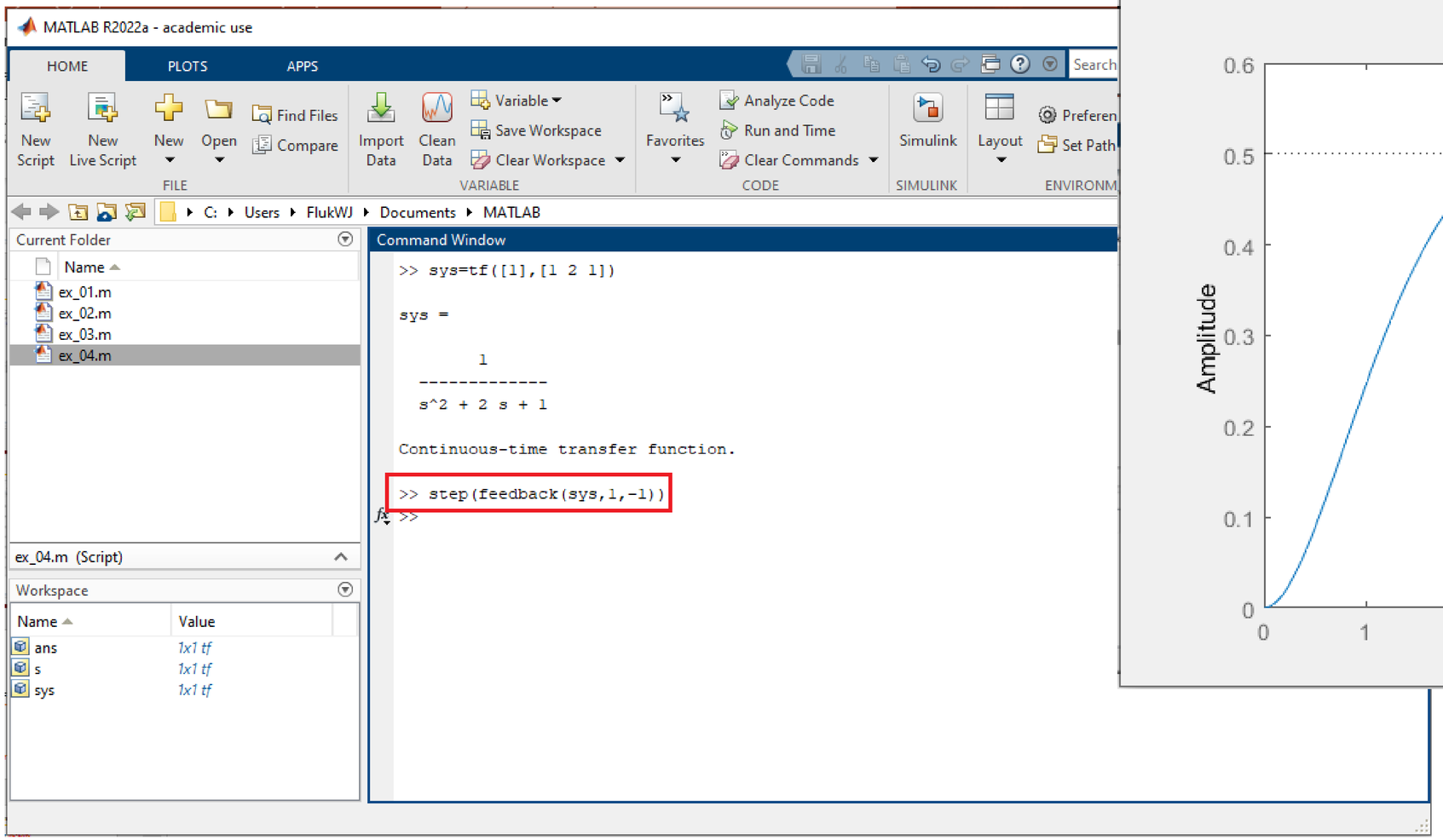


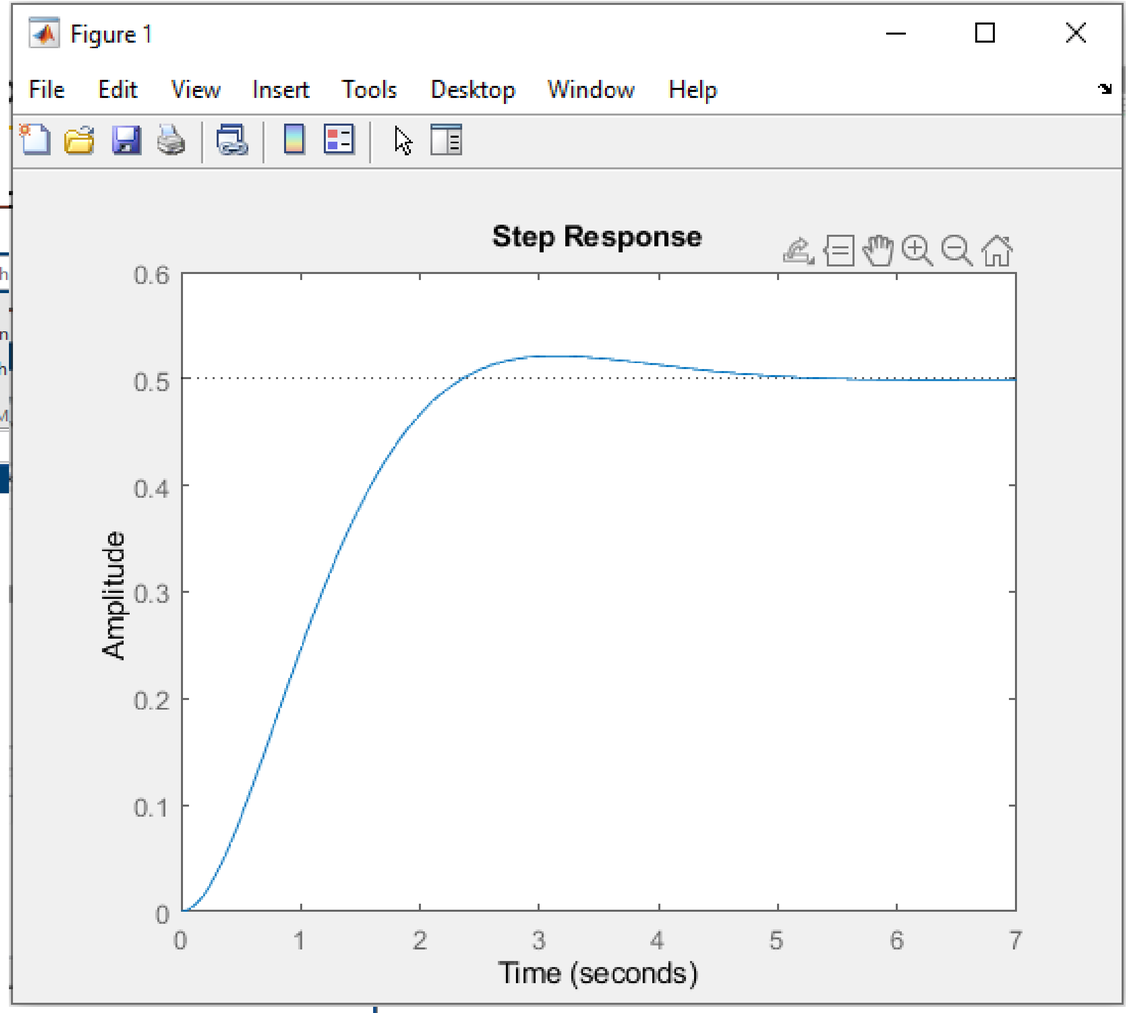


1.2 พื้นฐานการใช้โปรแกรม MATLAB สำหรับระบบควบคุม

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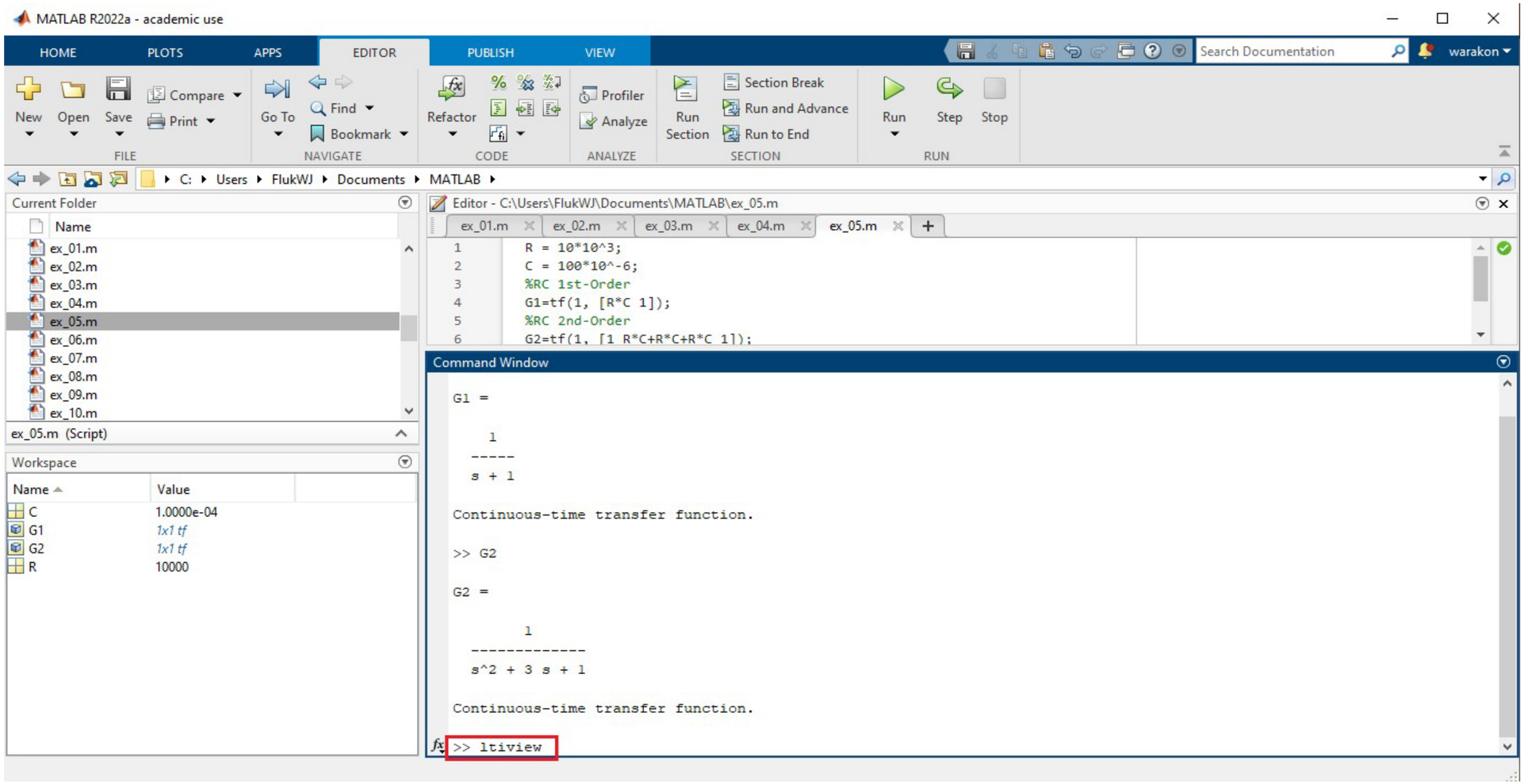
- Closed-loop





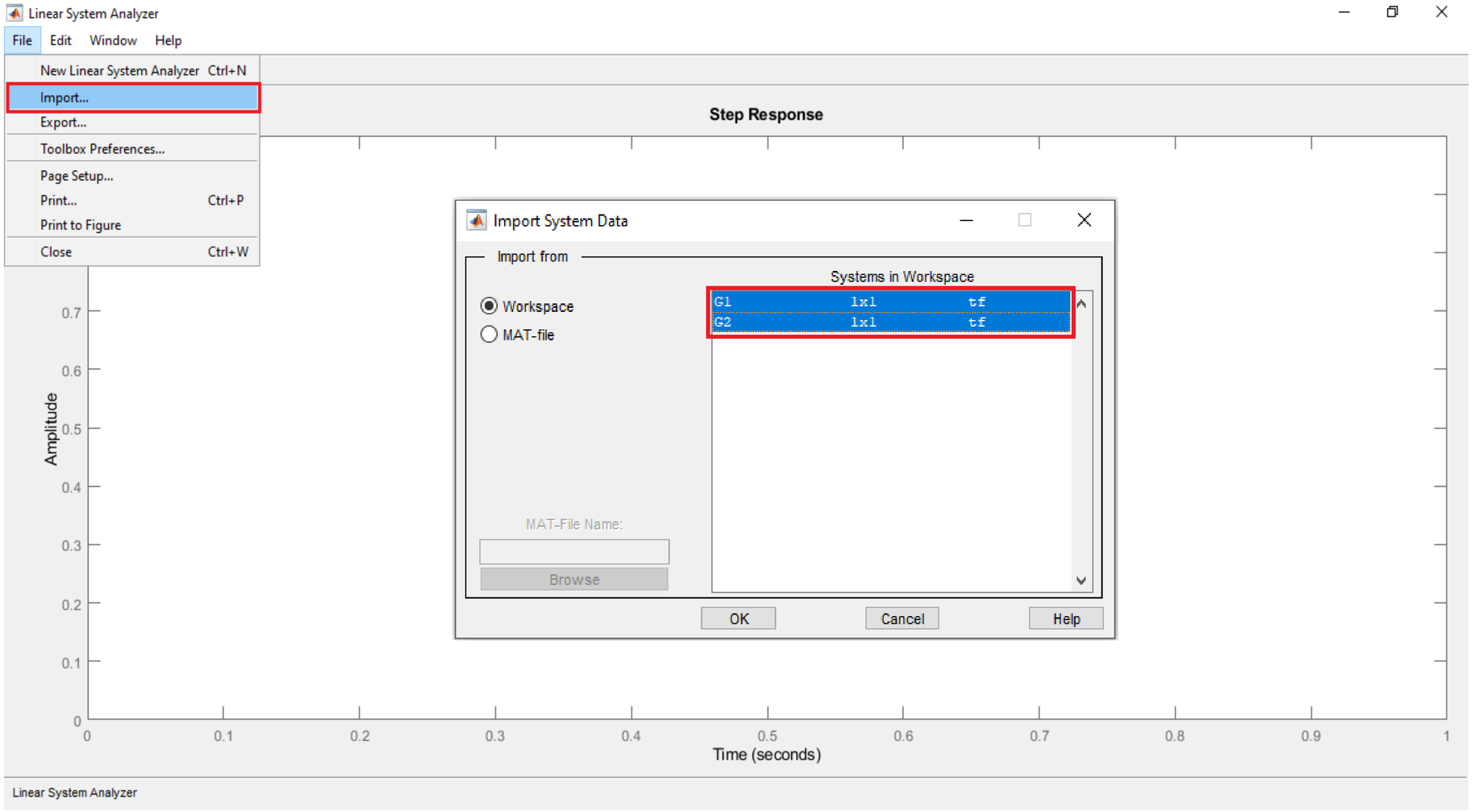
1.3 การวิเคราะผลตอบสนองโดยใช้เครื่องมือ LTIVIEW

- LTIVIEW



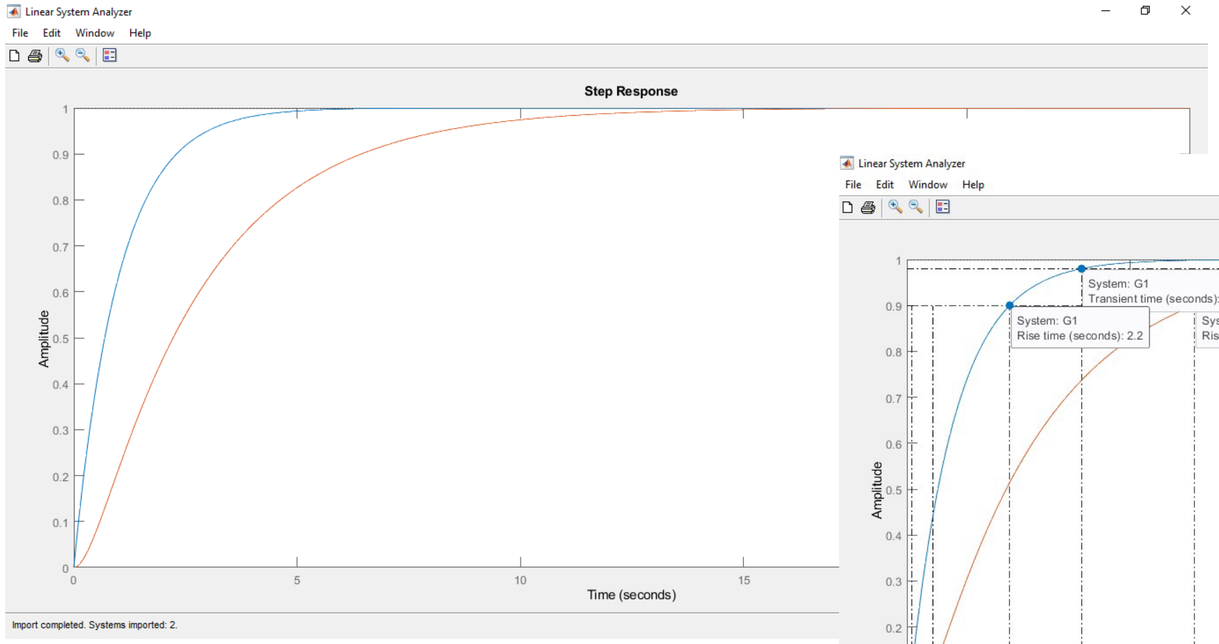
1.3 การวิเคราะผลตอบสนองโดยใช้เครื่องมือ LTIVIEW

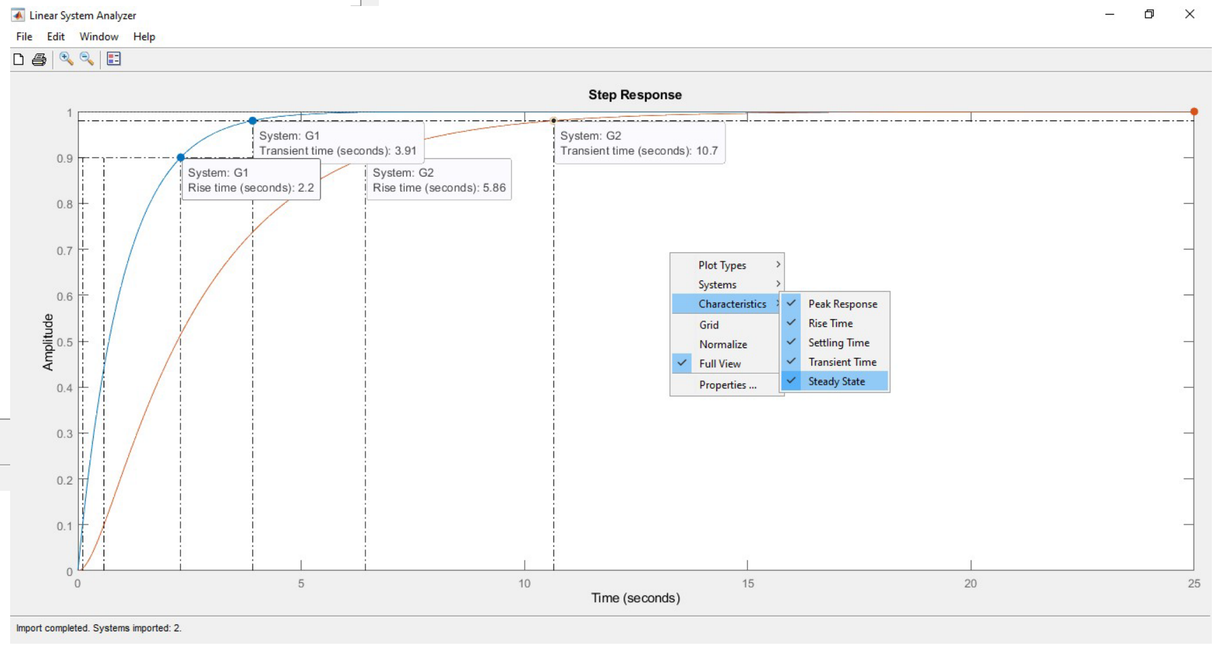
- LTIVIEW



1.3 การวิเคราะผลตอบสนองโดยใช้เครื่องมือ LTIVIEW

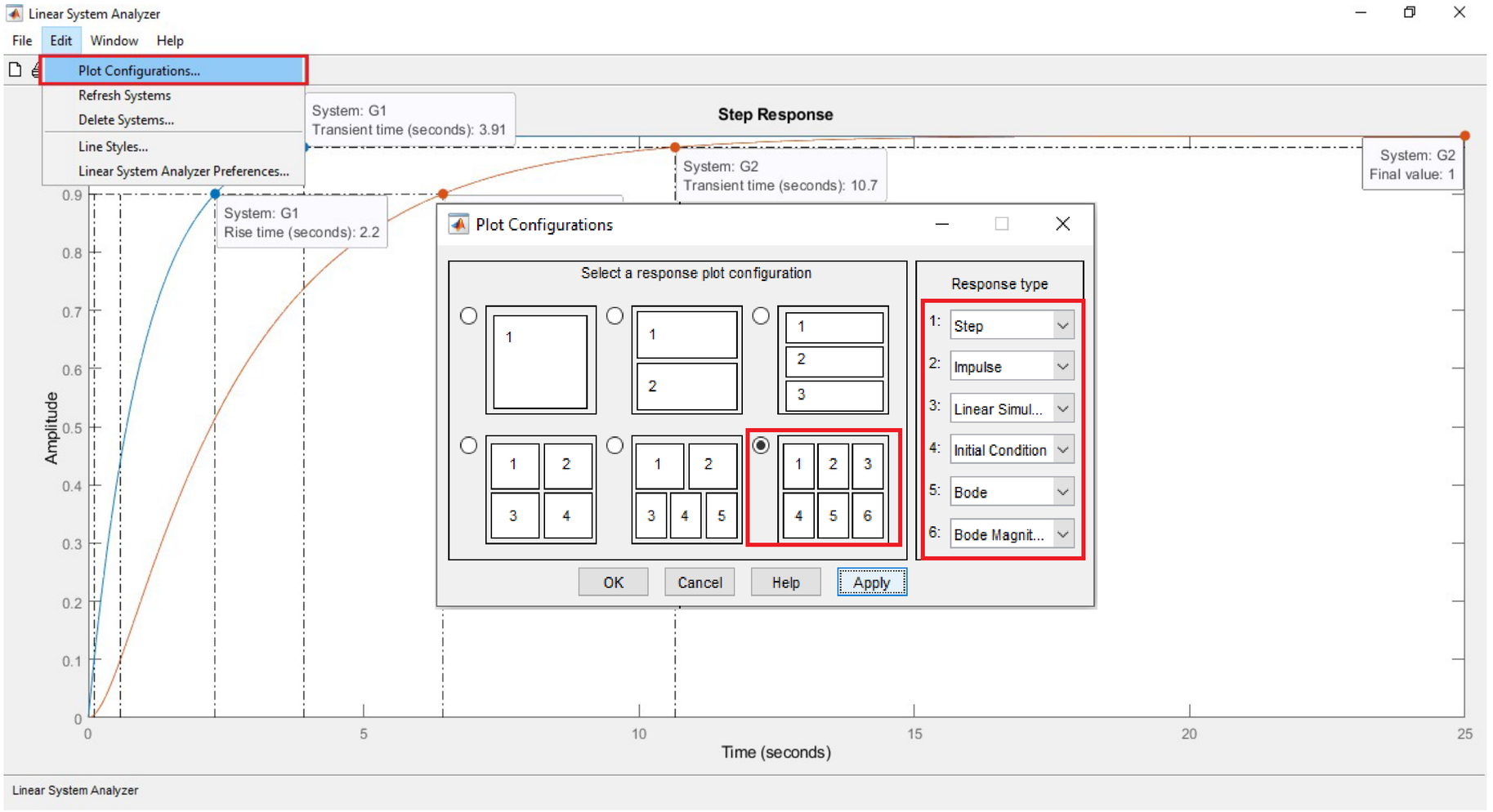
- LTIVIEW





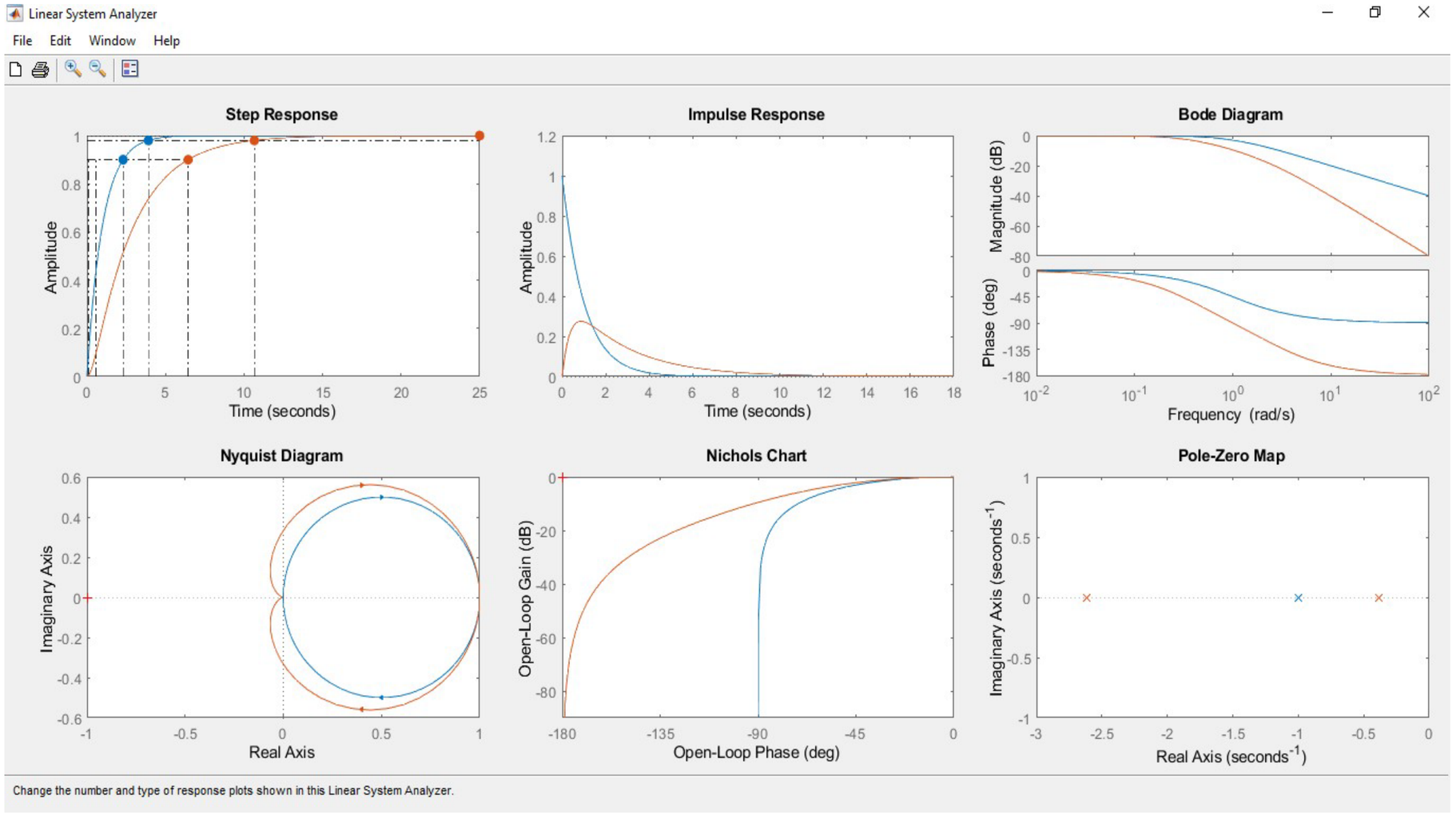
1.3 การวิเคราะผลตอบสนองโดยใช้เครื่องมือ LTIVIEW

- LTIVIEW



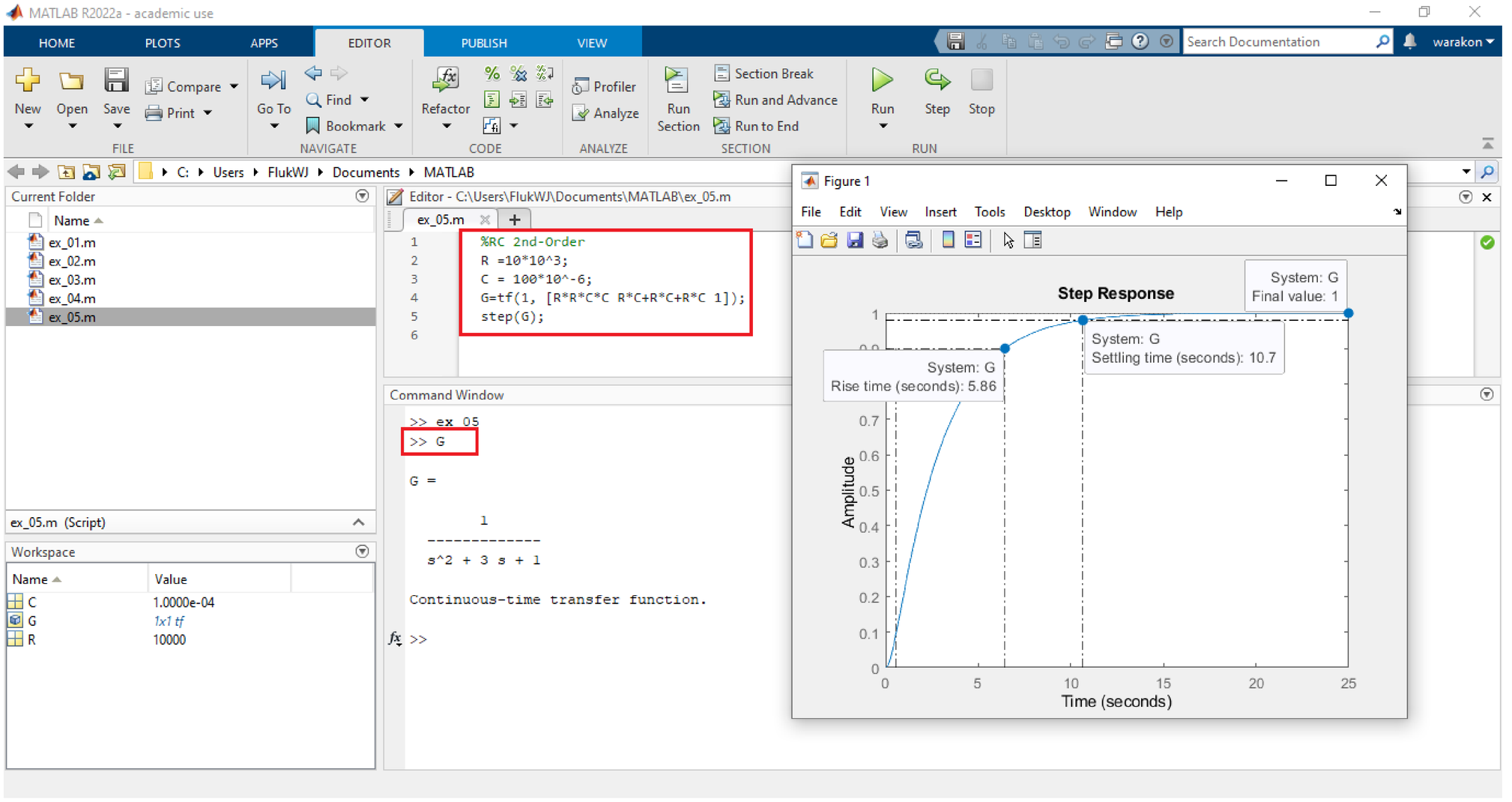
1.3 การวิเคราะผลตอบสนองโดยใช้เครื่องมือ LTIVIEW

- LTIVIEW



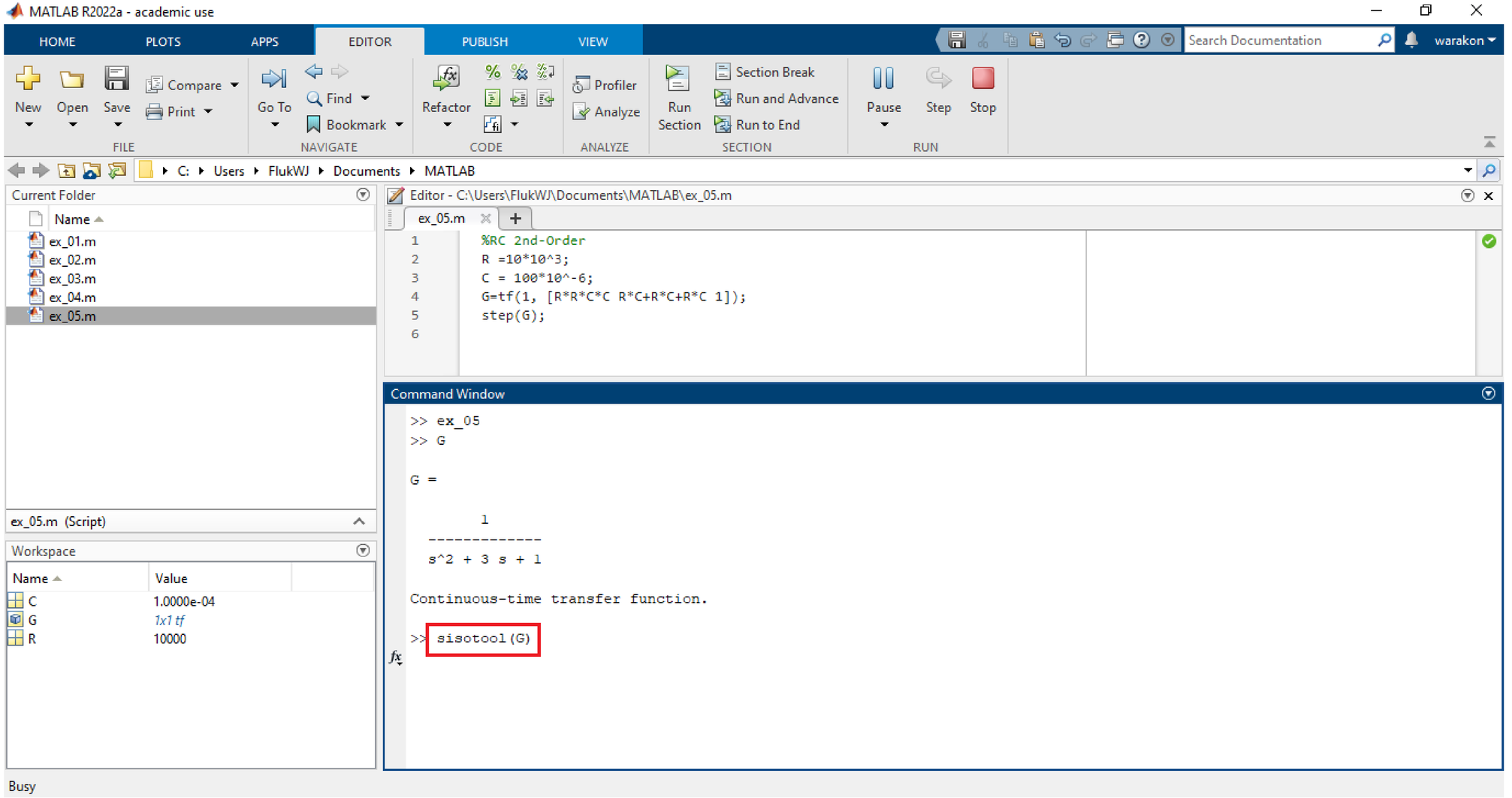
1.4 การออกแบบระบบควบคุมโดยใช้เครื่องมือ SISOTOOL

- Define G(s)



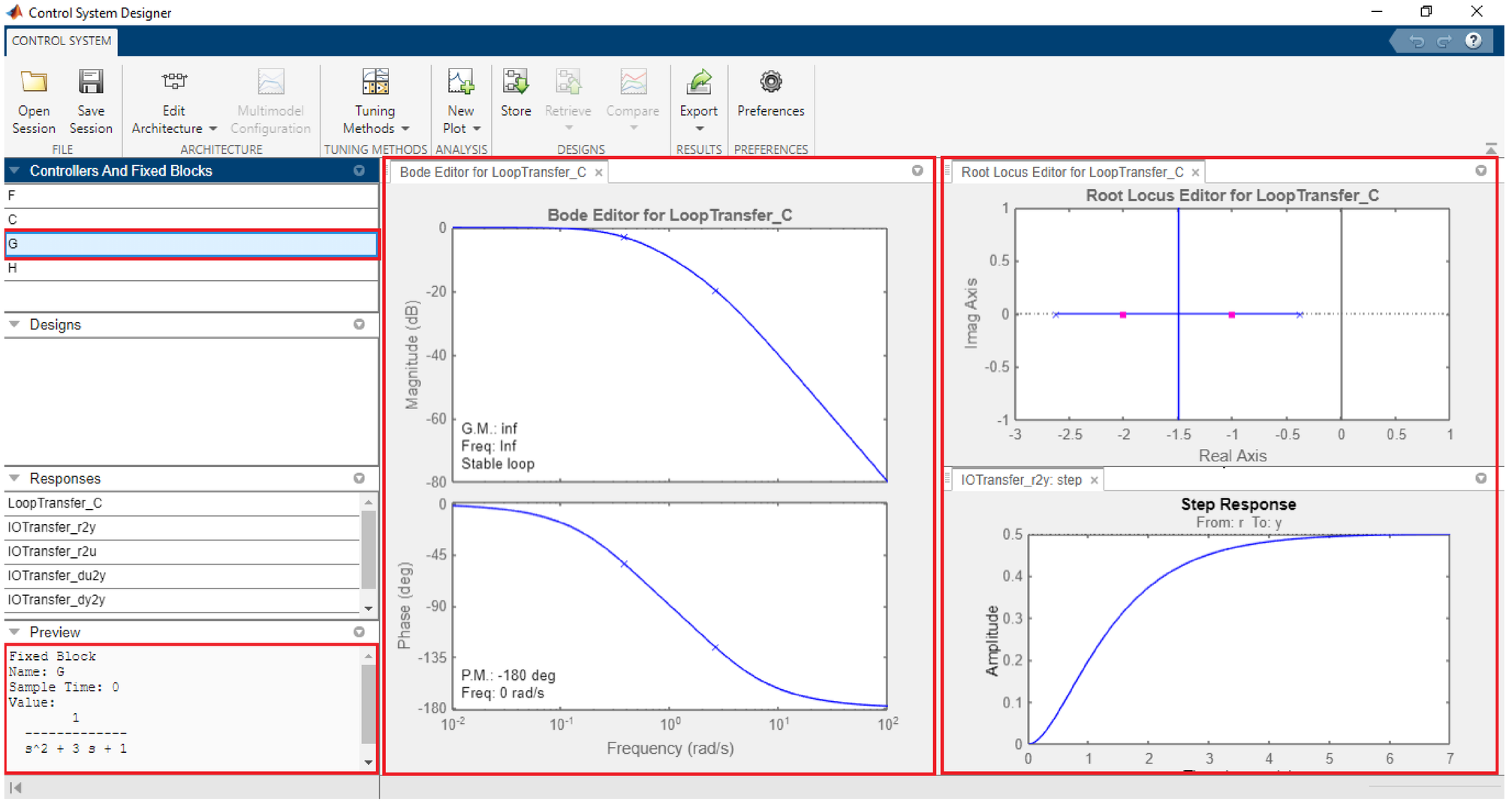
1.4 การออกแบบระบบควบคุมโดยใช้เครื่องมือ SISOTOOL

- SISOTOOL



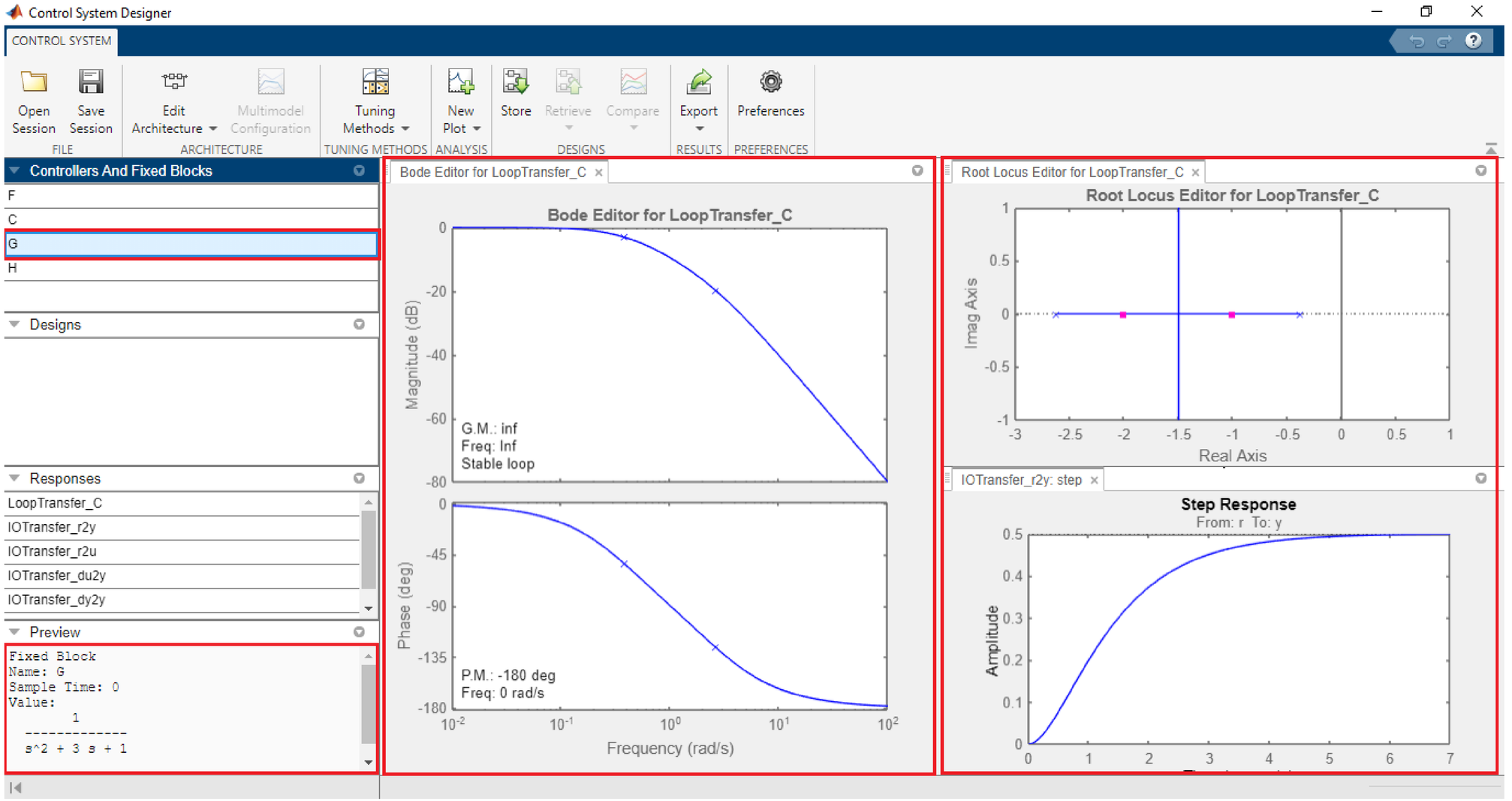
1.4 การออกแบบระบบควบคุมโดยใช้เครื่องมือ SISOTOOL

- SISOTOOL



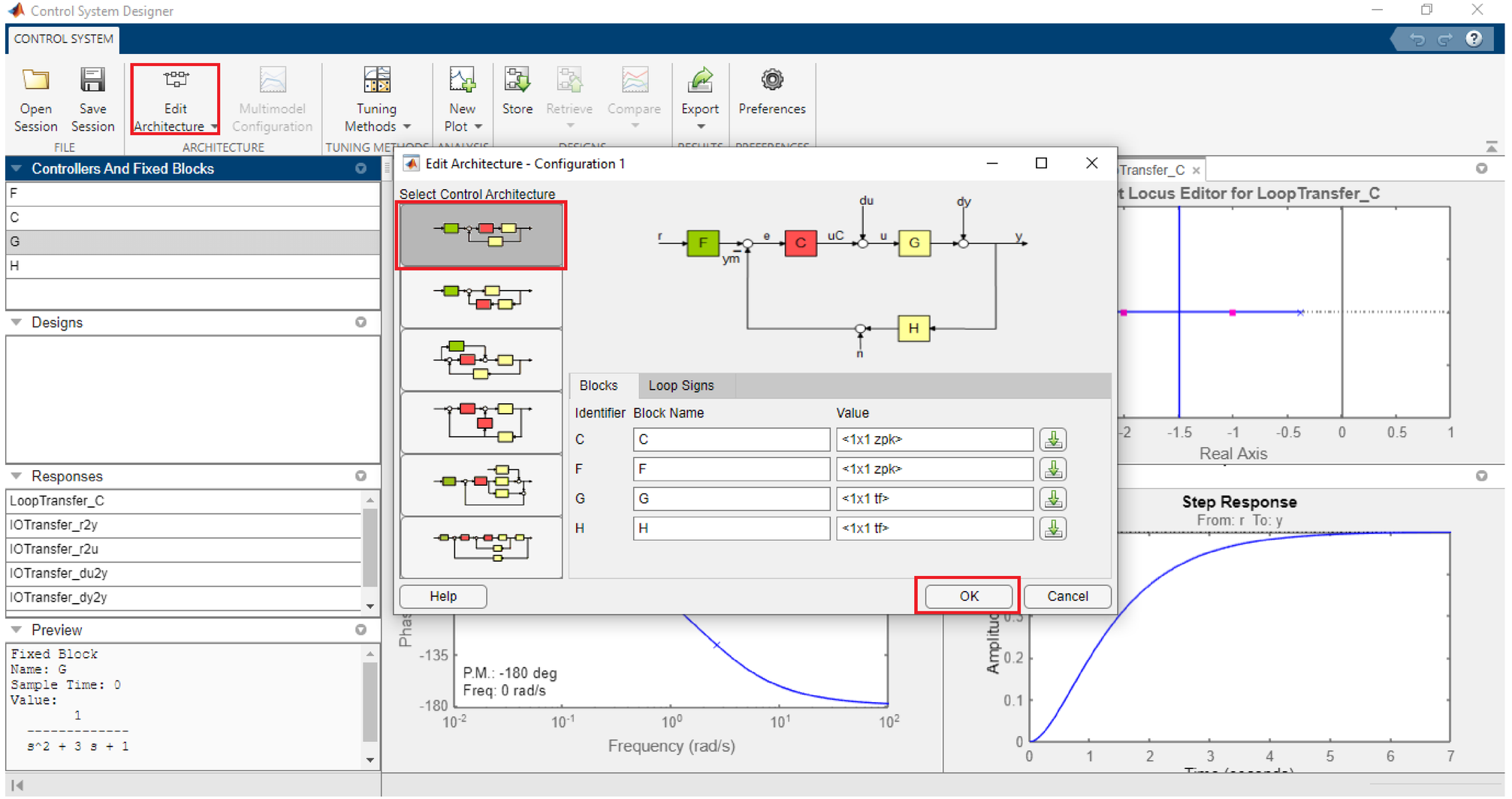
1.4 การออกแบบระบบควบคุมโดยใช้เครื่องมือ SISOTOOL

- Control System Designer



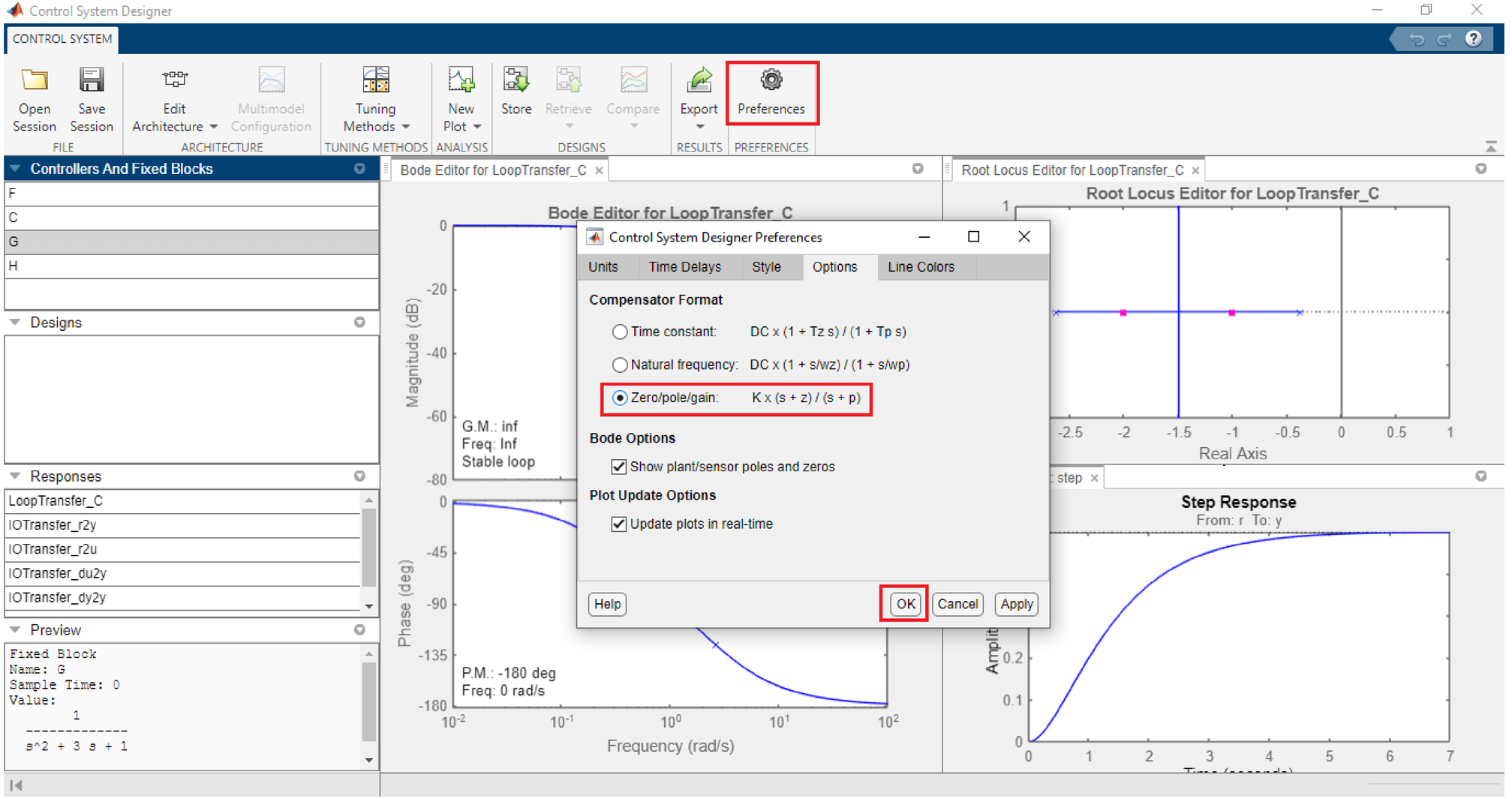
1.4 การออกแบบระบบควบคุมโดยใช้เครื่องมือ SISOTOOL

- Architecture



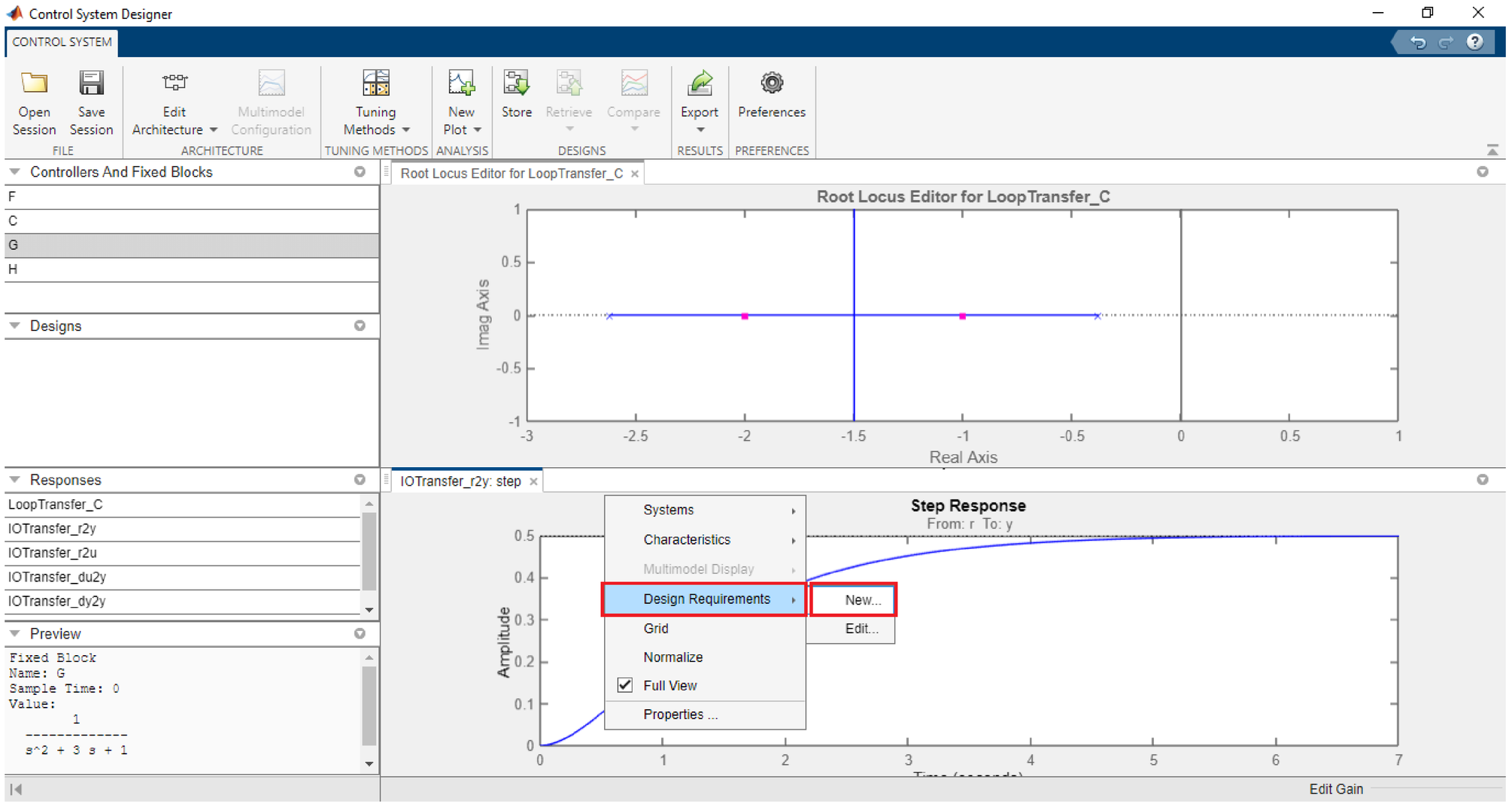
1.4 การออกแบบระบบควบคุมโดยใช้เครื่องมือ SISOTOOL

- Compensator Format



1.4 การออกแบบระบบควบคุมโดยใช้เครื่องมือ SISOTOOL

- Time domain (Rise time < 2 s, Setting time < 3 s, %Overshoot < 5%)

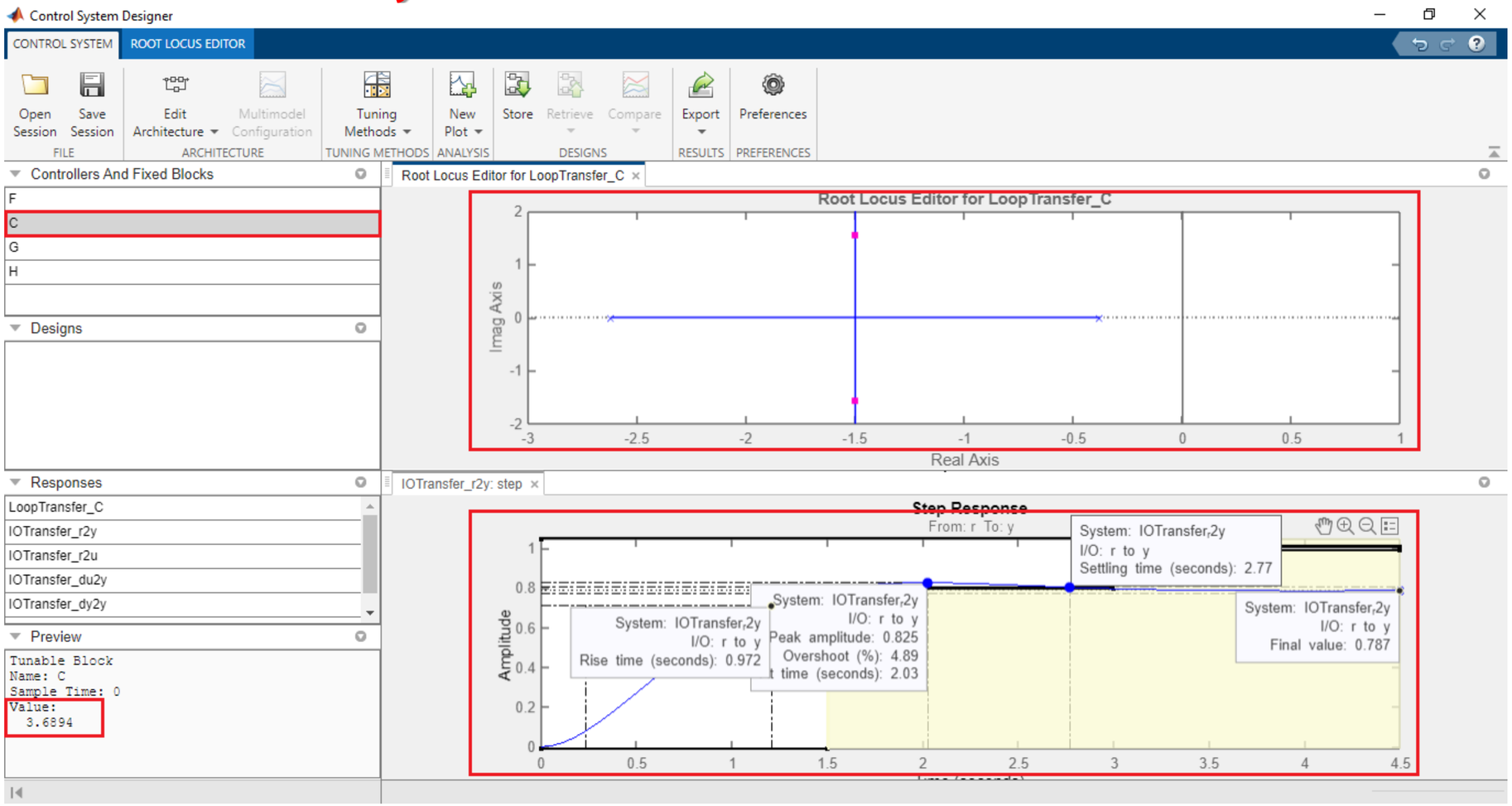


1.4 การออกแบบระบบควบคุมโดยใช้เครื่องมือ SISOTOOL

SCHOOL OF ENGINEERING KMITL

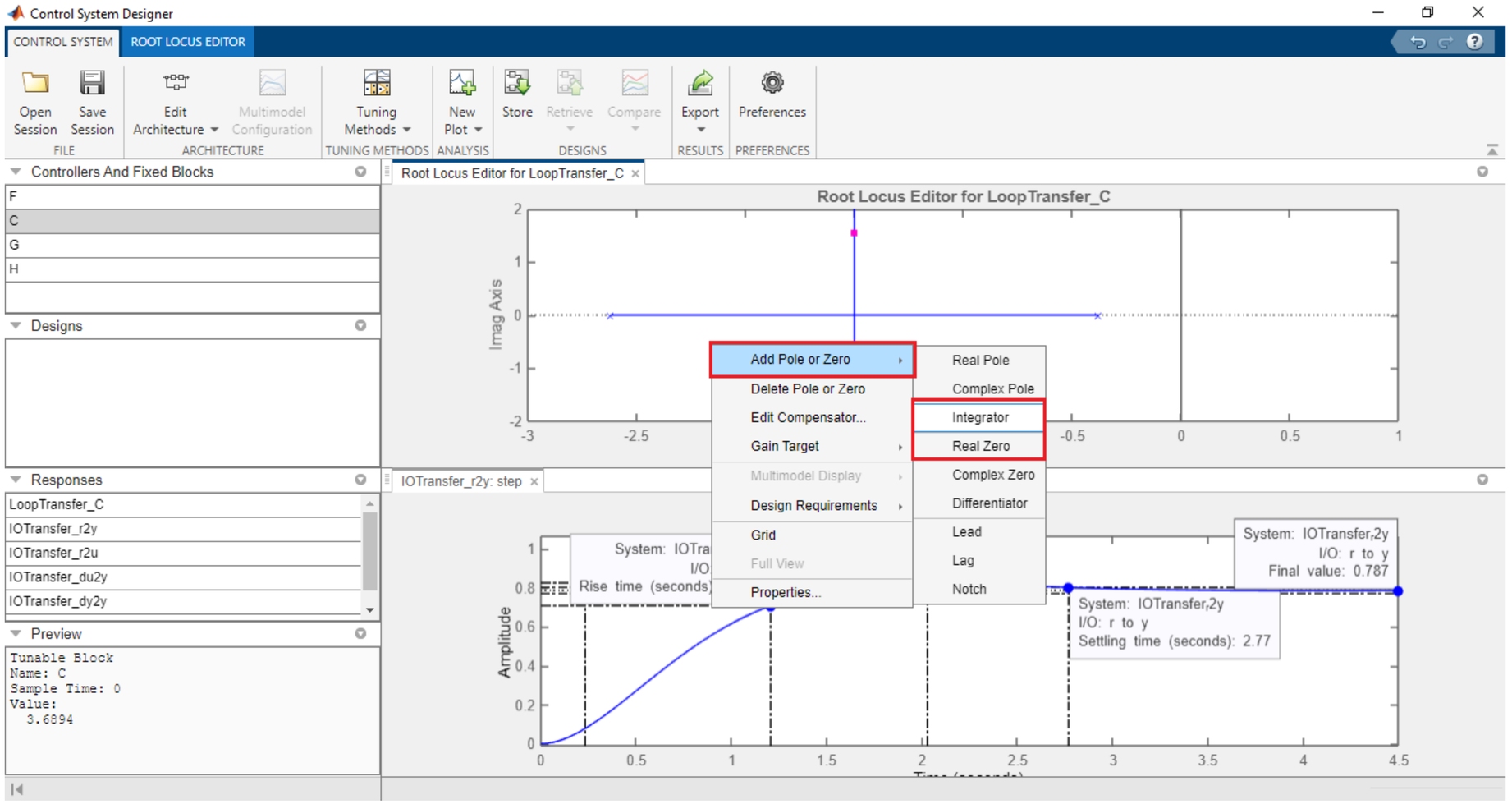
- P Control (Rise time =0.972 s, Setting time = 2.77 s, %Overshoot = 4.89%,

Error steady-state = 0.213)



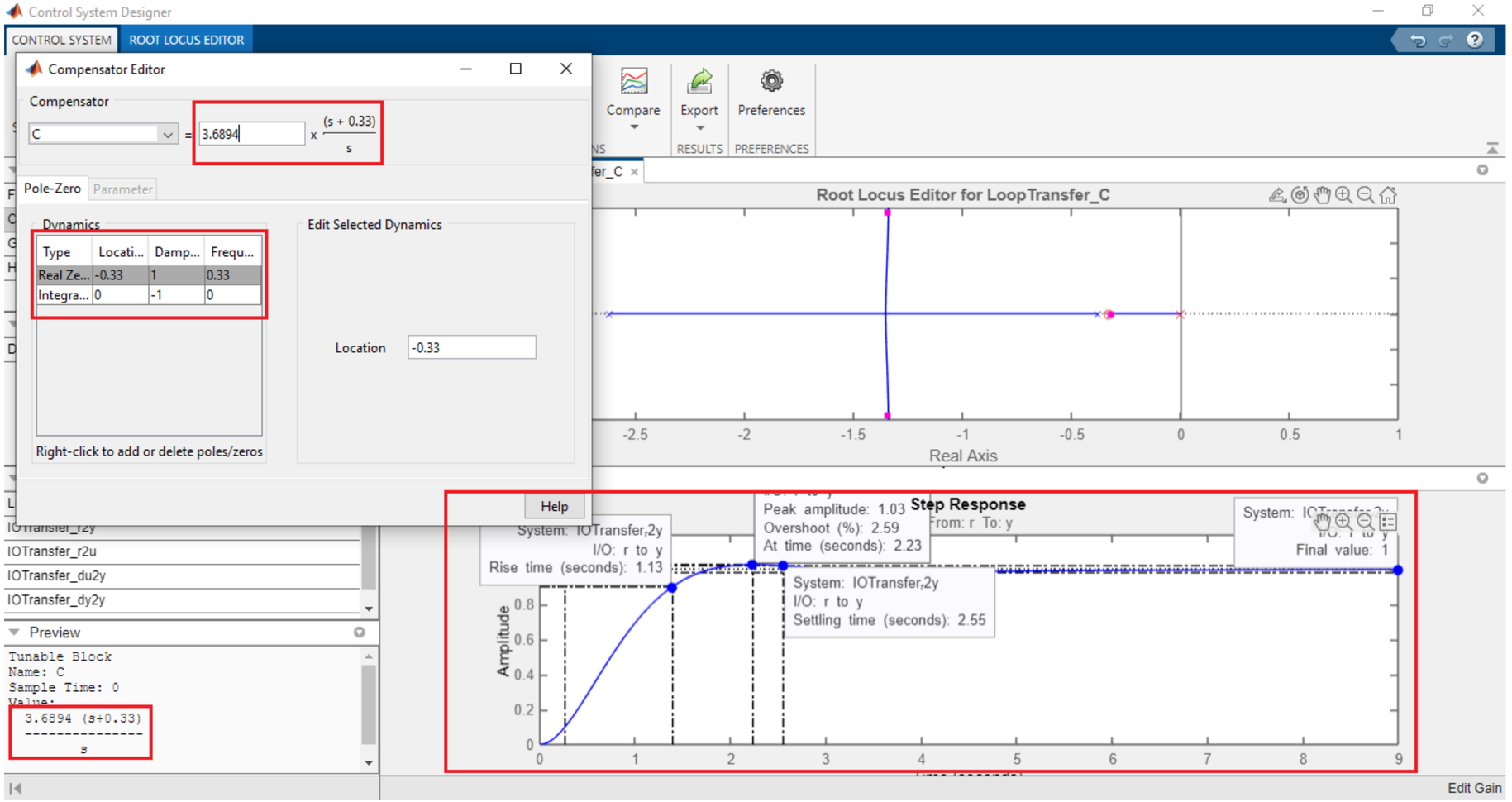
1.4 การออกแบบระบบควบคุมโดยใช้เครื่องมือ SISOTOOL

- PI Control (Rise time =0.972 s, Setting time = 2.77 s, %Overshoot = 4.89%,



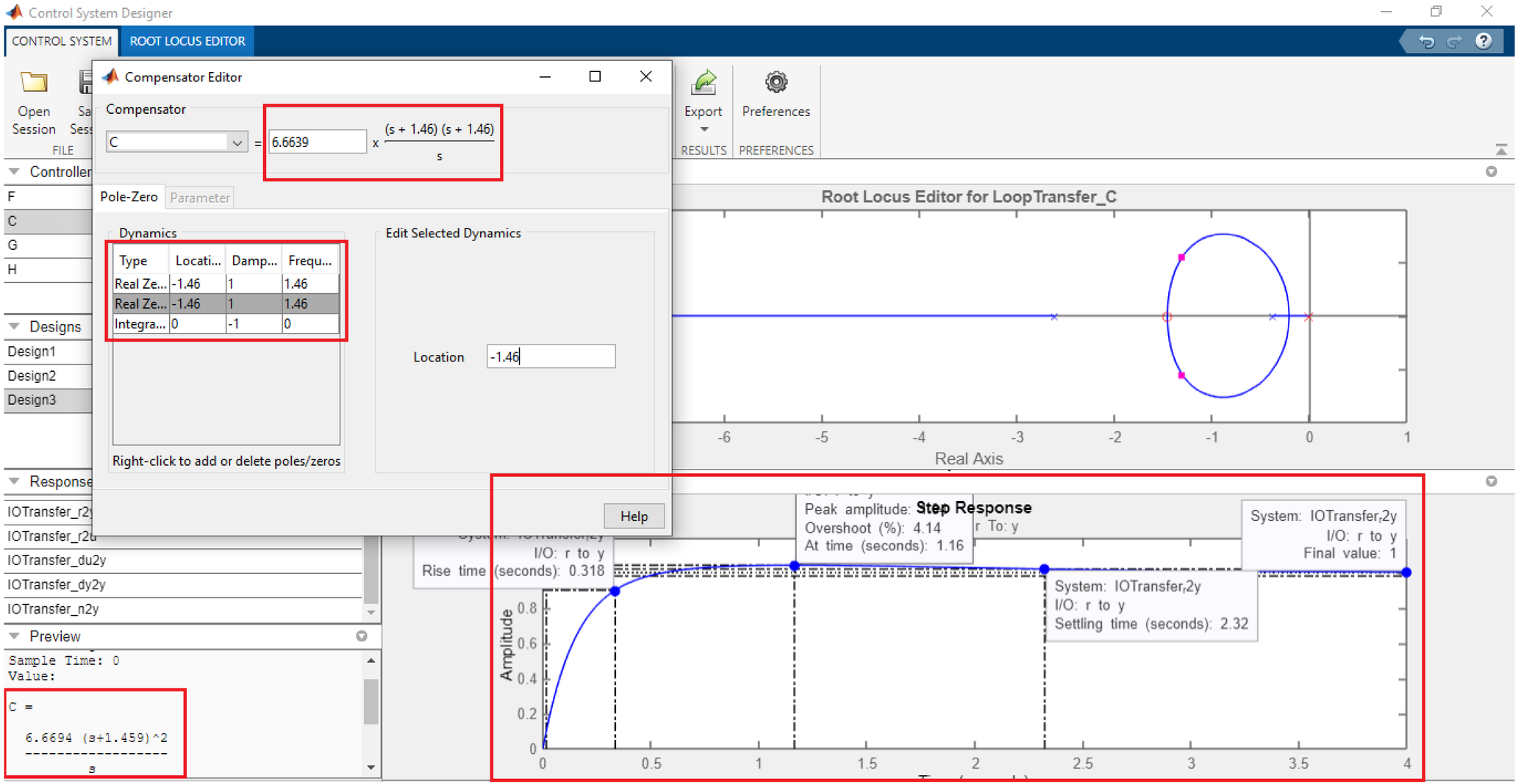
1.4 การออกแบบระบบควบคุมโดยใช้เครื่องมือ SISOTOOL

- PI Control (Rise time =1.13 s, Setting time = 2.55 s, %Overshoot = 2.59%)



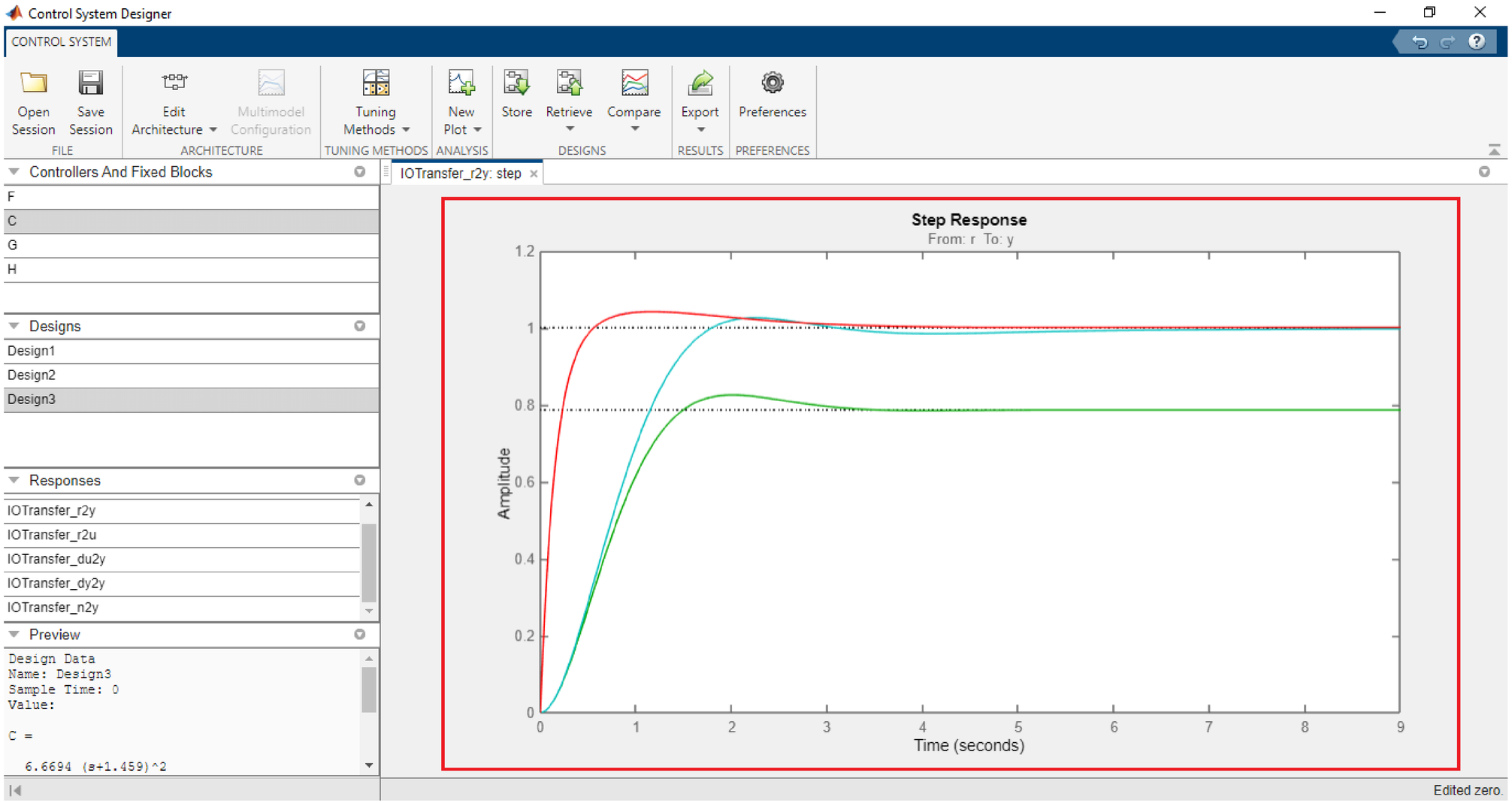
1.4 การออกแบบระบบควบคุมโดยใช้เครื่องมือ SISOTOOL

- PID Control (Rise time =0.318 s, Setting time = 2.32 s, %Overshoot = 4.14%)



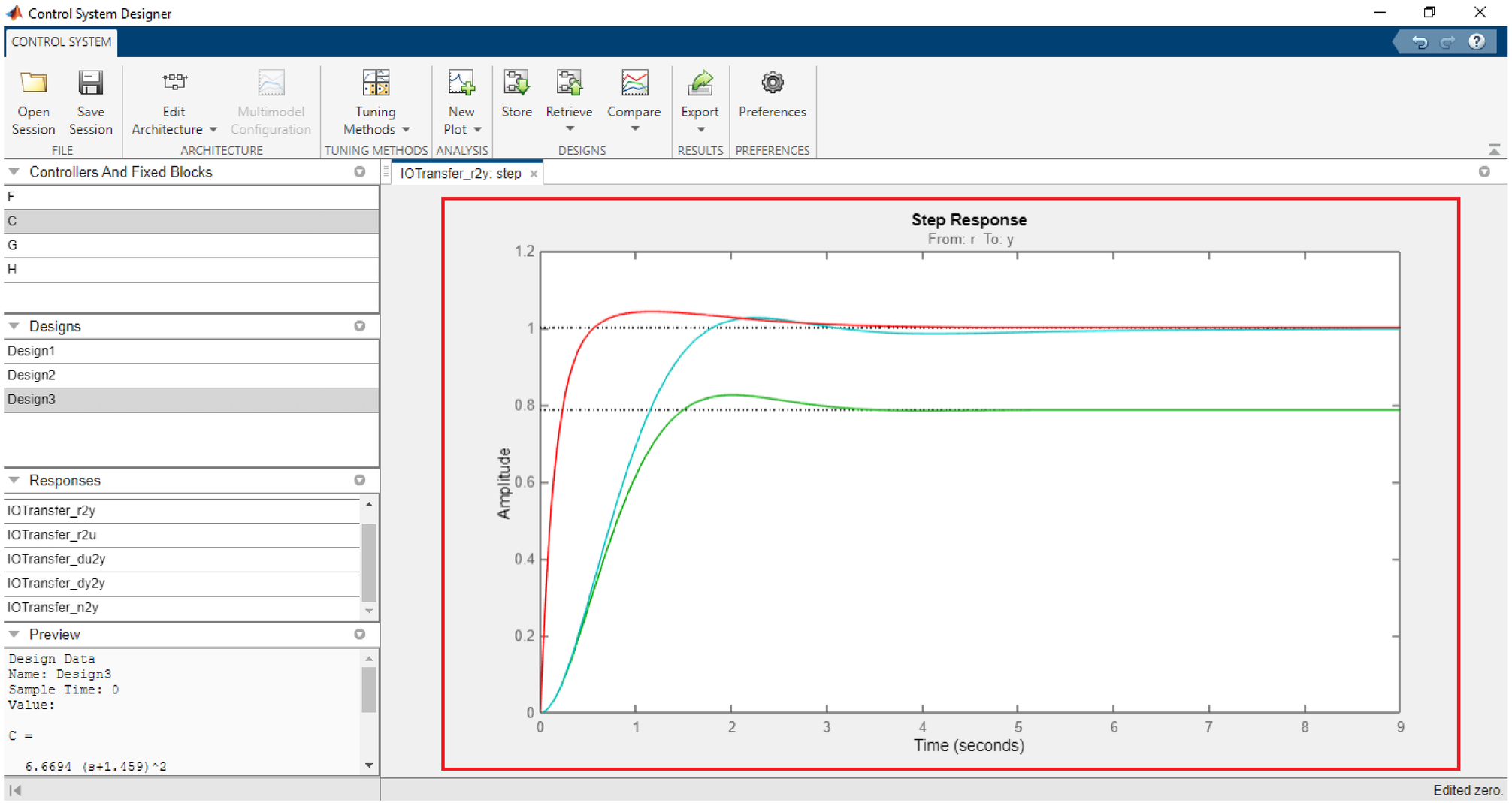
1.4 การออกแบบระบบควบคุมโดยใช้เครื่องมือ SISOTOOL

- P/PI/PID Compare



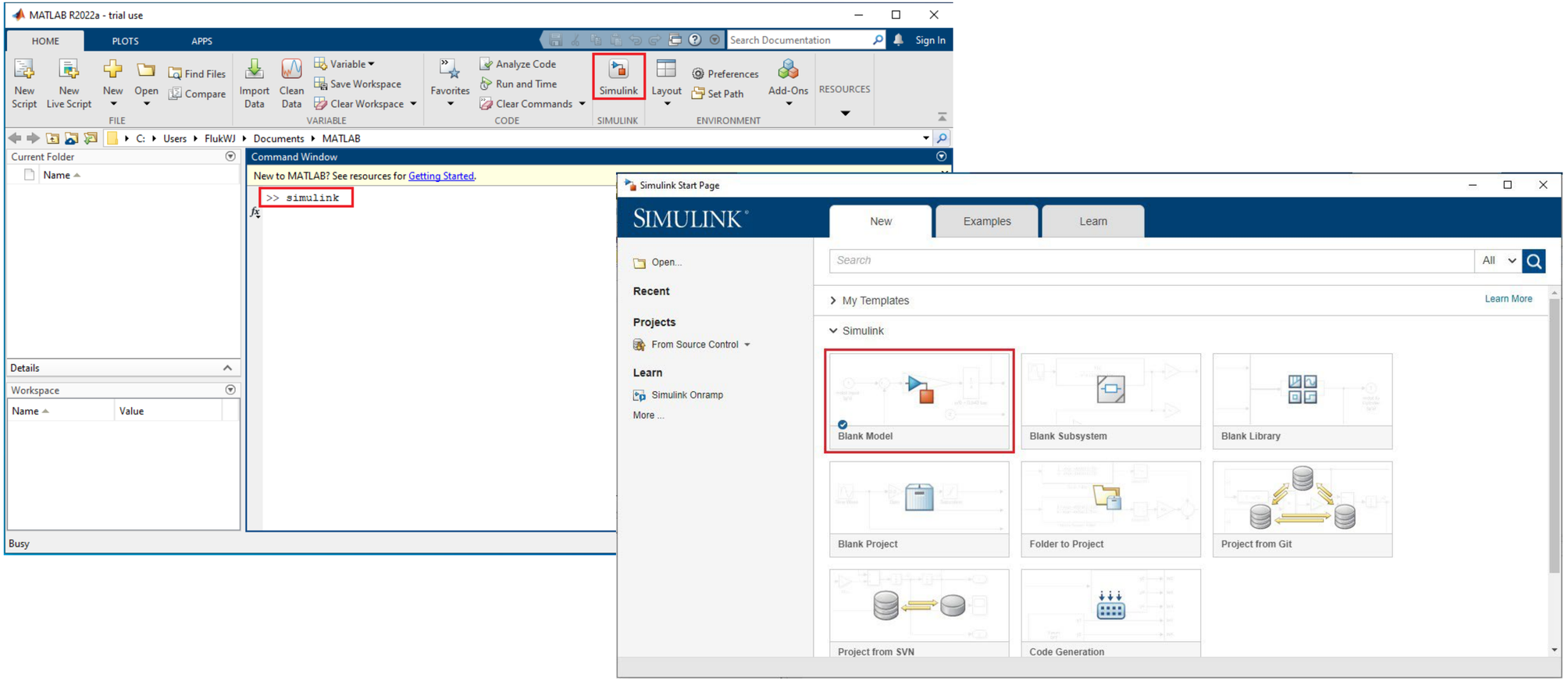
1.4 การออกแบบระบบควบคุมโดยใช้เครื่องมือ SISOTOOL

- P/PI/PID Compare



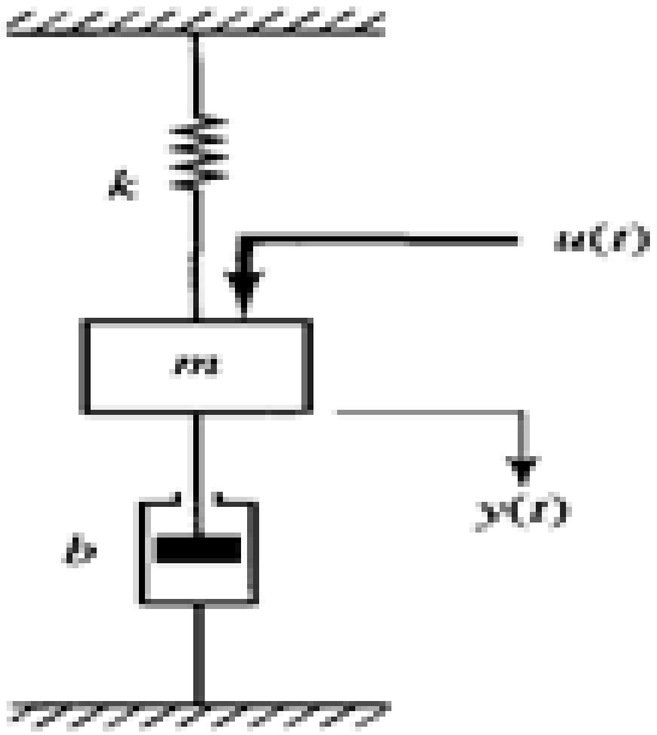
1.5 การใช้เครื่องมือ Simulink

- Open the Simulink



1.5 การใช้เครื่องมือ Simulink

- การสร้างแบบจำลองกระบวนการ
  - Differential equation
  - Transfer function
  - State-space

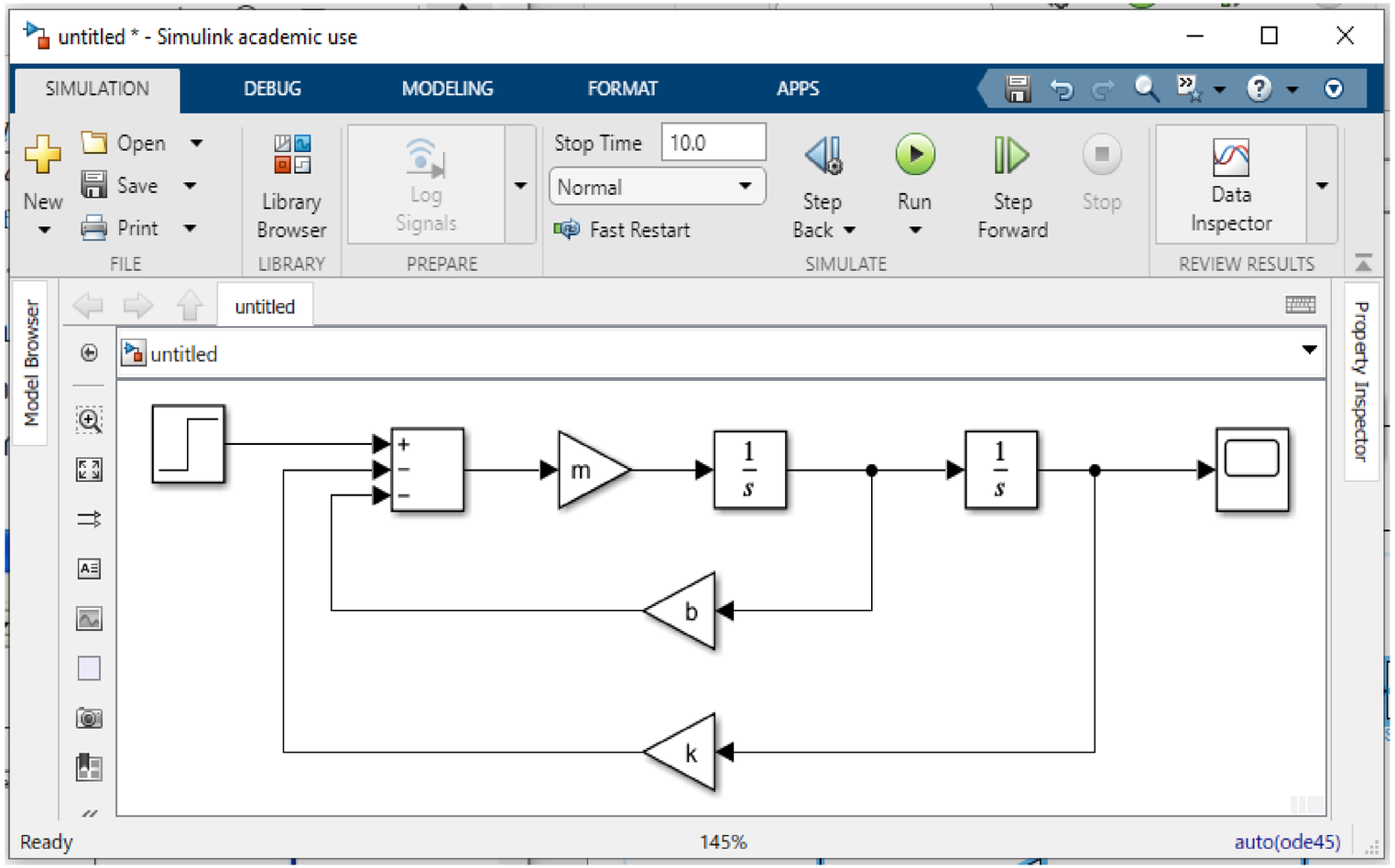


$m = 1, b = 0.5$

$k = 0.8$

1.5 การใช้เครื่องมือ Simulink

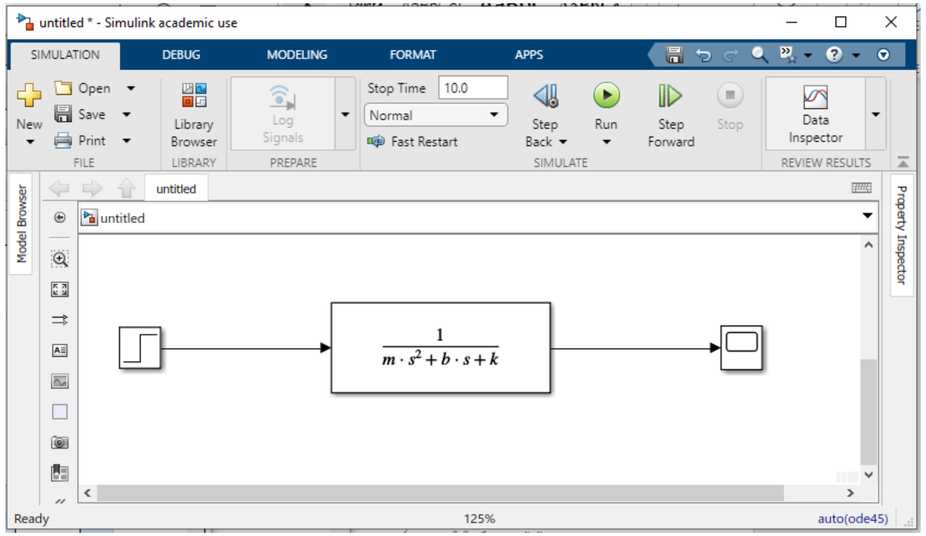
- Differential Equation $m\frac{d^2y}{dt} + b\frac{dy}{dt} + ky = u$ $\frac{d^2y}{dt} + 0.5\frac{dy}{dt} + 0.8y = u$

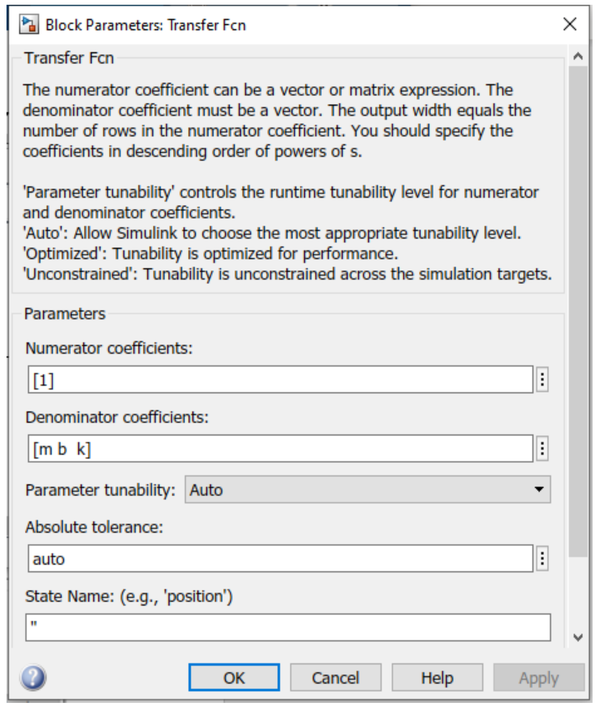


1.5 การใช้เครื่องมือ Simulink

- Transfer Function

$$\frac{Y(s)}{U(s)} = \frac{1}{ms^2 + bs + k} \qquad \frac{Y(s)}{U(s)} = \frac{1}{s^2 + 0.5s + 0.8}$$





1.5 การใช้เครื่องมือ Simulink

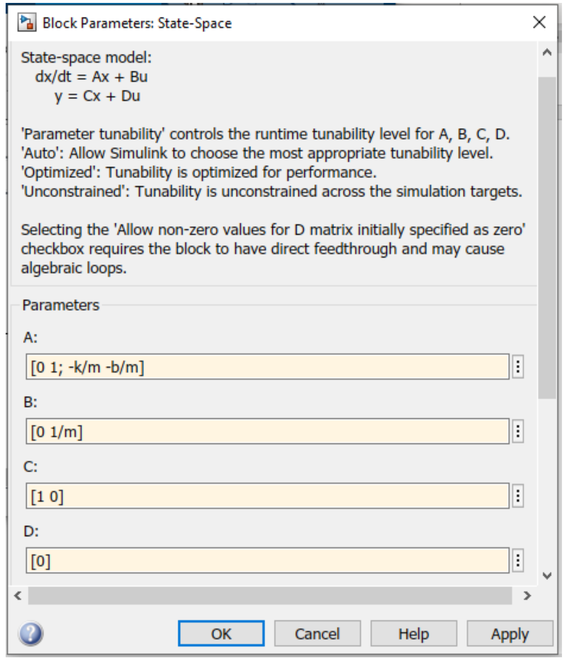
- State-Space

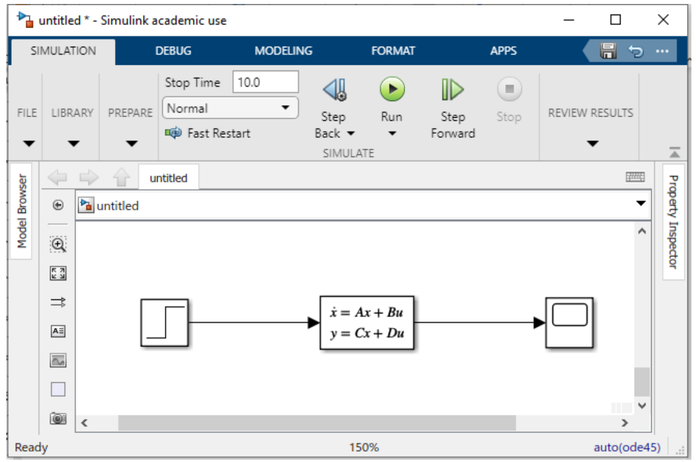
$$\begin{bmatrix} \dot{y} \\ \ddot{y} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{-k}{m} & \frac{-b}{m} \end{bmatrix} \begin{bmatrix} y \\ \dot{y} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} y \\ \dot{y} \end{bmatrix}$$

$$\begin{bmatrix} \dot{y} \\ \ddot{y} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -0.8 & -0.5 \end{bmatrix} \begin{bmatrix} y \\ \dot{y} \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} y \\ \dot{y} \end{bmatrix}$$





1.6 PID tuning

- Ziegler – Nichols method

3.1.1 Ziegler- Nichols method:

The Proportional-Integral-Derivative controller is realised as follows:

$$C(s) = K_p + \frac{K_i}{s} + K_d s$$

Where K_p = proportional gain, K_i = integral gain, and K_d = derivative gain.

In this Ziegler-Nichols it is only valid to open loop plants which are stable [20] as it is an open-loop tuning done by experimentation. In this our prior thing is to find the parameters A and L which we can get it through the plants step response as shown in Fig. 1.8. Firstly we should determine the point where it shows the maximum slope and draw a tangent, this tangent intersects with the vertical axis produces A and intersection with the horizontal axis produces L. By now after we find A and L we can find the Proportional-Integral-Derivative parameters to control.

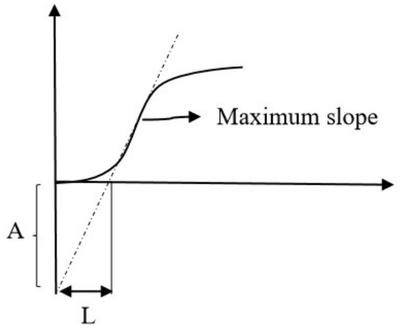


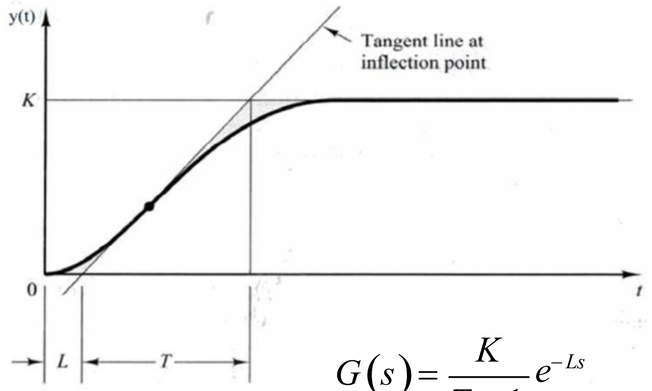
Fig.3.1. plant step response to get A and L.

Empirically obtained formulas are there to produce Proportional-Integral-Derivative control parameters from which we observe that after one oscillation there is a decay in its first overshoot of 0.25 times the original value.

- Step response Method

Tuning formula

Controller type	For step response			For frequency response		
	K_p	τ_i	τ_d	K_p	τ_i	τ_d
P	1/a			0.5Kc		
PI	0.9/a	3L		0.4Kc	0.8Tc	
PID	1.2/a	2L	L/2	0.6Kc	0.5Tc	0.12Tc



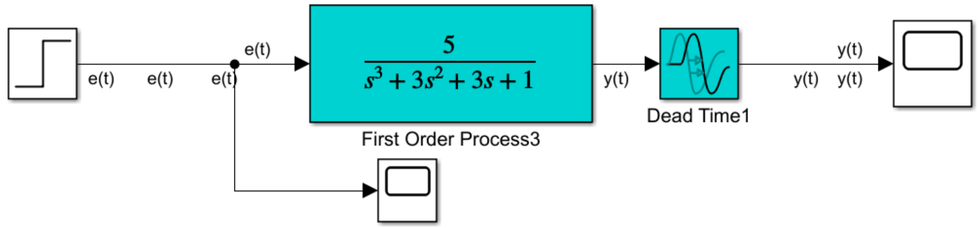
$$G(s) = \frac{K}{Ts+1} e^{-Ls}$$

K=gain; L= time delay; T= time constant $\quad a = \frac{KL}{T}$

Ref: Ruben Kandulna “Design of PID Controller for FOPDT and IPDT System”

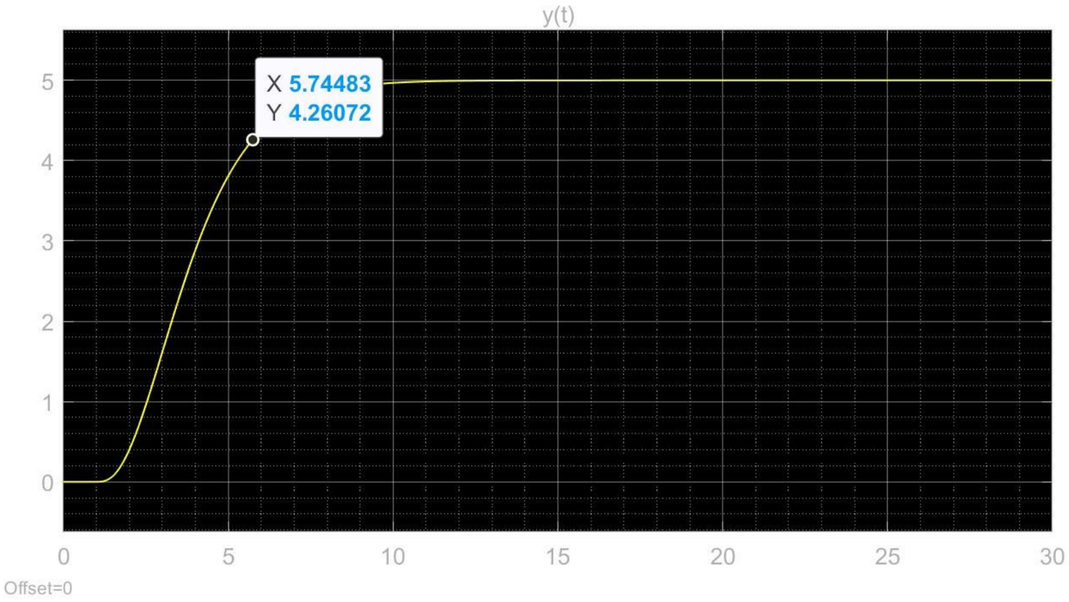
1.6 PID tuning

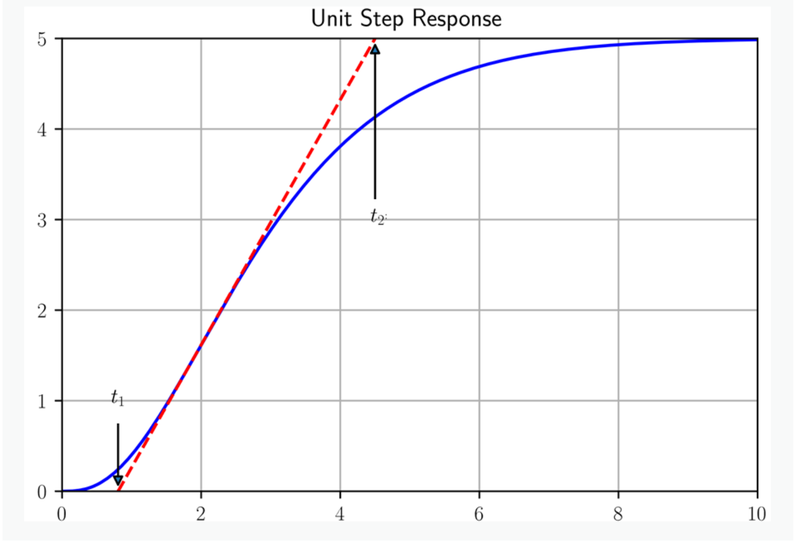
- Ziegler – Nichols method



L=1.15

T=5.74-1.15 =(t2-t1) =4.59

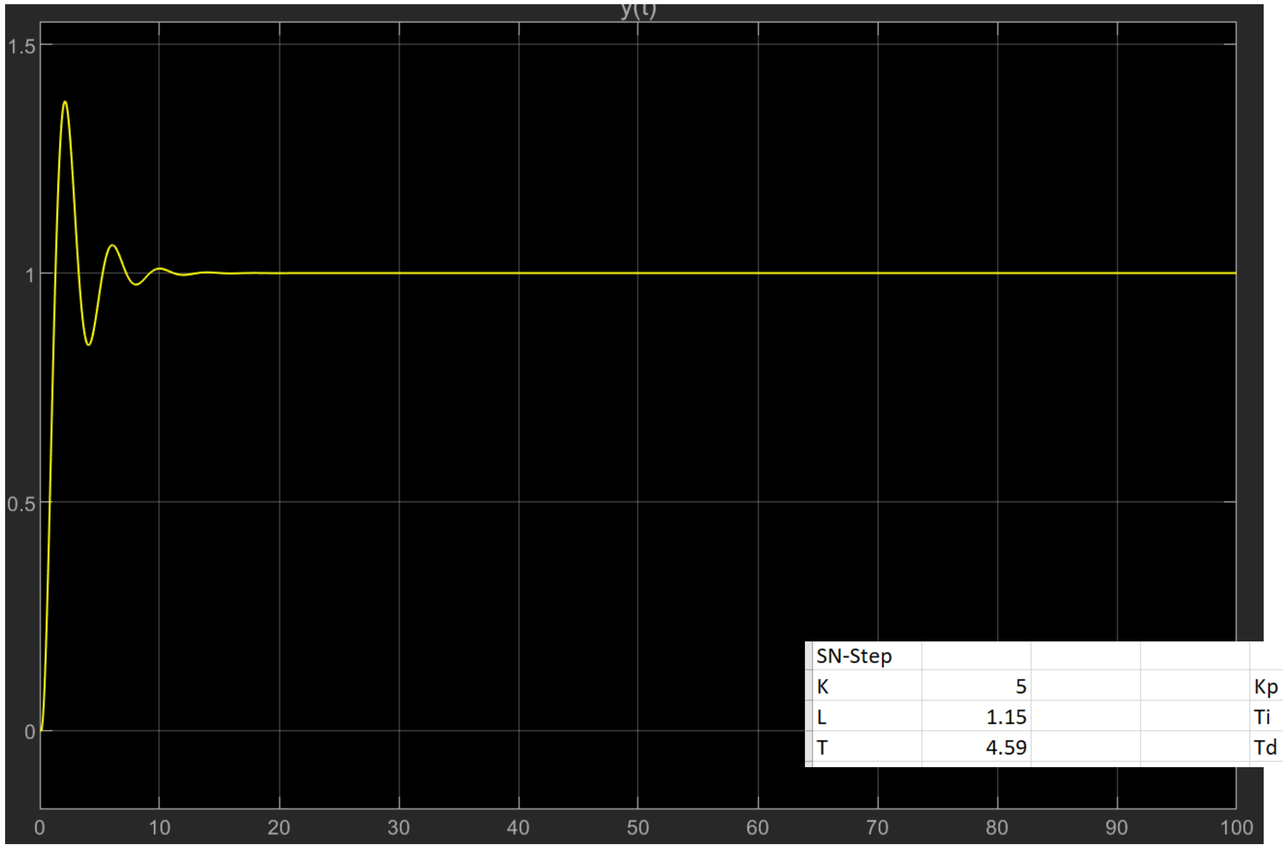




Ref: Ruben Kandulna “Design of PID Controller for FOPDT and IPDT System”

1.6 PID tuning

- Ziegler – Nichols method



SN-Step							
K	5			Kp	0.957913		
L	1.15			Ti	2.3	Ki	0.416484
T	4.59			Td	0.575	Kd	0.5508

Ref: Ruben Kandulna “Design of PID Controller for FOPDT and IPDT System”

1.6 PID tuning

- Self-Oscillation method

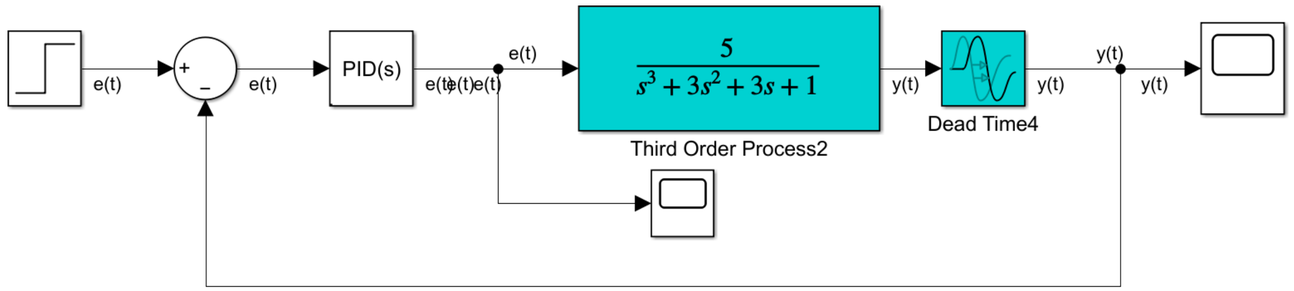
  - Self-Oscillation Method

- Set the integrator and derivative gain to 0.
- Increase the proportional gain K_p from 0, until the closed-loop system starts oscillation.
- Record the critical propotional gain K_{cr} and the oscillation period P_{cr}.
- Notice that if the open-loop transfer function of the plant is given, then
  - The critical gain is the gain margin, i.e., $K_{cr} = K_g$.
  - The critical oscillation period is related to the phase crossover frequency:

$$P_{cr} = 2\pi/\omega_\phi.$$

- Set the PID controller according to

Controller	K_p	T_i	T_d
P	$0.5K_{cr}$		
PI	$0.45K_{cr}$	$P_{cr}/1.2$	
PID	$0.6K_{cr}$	$0.5P_{cr}$	$0.125P_{cr}$

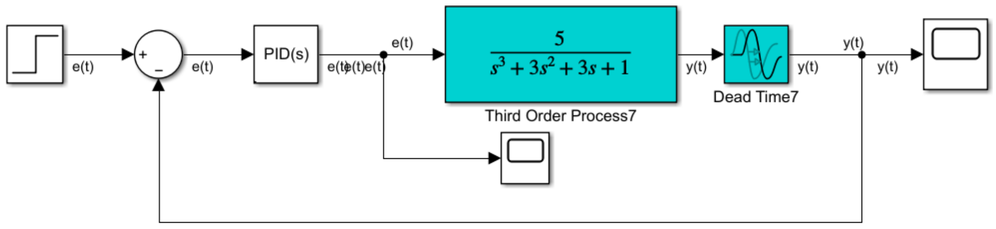


Ziegler-Nichols tuning: Self-Oscillation Method

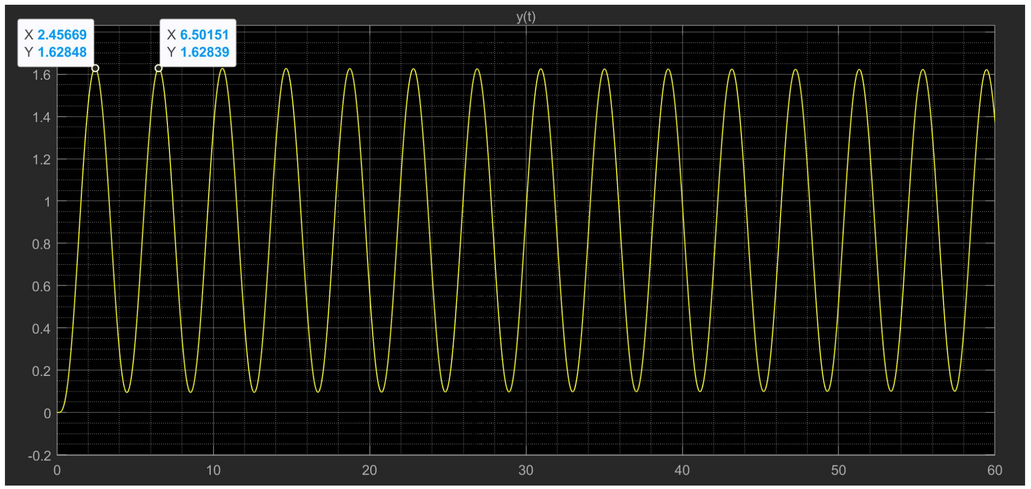
Ref: Ruben Kandulna “Design of PID Controller for FOPDT and IPDT System”

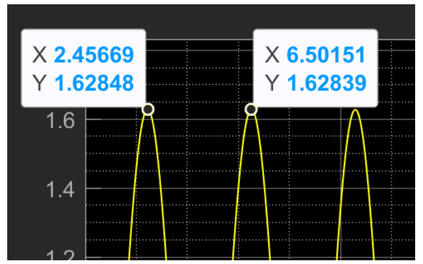
1.6 PID tuning

- Self-Oscillation method



Ziegler-Nichols tuning: Self-Oscillation Method



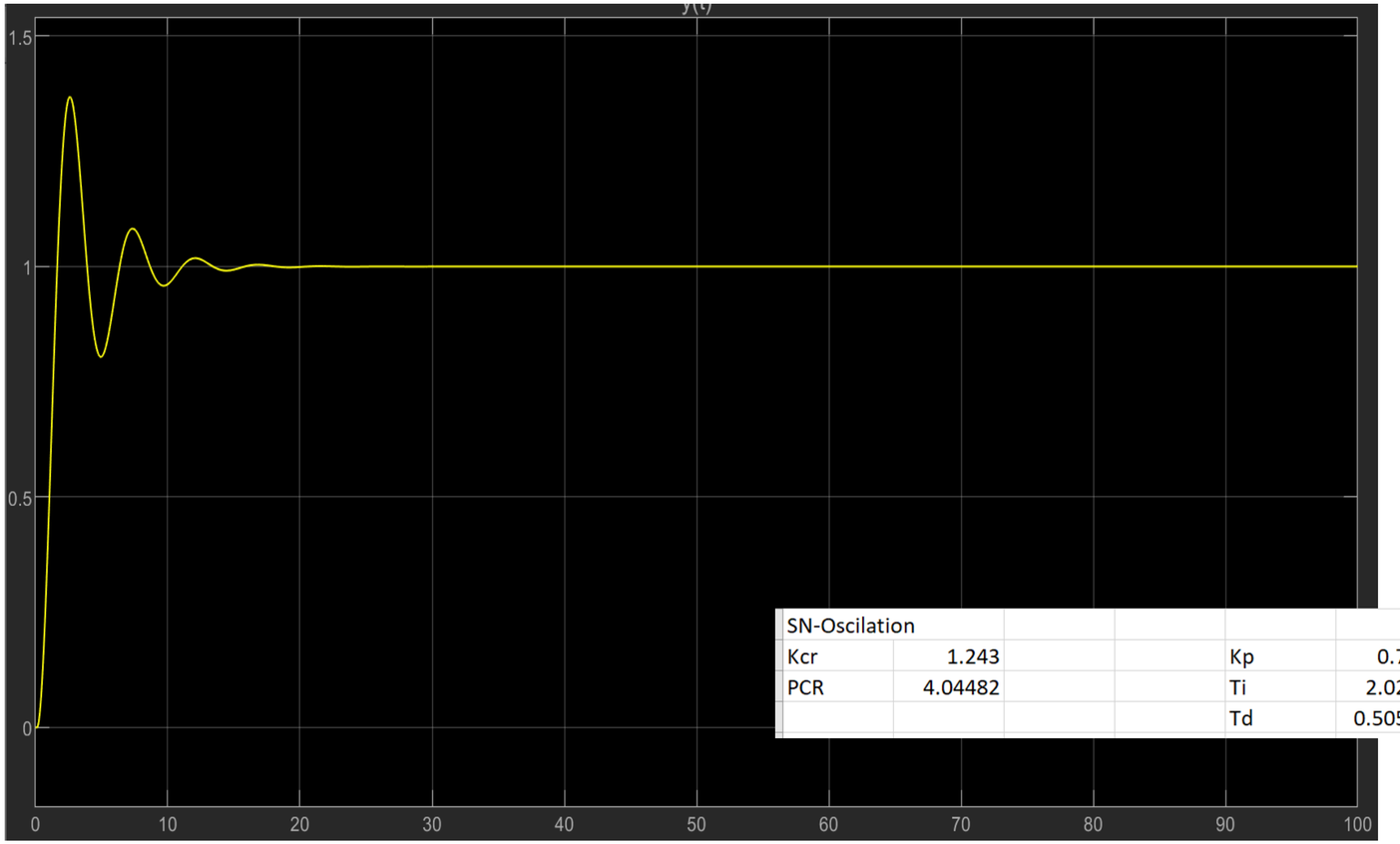


Kcr	1.243
PCR	4.04482

Ref: Ruben Kandulna “Design of PID Controller for FOPDT and IPDT System”

1.6 PID tuning

- Self-Oscillation method



SN-Oscilation							
Kcr	1.243		Kp	0.7458			
PCR	4.04482		Ti	2.02241		Ki	0.368768
			Td	0.505603		Kd	0.377078

Ref: Ruben Kandulna “Design of PID Controller for FOPDT and IPDT System”

1.6 PID tuning

- Chine-Hrones-Reswick method

3.1.2 Chine-Hrones-Reswick tuning:

This method focus on the main problem consisting of how to regulate set-point and how to reject the disturbances. Also regarding speed of response and overshoot an additional comment comparable with the Ziegler–Nichols tuning formula, the time constant T is been used clearly in this CHR method [19].

Closed-loop response which is more heavily damped, guarantees for an ideal plant and the one which is having high response speed without overshoot is considered as overshoot of 0% and other with good response speed with 20% overshoot is considered as overshoot of 20%.

Set point regulation

Controller type	Overshoot of 0%			Overshoot of 20%		
	K_p	τ_i	τ_d	K_p	τ_i	τ_d
P	0.3/a			0.7/a		
PI	0.35/a	1.2T		0.6/a	T	
PID	0.6/a	T	0.5L	0.95/a	1.4T	0.47L

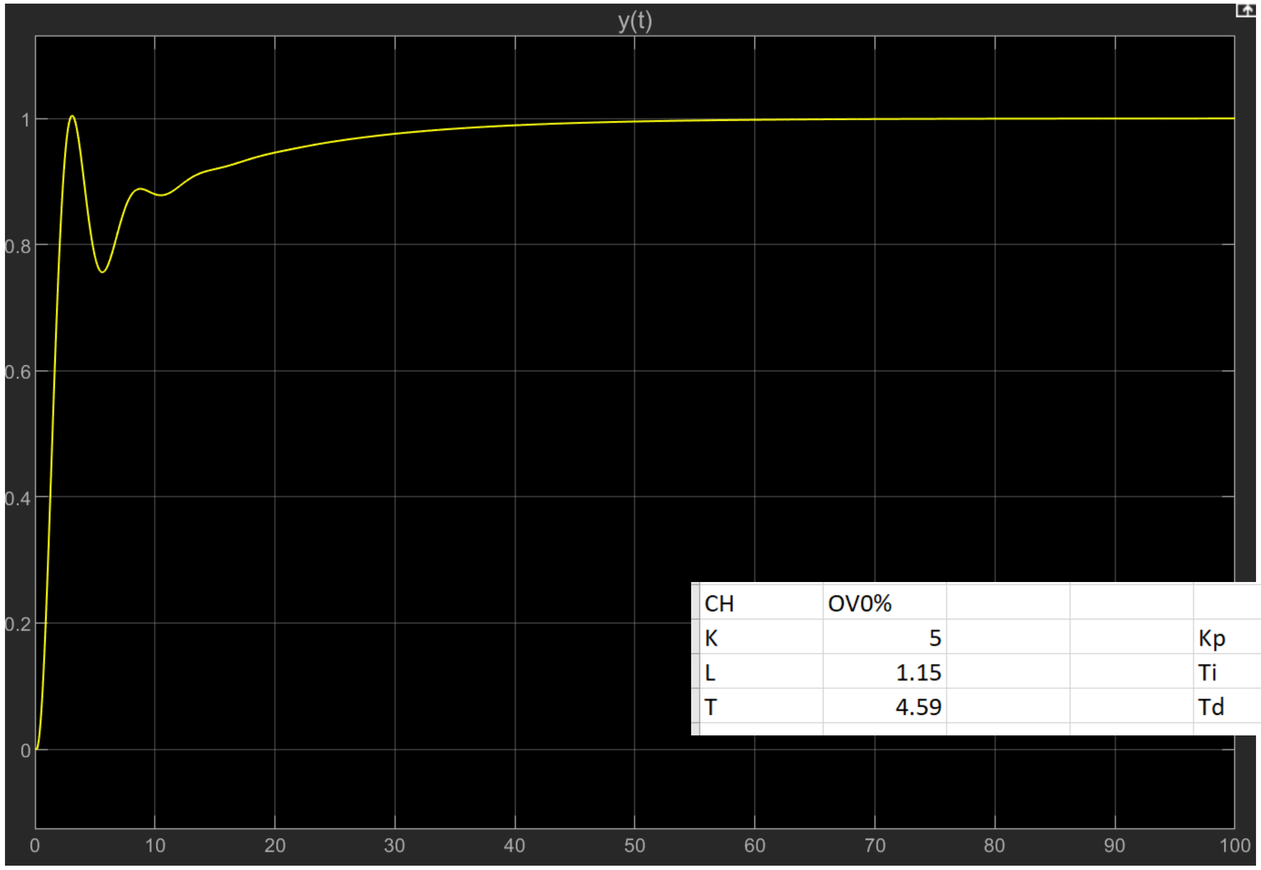
Disturbance rejection

Controller type	Overshoot of 0%			Overshoot of 20%		
	K_p	τ_i	τ_d	K_p	τ_i	τ_d
P	0.3/a			0.7/a		
PI	0.6/a	4L		0.7/a	2.3L	
PID	0.95/a	2.4L	0.42L	1.2/a	2L	0.42L

Ref: Ruben Kandulna “Design of PID Controller for FOPDT and IPDT System”

1.6 PID tuning

- Chine-Hrones-Reswick method



CH	OV0%						
K	5			Kp	0.478957		
L	1.15			Ti	4.59	Ki	0.104348
T	4.59			Td	0.575	Kd	0.2754

Ref: Ruben Kandulna “Design of PID Controller for FOPDT and IPDT System”

1.6 PID tuning

- Cohen-Coon method

3.1.3 Cohen-Coon Tuning algorithm:

Cohen-Coon method [18] is a dominant pole design method and tries to locate some poles to attain definite performance control. The. It is based on first –order plus dead time model:

$$G(s) = \frac{K}{Ts+1} e^{-Ls}$$

This tuning method approach was to decay the amplitude ratio for load disturbance so, the load disturbance is rejected also to minimize the integrator error. This gives good load disturbance rejection, Proportional-Integral-Derivative parameters in relation to K, T, and L:

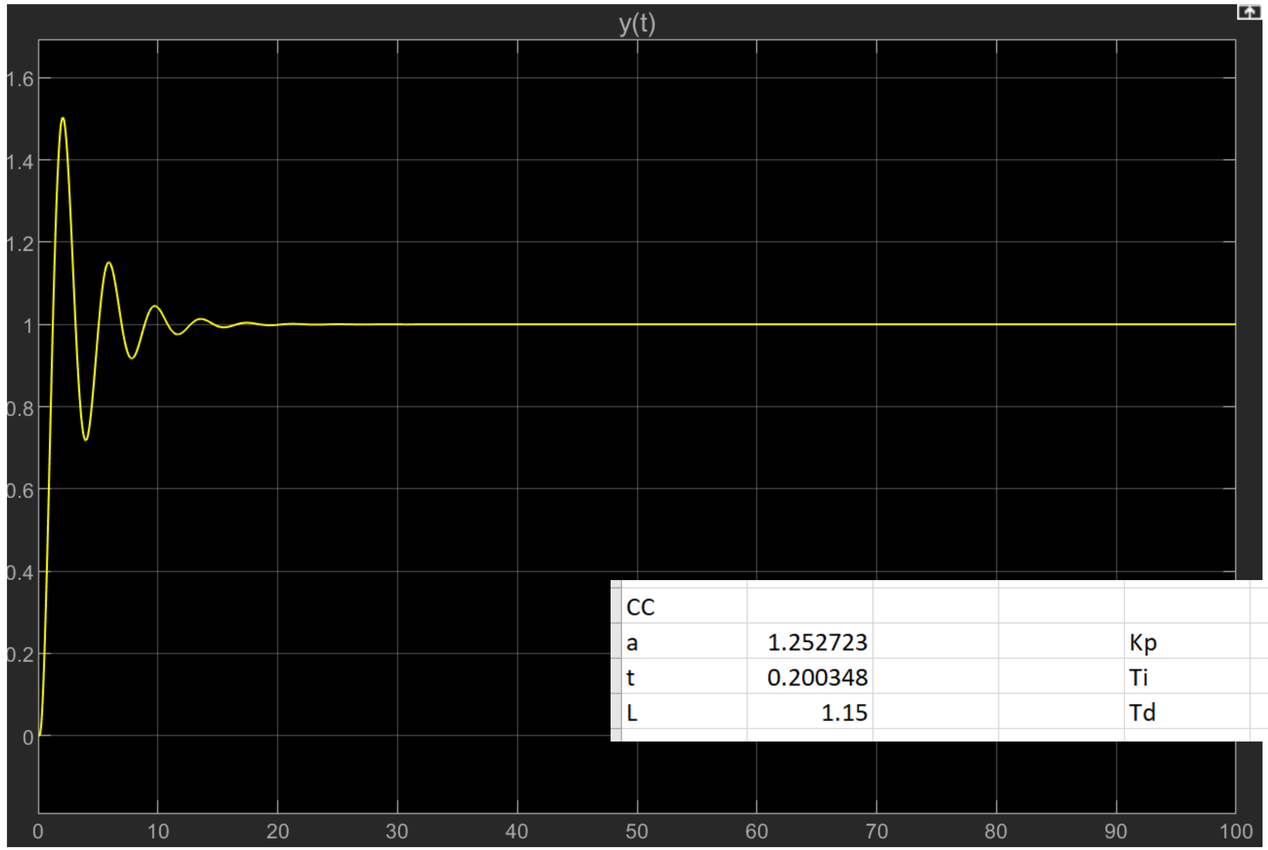
$$a = \frac{KL}{T},\ \tau = \frac{L}{L+T}$$

Controller type	K_p	τ_i	τ_d
P	$\frac{1}{a}\left(1 + \frac{0.35\tau}{1-\tau}\right)$		
PI	$\frac{0.9}{a}\left(1 + \frac{0.92\tau}{1-\tau}\right)$	$\frac{3.3-3\tau}{1+1.2\tau}L$	
PD	$\frac{1.24}{a}\left(1 + \frac{0.13\tau}{1-\tau}\right)$		$\frac{0.27-0.36\tau}{1-0.87\tau}L$
PID	$\frac{1.35}{a}\left(1 + \frac{0.18\tau}{1-\tau}\right)$	$\frac{2.5-2\tau}{1-0.39\tau}L$	$\frac{0.37-0.37\tau}{1-0.81\tau}L$

Ref: Ruben Kandulna “Design of PID Controller for FOPDT and IPDT System”

1.6 PID tuning

- Cohen-Coon method



CC					
a	1.252723	Kp	1.126252		
t	0.200348	Ti	2.618823	Ki	0.43006
L	1.15	Td	0.406165	Kd	0.457444

Ref: Ruben Kandulna “Design of PID Controller for FOPDT and IPDT System”

1.6 PID tuning

- Wang-Juang-Chan method

3.1.4. Wang-Juang-Chan method of tuning:

Name itself says that this tuning method is suggested by Wang, Juang, and Chan [9]. For choosing the Proportional-Integral-Derivative control parameters it is an easy & effective method which is built on the optimum Integral-Time-Absolute-Error criterion. The controller parameters can give by, if the parameters K, L & T of the plant are known

$$K_p = \frac{(0.73 + 0.53T/L)(T + L/2)}{K(T+L)} \tag{21}$$

$$\tau_i = T + \frac{L}{2}$$

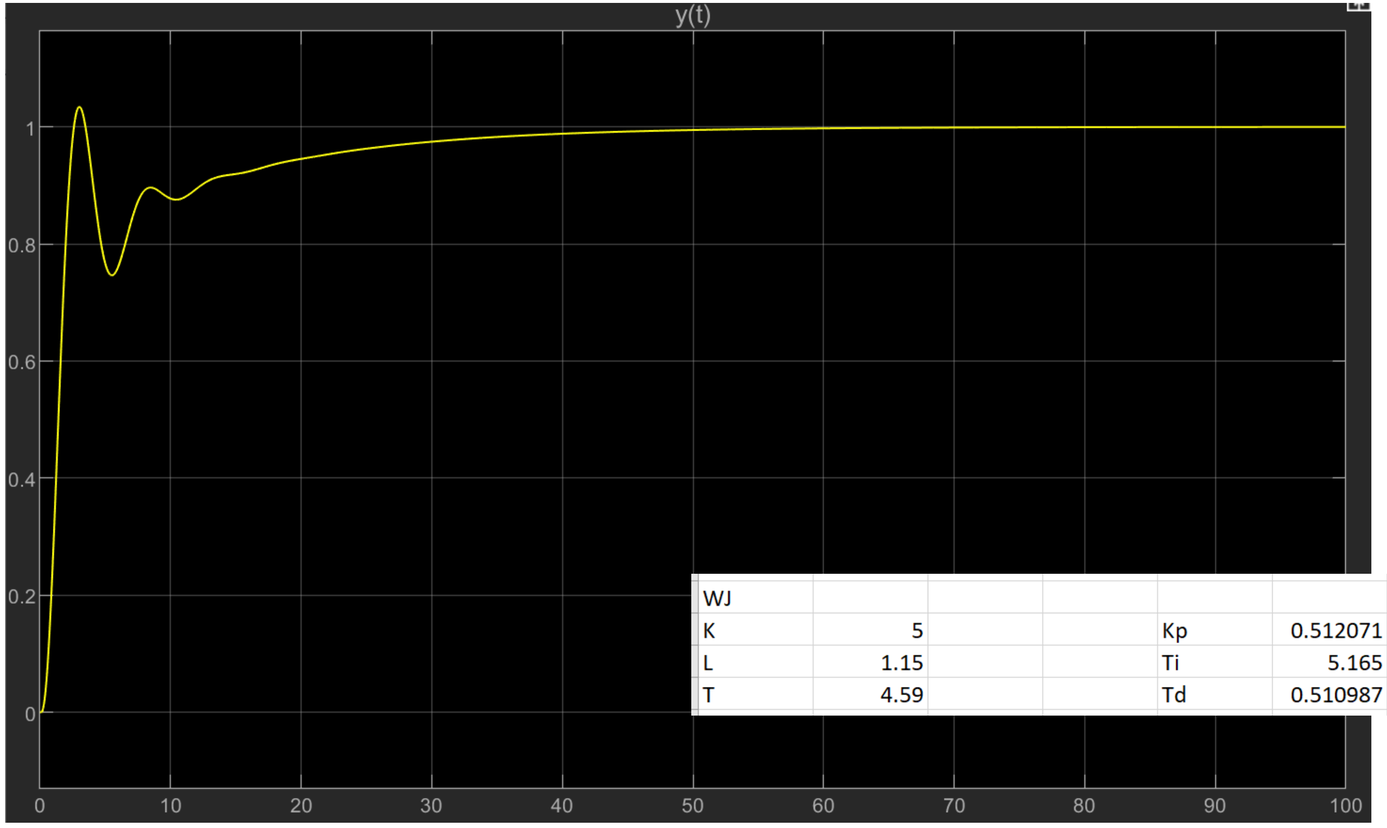
$$\tau_d = \frac{\frac{L}{2}T}{T + \frac{L}{2}}$$

$$K_p = 3.0568, \tau_i = 1.7925, \tau_d = 0.3255$$

Ref: Ruben Kandulna “Design of PID Controller for FOPDT and IPDT System”

1.6 PID tuning

- Wang-Juang-Chan method

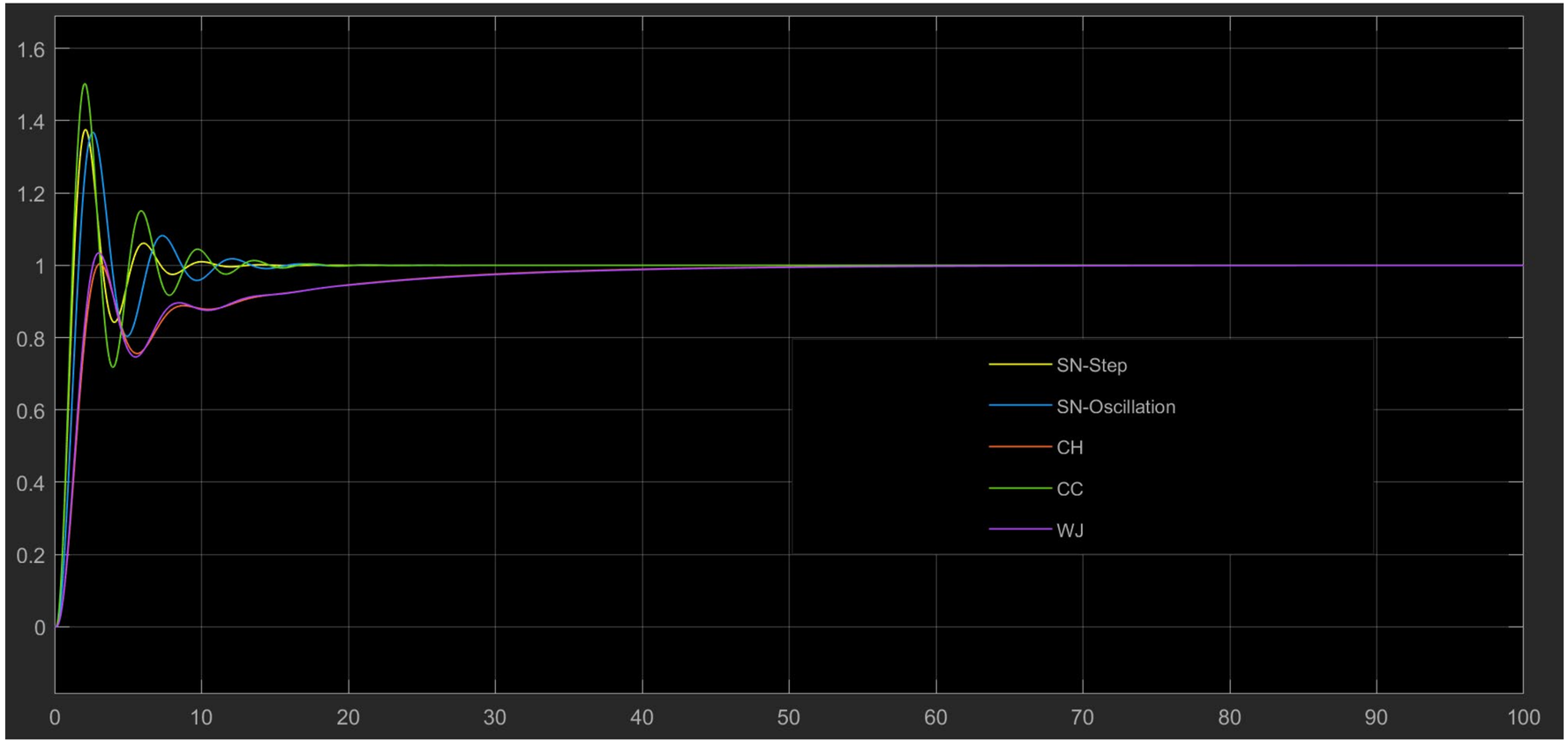


WJ					
K	5	Kp	0.512071		
L	1.15	Ti	5.165	Ki	0.099143
T	4.59	Td	0.510987	Kd	0.261662

Ref: Ruben Kandulna “Design of PID Controller for FOPDT and IPDT System”

1.6 PID tuning

- Wang-Juang-Chan method



Ref: Ruben Kandulna “Design of PID Controller for FOPDT and IPDT System”

1.7 คำถามท้ายการทดลอง

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- นักศึกษาจงยกตัวอย่างการเขียนโปรแกรมแก้ปัญหาเชิงวิศวกรรมมา 1 ตัวอย่าง
- การเขียนฟังก์ชันและสคริปมีข้อแตกต่างกันอย่างไร
- การลดรูประบบควบคุมมีคำสั่งที่จำเป็นอะไรบ้าง
- การสร้างอินพุตให้ระบบควบคุมมีวิธีการสร้างอย่างไรบ้าง
- การพล็อตกราฟในโปรแกรมมีข้อกำหนดอย่างไรบ้าง
- What are 5 Sentences Key takeaway for LAB1?

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